

Intro2SE – Requirements Analysis – 7

MeoArc

The student team is required to complete the Software Requirements Specification (SRS) document for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
VNU-HCMUS

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Software Requirements Specification

Objectives

This document focus on the following topics:

- ✓ Completing the Software Requirements Specification (SRS) document with the following contents:
 - Elaborate on the Problem Statement
 - Overview of Requirements (Functional and Non-Functional), Stakeholders
 - Use Case Model
 - Use Case Specifications
 - Create Prototype and Mockup Diagrams of the System Interface
- ✓ Understanding the Project Requirement Analysis document.
- ✓ This document will be used as input for AI tools to verify the quality of subsequent project artifacts.

All project artifacts must remain consistent and synchronized.

For example, if the project proposal is modified during this phase, a new version of the proposal must be documented.

By the conclusion of the project, all versions of every artifact must be submitted to demonstrate the evolution of your work.

1 Member Contribution Assessment

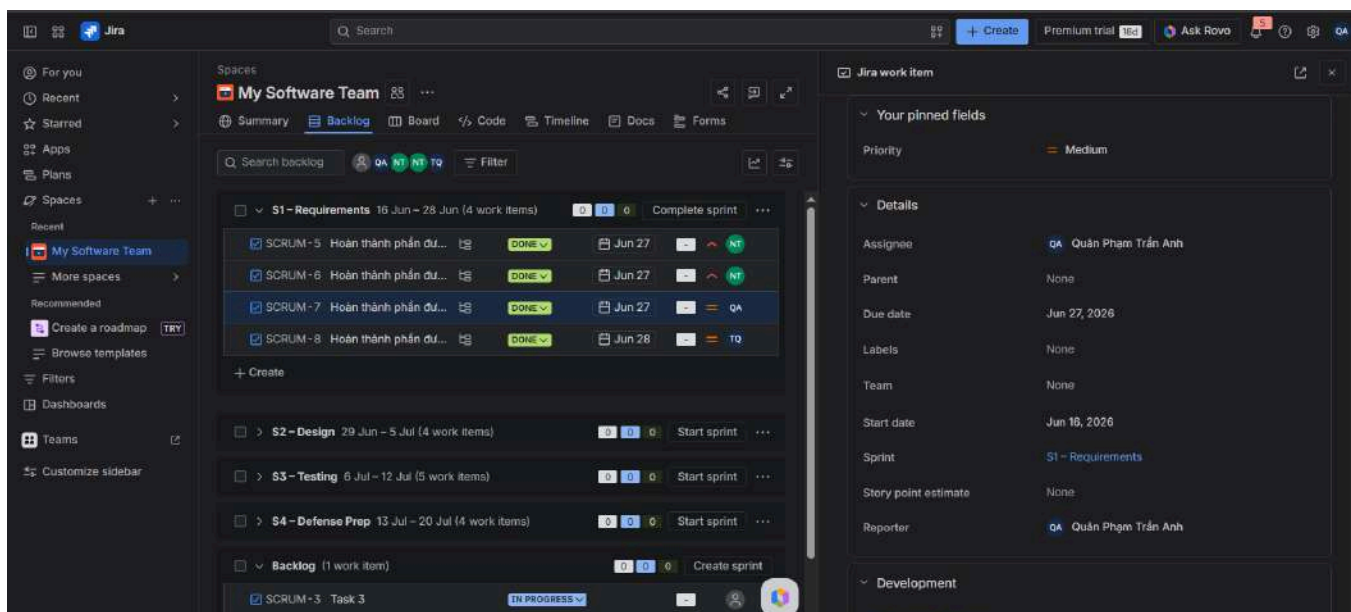
Written by: 24127226 – Phạm Trần Anh Quân

Edited by: 24127250 – Phan Quang Tiến

Reviewed by: 24127529 – Nguyễn Chí Tài

24127226 – Phạm Trần Anh Quân – 25%

- Task Evidence:



The screenshot displays the Jira interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with two sprints: "S1 - Requirements" (16 Jun - 28 Jun) and "S2 - Design" (29 Jun - 5 Jul). The "S1 - Requirements" sprint contains four work items (SCRUM-5 to SCRUM-8), all of which are "IN PROGRESS" and due by June 27 or 28. The "S2 - Design" sprint contains four work items (SCRUM-9 to SCRUM-12), all of which are "TO DO" and due by June 27 or 28.

The right sidebar shows the detailed view of a work item, "Viết Mục 3.1 - Stakeholders" (SCRUM-7). The work item is assigned to "QA" and is in the "IN PROGRESS" state. The description field is empty, and the "Linked work items" section is also empty. The "Agents" section shows a "Start work" button. The "Details" section shows the assignee as "QA" and the name "Quản Phạm Trần Anh".

The image displays two screenshots of the Jira interface, showing a backlog and a work item detail view.

Top Screenshot:

- Left Panel (Backlog):** Shows the "My Software Team" space. The "Backlog" tab is selected. The backlog is organized into two sprints: "S1 - Requirements" (16 Jun - 28 Jun, 4 work items) and "S2 - Design" (29 Jun - 5 Jul, 4 work items). The "S1 - Requirements" sprint is currently active, showing items SCRUM-5, SCRUM-6, SCRUM-7, and SCRUM-8, all in "IN PROGRESS" status. The "S2 - Design" sprint is upcoming, showing items SCRUM-9, SCRUM-10, SCRUM-11, and SCRUM-12, all in "TO DO" status.
- Right Panel (Jira work item):** Shows the details for "Edit Mục 4.2 - Use Case Specification (tồn bộ UC001-UC014)". The item is assigned to "QA" and is in "IN PROGRESS" status. The "Description" field is empty. The "Linked work items" section is empty. The "Agents" section shows a "Start work" button.

Bottom Screenshot:

- Left Panel (Backlog):** Similar to the top screenshot, but the "S2 - Design" sprint is now active, showing items SCRUM-9, SCRUM-10, SCRUM-11, and SCRUM-12, all in "TO DO" status. The "S1 - Requirements" sprint is now completed.
- Right Panel (Jira work item):** Shows the details for "Viết Mục 6 - Presentation". The item is assigned to "QA" and is in "IN PROGRESS" status. The "Description" field is empty. The "Linked work items" section is empty. The "Agents" section shows a "Start work" button.

The screenshot displays the Jira interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with two sections: "S1 - Requirements" and "S2 - Design". The "S1 - Requirements" section lists four items: SCRUM-5, SCRUM-6, SCRUM-7, and SCRUM-8, all in "IN PROGRESS" status. The "S2 - Design" section lists four items: SCRUM-9, SCRUM-10, SCRUM-11, and SCRUM-12, all in "TO DO" status. The right sidebar shows the "Jira work item" detail view for "SCRUM-7 / SCRUM-81". The title is "Edit Mục 3.2.1 - Functional Requirements". The description field is empty. The "Linked work items" section is empty. The "Agents" section shows a "Start work" button. The "Details" section shows the assignee as "QA - Quân Phạm Trần Anh" with an "Assign to me" link.

Spaces

My Software Team

Summary Backlog Board Code Timeline Docs Forms +

Search backlog

Filter

S1 - Requirements 16 Jun - 28 Jun (4 work items) Complete sprint

SCRUM-5 Hoàn thành phần được gia... IN PROGRESS Jun 27 NT

SCRUM-6 Hoàn thành phần được gia... IN PROGRESS Jun 27 NT

SCRUM-7 Hoàn thành phần được gia... IN PROGRESS Jun 27 QA

SCRUM-8 Hoàn thành phần được gia... IN PROGRESS Jun 28 TQ

+ Create

S2 - Design 29 Jun - 5 Jul (4 work items) Start sprint

SCRUM-9 Viết Architecture + DB schema ... TO DO NT

SCRUM-10 Implement các tool/skill: tìm ... TO DO NT

SCRUM-11 Viết UI/UX Design + Screen S... TO DO QA

SCRUM-12 Dựng CI pipeline (auto-test k... TO DO TQ

Jira work item

SCRUM-7 / SCRUM-81

Edit Mục 3.2.1 - Functional Requirements

Done Agents Done Improve Subtask

Description

Add a description...

Linked work items

Add linked work item

Agents

Start work

Details

Assignee QA - Quân Phạm Trần Anh Assign to me

The screenshot displays the Jira interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with two sprints: "S1 - Requirements" (16 Jun - 28 Jun) and "S2 - Design" (29 Jun - 5 Jul). The "S1 - Requirements" sprint is currently active, showing four work items (SCRUM-5 to SCRUM-8) all in "IN PROGRESS" status. The "S2 - Design" sprint is upcoming, showing four work items (SCRUM-9 to SCRUM-12) in "TO DO" status.

The right sidebar shows the details of a selected work item, "SCRUM-7 / SCRUM-83", titled "Self-training: Figma - wireframe màn hình chính (chat, canvas, email cards)". The work item is assigned to "QA" and is in "IN PROGRESS" status. The "Description" field is empty, and the "Linked work items" section is also empty. The "Agents" section shows a "Start work" button. The "Details" section shows the assignee as "QA - Quân Phạm Trần Anh".

The image displays two screenshots of the Jira interface. The top screenshot shows a backlog view for 'My Software Team' with two sprints: 'S1 - Requirements' and 'S2 - Design'. The bottom screenshot shows a detailed view of a Jira work item titled 'Tổng hợp Mục 1 – Contribution & evidence (Sprint 1)'.

Top Screenshot: Jira Backlog

Spaces
My Software Team

Summary | **Backlog** | Board | Code | Timeline | Docs | Forms

Search backlog | Filter

S1 - Requirements 16 Jun – 28 Jun (4 work items) | Complete sprint

- SCRUM-5 Hoàn thành phần được gia... | IN PROGRESS | Jun 27 | NT
- SCRUM-6 Hoàn thành phần được gia... | IN PROGRESS | Jun 27 | NT
- SCRUM-7 Hoàn thành phần được gia... | IN PROGRESS | Jun 27 | QA
- SCRUM-8 Hoàn thành phần được gia... | IN PROGRESS | Jun 28 | TQ

S2 - Design 29 Jun – 5 Jul (4 work items) | Start sprint

- SCRUM-9 Viết Architecture + DB schema ... | TO DO | NT
- SCRUM-10 Implement các tool/skill: tìm ... | TO DO | NT
- SCRUM-11 Viết UI/UX Design + Screen S... | TO DO | QA
- SCRUM-12 Dựng CI pipeline (auto-test k... | TO DO | TQ

Bottom Screenshot: Jira Work Item Detail

Jira work item

SCRUM-7 / SCRUM-85

Tổng hợp Mục 1 – Contribution & evidence (Sprint 1)

Done | Agents | Done | Improve Subtask

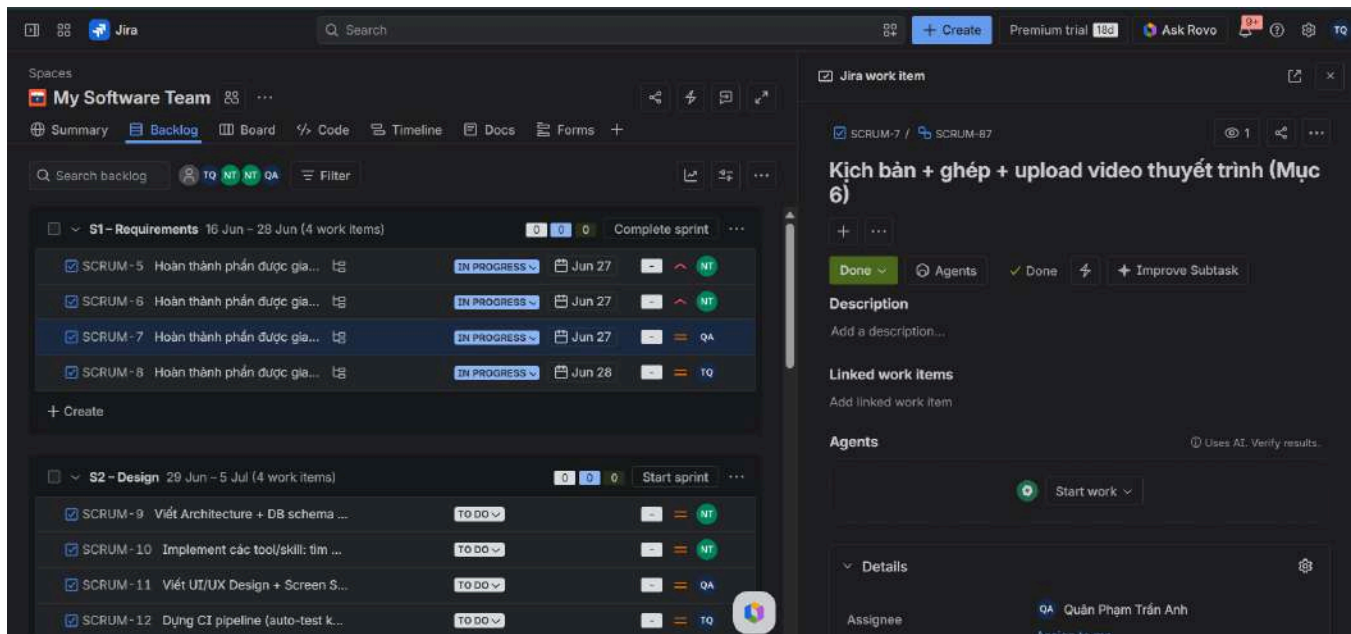
Description
Add a description...

Linked work items
Add linked work item

Agents
Uses AI. Verify results.
Start work

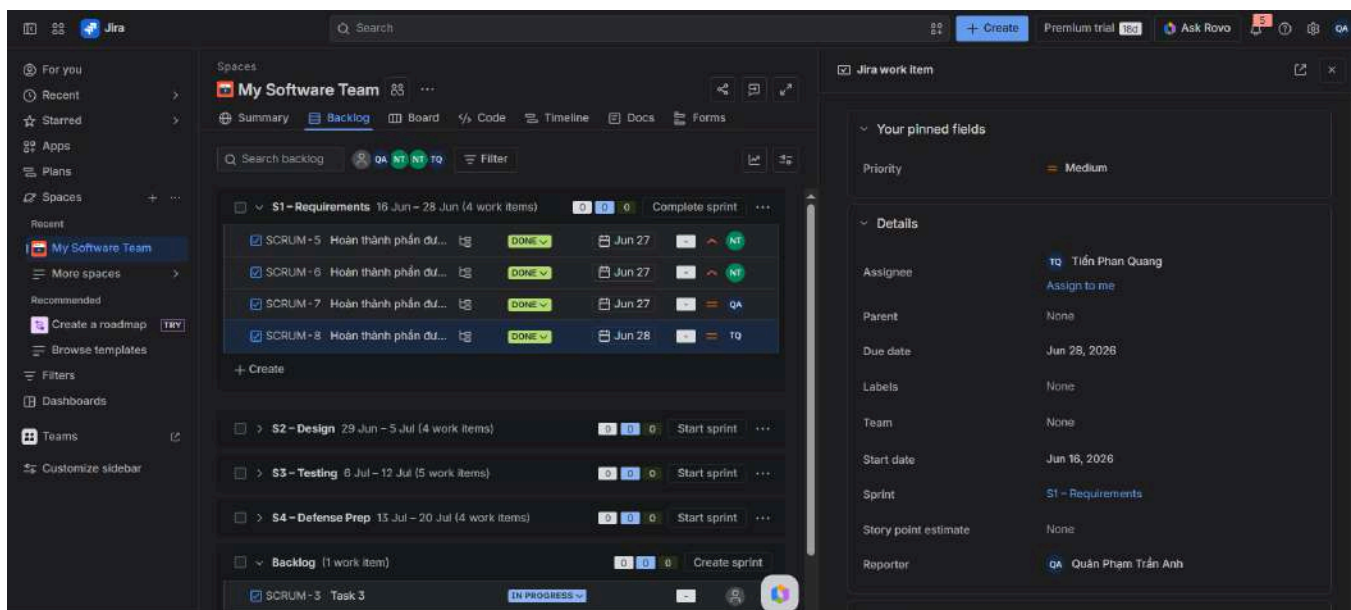
Details

Assignee: QA Quân Phạm Trần Anh
Assign to me



24127250 - Phan Quang Tiến - 25%

- Task Evidence:



The screenshot displays the Jira Software interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with two sprints: "S1 - Requirements" (16 Jun - 28 Jun) and "S2 - Design" (29 Jun - 5 Jul). The "S1 - Requirements" sprint is currently active, showing four work items: SCRUM-5, SCRUM-6, SCRUM-7, and SCRUM-8. SCRUM-8 is selected, and its details are shown on the right. The work item "SCRUM-8" is titled "Viết Mục 7 - Reflective Report" and is in the "IN PROGRESS" state. The right sidebar shows the "Jira work item" details for SCRUM-8, including a description field, linked work items, and a details section with assignee "TQ Tiến Phan Quang" and labels "None".

The screenshot displays the Jira Software interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with two sprints: "S1 - Requirements" (16 Jun - 28 Jun) and "S2 - Design" (29 Jun - 5 Jul). The "S1 - Requirements" sprint is currently active, showing four work items: SCRUM-5, SCRUM-6, SCRUM-7, and SCRUM-8. SCRUM-8 is selected, and its details are shown on the right. The work item "SCRUM-8" is titled "Edit Mục 2 - Problem Statement" and is in the "IN PROGRESS" state. The right sidebar shows the "Jira work item" details for SCRUM-8, including a description field, linked work items, and a details section with assignee "TQ Tiến Phan Quang" and labels "None".

The screenshot displays the Jira interface for a team named "My Software Team". The left sidebar shows the "Backlog" tab selected. The main area shows a list of work items grouped by sprint. The right sidebar shows the details of a selected work item.

Backlog Items:

Sprint	Item ID	Item Name	Status	Due Date	Assignee
S1 - Requirements (16 Jun - 28 Jun)	SCRUM-5	Hoàn thành phần được gia...	IN PROGRESS	Jun 27	NT
	SCRUM-6	Hoàn thành phần được gia...	IN PROGRESS	Jun 27	NT
	SCRUM-7	Hoàn thành phần được gia...	IN PROGRESS	Jun 27	QA
	SCRUM-8	Hoàn thành phần được gia...	IN PROGRESS	Jun 28	TQ
S2 - Design (29 Jun - 5 Jul)	SCRUM-9	Viết Architecture + DB schema ...	TO DO		NT
	SCRUM-10	Implement các tool/skill; tìm ...	TO DO		NT
	SCRUM-11	Viết UI/UX Design + Screen S...	TO DO		QA
	SCRUM-12	Dựng CI pipeline (auto-test k...	TO DO		TQ

Work Item Details (SCRUM-8 / SCRUM-91):

Review Mục 3.2.2 – Non-Functional Requirements

Description: Add a description...

Linked work items: Add linked work item

Agents: Uses AI. Verify results.

Details:

- Assignee: TQ Tiến Phan Quang
- Labels: None

The screenshot displays the Jira interface for a team named "My Software Team". The left sidebar shows the "Backlog" tab selected. The main area shows a list of work items grouped by sprint. The right sidebar shows the details of a selected work item.

Backlog Items:

Sprint	Item ID	Item Name	Status	Due Date	Assignee
S1 - Requirements (16 Jun - 28 Jun)	SCRUM-5	Hoàn thành phần được gia...	IN PROGRESS	Jun 27	NT
	SCRUM-6	Hoàn thành phần được gia...	IN PROGRESS	Jun 27	NT
	SCRUM-7	Hoàn thành phần được gia...	IN PROGRESS	Jun 27	QA
	SCRUM-8	Hoàn thành phần được gia...	IN PROGRESS	Jun 28	TQ
S2 - Design (29 Jun - 5 Jul)	SCRUM-9	Viết Architecture + DB schema ...	TO DO		NT
	SCRUM-10	Implement các tool/skill; tìm ...	TO DO		NT
	SCRUM-11	Viết UI/UX Design + Screen S...	TO DO		QA
	SCRUM-12	Dựng CI pipeline (auto-test k...	TO DO		TQ

Work Item Details (SCRUM-8 / SCRUM-92):

Review Mục 4.1 – Use Case Diagram

Description: Add a description...

Linked work items: Add linked work item

Agents: Uses AI. Verify results.

Details:

- Assignee: TQ Tiến Phan Quang
- Labels: None

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' view with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The 'S1 - Requirements' sprint contains four work items (SCRUM-5 to SCRUM-8) all in 'IN PROGRESS' status. The 'S2 - Design' sprint contains four work items (SCRUM-9 to SCRUM-12) all in 'TO DO' status. The right sidebar shows a 'Jira work item' for 'SCRUM-8 / SCRUM-93' titled 'Review Mục 4.2 - Use Case Specification (t toàn bộ UC001-UC014)'. The work item has a 'Done' status, a description field, and an assignee 'TQ Tiến Phan Quang'.

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' view with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The 'S1 - Requirements' sprint contains four work items (SCRUM-5 to SCRUM-8) all in 'IN PROGRESS' status. The 'S2 - Design' sprint contains four work items (SCRUM-9 to SCRUM-12) all in 'TO DO' status. The right sidebar shows a 'Jira work item' for 'SCRUM-8 / SCRUM-98' titled 'Viết Mục 4.2 - Use Case Specification UC005 (Search & Filter Emails)'. The work item has a 'Done' status, a description field, and an assignee 'TQ Tiến Phan Quang'.

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the backlog with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The right sidebar shows a work item for 'Viết Mục 4.2 – Use Case Specification UC006 (Manage Emails)'. The work item is in the 'IN PROGRESS' state, assigned to 'TQ' (Tiến Phan Quang), and has a due date of 'Jun 28'. The description field is empty, and the 'Agents' section shows a 'Start work' button.

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the backlog with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The right sidebar shows a work item for 'Viết Mục 4.2 – Use Case Specification UC012 (MCP Client Access)'. The work item is in the 'IN PROGRESS' state, assigned to 'TQ' (Tiến Phan Quang), and has a due date of 'Jun 28'. The description field is empty, and the 'Agents' section shows a 'Start work' button.

The screenshot displays the Jira interface for 'My Software Team'. The left sidebar shows the 'Backlog' view with two sections: 'S1 - Requirements' and 'S2 - Design'. The 'S1 - Requirements' section lists four items (SCRUM-5 to SCRUM-8) with status 'IN PROGRESS' and due dates. The 'S2 - Design' section lists four items (SCRUM-9 to SCRUM-12) with status 'TO DO'. The right sidebar shows the 'Jira work item' detail view for 'SCRUM-8 / SCRUM-100'. The title is 'Viết Mục 4.2 – Use Case Specification UC013 (Manage Settings)'. The description field is empty. The 'Linked work items' section is empty. The 'Agents' section shows a 'Start work' button. The 'Details' section shows the assignee 'TQ Tiến Phan Quang'.

Backlog Items:

Item	Status	Due Date	Assignee
SCRUM-5	IN PROGRESS	Jun 27	NT
SCRUM-6	IN PROGRESS	Jun 27	NT
SCRUM-7	IN PROGRESS	Jun 27	QA
SCRUM-8	IN PROGRESS	Jun 28	TQ
SCRUM-9	TO DO		NT
SCRUM-10	TO DO		NT
SCRUM-11	TO DO		QA
SCRUM-12	TO DO		TQ

Jira Work Item Detail:

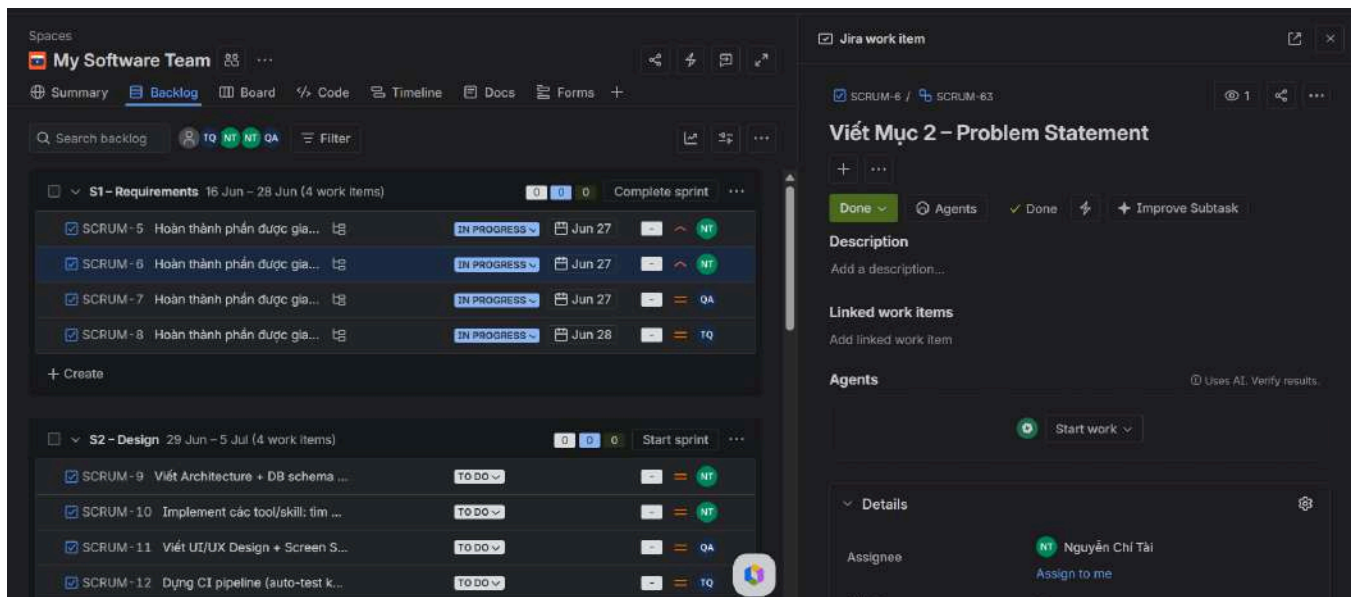
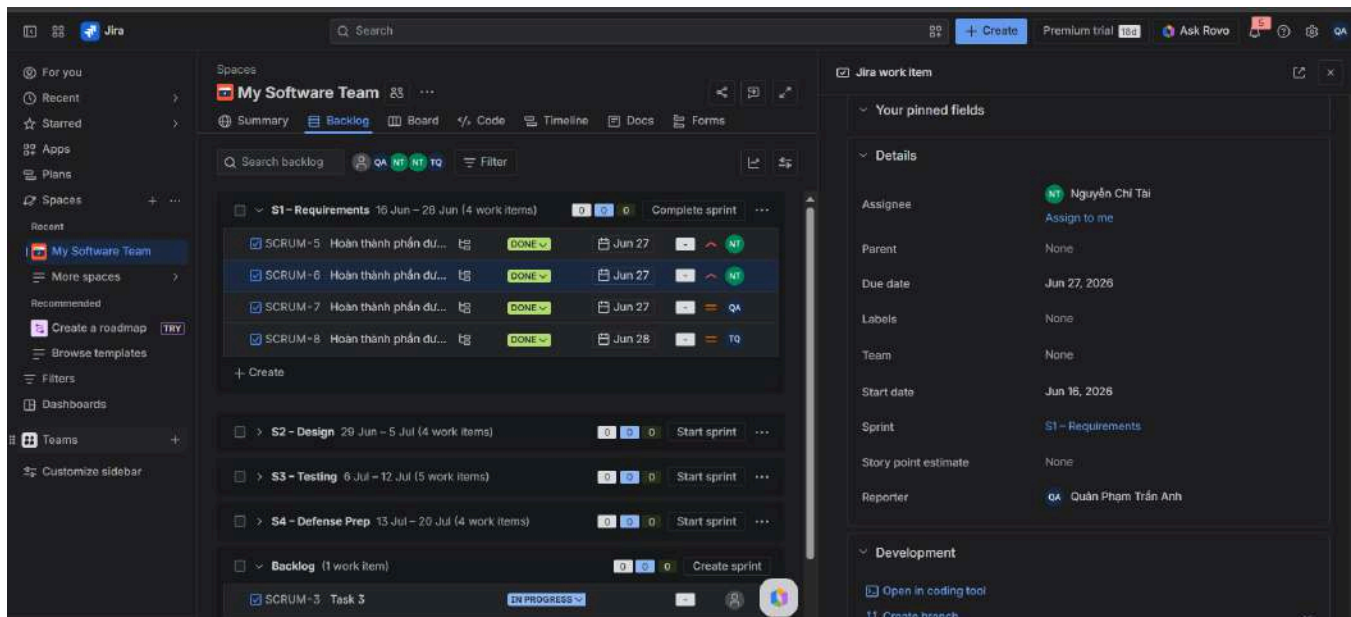
- Title:** Viết Mục 4.2 – Use Case Specification UC013 (Manage Settings)
- Description:** Add a description...
- Linked work items:** Add linked work item
- Agents:** Start work
- Details:** Assignee: TQ Tiến Phan Quang

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the 'Backlog' view with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The main panel shows the 'Jira work item' detail for 'SCRUM-8 / SCRUM-95'. The title is 'Phác thảo Test Plan sơ bộ dựa trên 14 UC'. The status is 'IN PROGRESS' with a due date of 'Jun 28'. The assignee is 'TQ' (Tiến Phan Quang). The description field is empty, and the 'Agents' section shows 'Start work'.

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the 'Backlog' view with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The main panel shows the 'Jira work item' detail for 'SCRUM-8 / SCRUM-95'. The title is 'Quay phần thuyết trình của mình (Mục 7)'. The status is 'IN PROGRESS' with a due date of 'Jun 28'. The assignee is 'TQ' (Tiến Phan Quang). The description field is empty, and the 'Agents' section shows 'Start work'.

24127529 - Nguyễn Chí Tài - 25%

- Task Evidence:



The screenshot displays the Jira interface for 'My Software Team'. The left sidebar shows the 'Backlog' view with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The 'S1 - Requirements' sprint contains four work items (SCRUM-5 to SCRUM-8) all in 'IN PROGRESS' status. The 'S2 - Design' sprint contains four work items (SCRUM-9 to SCRUM-12) all in 'TO DO' status. The right sidebar shows the detailed view of a Jira work item, specifically 'Viết Mục 3.2.2 - Non-Functional Requirements' (SCRUM-6). The work item is assigned to 'Nguyễn Chí Tài' and has a status of 'IN PROGRESS'.

This screenshot is identical to the one above, showing the Jira interface for 'My Software Team'. It displays the backlog with sprints 'S1 - Requirements' and 'S2 - Design', and the detailed view of the work item 'Viết Mục 3.2.2 - Non-Functional Requirements' (SCRUM-6) assigned to 'Nguyễn Chí Tài'.

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the backlog with two sections: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The 'S1 - Requirements' section lists four items: SCRUM-5, SCRUM-6, SCRUM-7, and SCRUM-8, all in 'IN PROGRESS' status. The 'S2 - Design' section lists four items: SCRUM-9, SCRUM-10, SCRUM-11, and SCRUM-12, all in 'TO DO' status. The right sidebar shows the details of a selected work item, SCRUM-6, titled 'Viết Mục 4.2 - Use Case Specification UC002 (Logout & Revoke Access)'. The work item is in 'IN PROGRESS' status and is assigned to 'NT' (Nguyễn Chí Tài). The description field is empty, and the 'Agents' section shows a 'Start work' button.

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the backlog with two sections: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The 'S1 - Requirements' section lists four items: SCRUM-5, SCRUM-6, SCRUM-7, and SCRUM-8, all in 'IN PROGRESS' status. The 'S2 - Design' section lists four items: SCRUM-9, SCRUM-10, SCRUM-11, and SCRUM-12, all in 'TO DO' status. The right sidebar shows the details of a selected work item, SCRUM-6, titled 'Viết Mục 4.2 - Use Case Specification UC003 (View Email List)'. The work item is in 'IN PROGRESS' status and is assigned to 'NT' (Nguyễn Chí Tài). The description field is empty, and the 'Agents' section shows a 'Start work' button.

The screenshot displays the Jira interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with two sprints: "S1 - Requirements" (16 Jun - 28 Jun) and "S2 - Design" (29 Jun - 5 Jul). The "S1 - Requirements" sprint contains four work items (SCRUM-5 to SCRUM-8) all in "IN PROGRESS" status. The "S2 - Design" sprint contains four work items (SCRUM-9 to SCRUM-12) all in "TO DO" status. The right sidebar shows a "Jira work item" for "SCRUM-6 / SCRUM-68" titled "Viết Mục 4.2 - Use Case Specification UC004 (View Email Details)". The work item is in "Done" status and is assigned to "Nguyễn Chí Tài". The "Description" field is empty, and the "Linked work items" and "Agents" sections are also empty.

Spaces

My Software Team

Summary Backlog Board Code Timeline Docs Forms +

Search backlog Filter

S1 - Requirements 16 Jun - 28 Jun (4 work items) Complete sprint

SCRUM-5 Hoàn thành phần được gia... IN PROGRESS Jun 27 NT

SCRUM-6 Hoàn thành phần được gia... IN PROGRESS Jun 27 NT

SCRUM-7 Hoàn thành phần được gia... IN PROGRESS Jun 27 QA

SCRUM-8 Hoàn thành phần được gia... IN PROGRESS Jun 28 TQ

+ Create

S2 - Design 29 Jun - 5 Jul (4 work items) Start sprint

SCRUM-9 Viết Architecture + DB schema ... TO DO NT

SCRUM-10 Implement các tool/skill: tìm ... TO DO NT

SCRUM-11 Viết UI/UX Design + Screen S... TO DO QA

SCRUM-12 Dựng CI pipeline (auto-test k... TO DO TQ

Jira work item

SCRUM-6 / SCRUM-68

Viết Mục 4.2 - Use Case Specification UC004 (View Email Details)

Done Agents Done Improve Subtask

Description

Add a description...

Linked work items

Add linked work item

Agents

Start work

Details

Assignee NT Nguyễn Chí Tài Assign to me

The image displays two screenshots of the Jira interface, showing a project named "My Software Team" with a backlog and a right-hand panel for editing a work item.

Top Screenshot: Edit Mục 4.1 – Use Case Diagram

- Backlog:**
 - S1 - Requirements (16 Jun – 28 Jun, 4 work items):**
 - SCRUM-5: Hoàn thành phần được gia... (IN PROGRESS, Jun 27, NT)
 - SCRUM-6: Hoàn thành phần được gia... (IN PROGRESS, Jun 27, NT)
 - SCRUM-7: Hoàn thành phần được gia... (IN PROGRESS, Jun 27, QA)
 - SCRUM-8: Hoàn thành phần được gia... (IN PROGRESS, Jun 28, TQ)
 - S2 - Design (29 Jun – 5 Jul, 4 work items):**
 - SCRUM-9: Viết Architecture + DB schema ... (TO DO, NT)
 - SCRUM-10: Implement các tool/skill: tìm ... (TO DO, NT)
 - SCRUM-11: Viết UI/UX Design + Screen S... (TO DO, QA)
 - SCRUM-12: Dựng CI pipeline (auto-test k... (TO DO, TQ)
- Right Panel (Edit Mục 4.1 – Use Case Diagram):**
 - Description:** Add a description...
 - Linked work items:** Add linked work item
 - Agents:** Uses AI. Verify results. Start work
 - Details:** Assignee: NT Nguyễn Chí Tài (Assign to me)

Bottom Screenshot: Review Mục 3.2.1 – Functional Requirements

- Backlog:** (Same as the top screenshot)
- Right Panel (Review Mục 3.2.1 – Functional Requirements):**
 - Description:** Add a description...
 - Linked work items:** Add linked work item
 - Agents:** Uses AI. Verify results. Start work
 - Details:** Assignee: NT Nguyễn Chí Tài (Assign to me)

The screenshot displays the Jira workspace for 'My Software Team'. The left sidebar shows the 'Backlog' view with two sprints: 'S1 - Requirements' (16 Jun - 28 Jun) and 'S2 - Design' (29 Jun - 5 Jul). The main panel shows the 'Jira work item' details for 'SCRUM-6 / SCRUM-75'.

Backlog View:

- S1 - Requirements (16 Jun - 28 Jun):** 4 work items. All are in 'IN PROGRESS' status.
 - SCRUM-5: Hoàn thành phần được gia... (Jun 27)
 - SCRUM-6: Hoàn thành phần được gia... (Jun 27)
 - SCRUM-7: Hoàn thành phần được gia... (Jun 27)
 - SCRUM-8: Hoàn thành phần được gia... (Jun 28)
- S2 - Design (29 Jun - 5 Jul):** 4 work items. All are in 'TO DO' status.
 - SCRUM-9: Viết Architecture + DB schema ...
 - SCRUM-10: Implement các tool/skill: tìm ...
 - SCRUM-11: Viết UI/UX Design + Screen S...
 - SCRUM-12: Dựng CI pipeline (auto-test k...

Jira work item details (SCRUM-6 / SCRUM-75):

- Title:** Self-training: thiết kế schema PostgreSQL sơ bộ từ Gmail API payload
- Status:** Done
- Description:** Add a description...
- Linked work items:** Add linked work item
- Agents:** Uses AI. Verify results. Start work
- Details:**
 - Assignee:** NT Nguyễn Chí Tài
 - Action:** Assign to me

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' tab selected. The main area shows a list of work items under the 'S1 - Requirements' sprint (16 Jun - 28 Jun, 4 work items). The items are:

- SCRUM-5: Hoàn thành phần được gia... (IN PROGRESS, Jun 27, NT)
- SCRUM-6: Hoàn thành phần được gia... (IN PROGRESS, Jun 27, NT)
- SCRUM-7: Hoàn thành phần được gia... (IN PROGRESS, Jun 27, QA)
- SCRUM-8: Hoàn thành phần được gia... (IN PROGRESS, Jun 28, TQ)

Below this, the 'S2 - Design' sprint (29 Jun - 5 Jul, 4 work items) is shown with items:

- SCRUM-9: Viết Architecture + DB schema ... (TO DO, NT)
- SCRUM-10: Implement các tool/skill: tìm ... (TO DO, NT)
- SCRUM-11: Viết UI/UX Design + Screen S... (TO DO, QA)
- SCRUM-12: Dựng CI pipeline (auto-test k... (TO DO, TQ)

The right sidebar shows the details of a Jira work item (SCRUM-6 / SCRUM-8) titled 'Quay video thuyết trình phần mình'. The assignee is 'NT Nguyễn Chí Tài'.

24127545 - Nguyễn Ngọc Thiên - 25%

- Task Evidence:

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' tab selected. The main area shows a list of work items under the 'S1 - Requirements' sprint (16 Jun - 28 Jun, 4 work items). The items are:

- SCRUM-5: Hoàn thành phần dự... (DONE, Jun 27, NT)
- SCRUM-6: Hoàn thành phần dự... (DONE, Jun 27, NT)
- SCRUM-7: Hoàn thành phần dự... (DONE, Jun 27, QA)
- SCRUM-8: Hoàn thành phần dự... (DONE, Jun 28, TQ)

Below this, the 'S2 - Design' sprint (29 Jun - 5 Jul, 4 work items) is shown with items:

- SCRUM-9: Viết Architecture + DB schema ... (TO DO, NT)
- SCRUM-10: Implement các tool/skill: tìm ... (TO DO, NT)
- SCRUM-11: Viết UI/UX Design + Screen S... (TO DO, QA)
- SCRUM-12: Dựng CI pipeline (auto-test k... (TO DO, TQ)

The right sidebar shows the details of a Jira work item (SCRUM-6 / SCRUM-8) titled 'Quay video thuyết trình phần mình'. The assignee is 'NT Nguyễn Ngọc Thiên'.

Spaces / My Software Team / SCRUM-5 / SCRUM-50

Viết Mục 3.2.1 – Functional Requirements

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

TQ

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Done Agents Done Improve Subtask

Details

AssigneeNT Nguyễn Ngọc Thiên

Assign to me

LabelsNone

ParentSCRUM-5 Hoàn thành phần được gia

SprintS1 – Requirements

Story point estimateNone

ReporterQA Quân Phạm Trần Anh

Development

More fieldsAtlassian Project

AutomationRule executions

Spaces / My Software Team / SCRUM-5 / SCRUM-51

Viết Mục 4.2 – Use Case Specification UC007 (Manage Mailbox via Natural Language)

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

TQ

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Done Agents Done Improve Subtask

Details

AssigneeNT Nguyễn Ngọc Thiên

Assign to me

LabelsNone

ParentSCRUM-5 Hoàn thành phần được gia

SprintS1 – Requirements

Story point estimateNone

ReporterQA Quân Phạm Trần Anh

Development

More fieldsAtlassian Project

AutomationRule executions

Spaces / My Software Team / SCRUM-5 / SCRUM-52

Viết Mục 4.2 – Use Case Specification UC008 (Summarize Email)

+ ...

Description
Add a description...

Linked work items
Add linked work item

Agents Uses AI. Verify results.

Start work

Activity
All Comments History Work log

TQ Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Details

Assignee: **NT** Nguyễn Ngọc Thiên
[Assign to me](#)

Labels: None

Parent: ☒ SCRUM-5 Hoàn thành phần được gia

Sprint: S1 – Requirements

Story point estimate: None

Reporter: **QA** Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Spaces / My Software Team / SCRUM-5 / SCRUM-53

Viết Mục 4.2 – Use Case Specification UC009 (Categorize Email)

+ ...

Description
Add a description...

Linked work items
Add linked work item

Agents Uses AI. Verify results.

Start work

Activity
All Comments History Work log

TQ Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Details

Assignee: **NT** Nguyễn Ngọc Thiên
[Assign to me](#)

Labels: None

Parent: ☒ SCRUM-5 Hoàn thành phần được gia

Sprint: S1 – Requirements

Story point estimate: None

Reporter: **QA** Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Spaces / My Software Team / SCRUM-5 / SCRUM-54

Viết Mục 4.2 – Use Case Specification UC010 (Compose & Send Email)

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

To

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Done Agents Done Improve Subtask

Details

Assignee NT Nguyễn Ngọc Thiên

Assign to me

Labels None

Parent SCRUM-5 Hoàn thành phần được gia

Sprint S1 - Requirements

Story point estimate None

Reporter QA Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Spaces / My Software Team / SCRUM-5 / SCRUM-55

Viết Mục 4.2 – Use Case Specification UC011 (Manage Conversation History)

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

To

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Done Agents Done Improve Subtask

Details

Assignee NT Nguyễn Ngọc Thiên

Assign to me

Labels None

Parent SCRUM-5 Hoàn thành phần được gia

Sprint S1 - Requirements

Story point estimate None

Reporter QA Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

HCMUS | SE Dept.

24

Spaces / My Software Team / SCRUM-5 / SCRUM-56

Viết Mục 4.2 – Use Case Specification UC014 (Run AI Skill)

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

TQ

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Done Agents Done Improve Subtask

Details

Assignee

NT Nguyễn Ngọc Thiên

Assign to me

Labels

None

Parent

SCRUM-5 Hoàn thành phần được gia

Sprint

S1 - Requirements

Story point estimate

None

Reporter

QA Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Created 4 days ago

Configure

Spaces / My Software Team / SCRUM-5 / SCRUM-57

Viết Mục 5 – AI Usage Declaration

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

TQ

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Done Agents Done Improve Subtask

Details

Assignee

NT Nguyễn Ngọc Thiên

Assign to me

Labels

None

Parent

SCRUM-5 Hoàn thành phần được gia

Sprint

S1 - Requirements

Story point estimate

None

Reporter

QA Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Created 4 days ago

Configure

HCMUS | SE Dept.

25

Spaces / My Software Team / SCRUM-5 / SCRUM-58

1

...

Edit Mục 2 – Problem Statement

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

TQ

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press M to comment

Done Agents Done Improve Subtask

Details

Assignee NT Nguyễn Ngọc Thiên Assign to me

Labels None

Parent SCRUM-5 Hoàn thành phần được gia

Sprint S1 – Requirements

Story point estimate None

Reporter QA Quân Phạm Trần Anh

Development

More fields

Atlassian Project

Automation

Rule executions

Created 4 days ago

Configure

Spaces / My Software Team / SCRUM-5 / SCRUM-59

1

...

Edit Mục 3.2.2 – Non-Functional Requirements

+ ...

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

TQ

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press M to comment

Done Agents Done Improve Subtask

Details

Assignee NT Nguyễn Ngọc Thiên Assign to me

Labels None

Parent SCRUM-5 Hoàn thành phần được gia

Sprint S1 – Requirements

Story point estimate None

Reporter QA Quân Phạm Trần Anh

Development

More fields

Atlassian Project

Automation

Rule executions

Created 4 days ago

Configure

Spaces / My Software Team / SCRUM-5 / SCRUM-60

Self-training: Gemini function calling – định nghĩa tool schema hạt mịn

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Details

Assignee NT Nguyễn Ngọc Thiên
Assign to me

Labels None

Parent SCRUM-5 Hoàn thành phần được gia

Sprint S1 – Requirements

Story point estimate None

Reporter QA Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Created 4 days ago

Spaces / My Software Team / SCRUM-5 / SCRUM-61

Nghiên cứu MCP server – phơi fine-grained tools cho agent ngoài

Description

Add a description...

Linked work items

Add linked work item

Agents

Uses AI. Verify results.

Start work

Activity

All Comments History Work log

Add a comment...

Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Details

Assignee NT Nguyễn Ngọc Thiên
Assign to me

Labels None

Parent SCRUM-5 Hoàn thành phần được gia

Sprint S1 – Requirements

Story point estimate None

Reporter QA Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Created 4 days ago

Spaces / My Software Team / SCRUM-5 / SCRUM-62

Quay video phần minh thuyết trình

+ ...

Description
Add a description...

Linked work items
Add linked work item

Agents Uses AI. Verify results.

Start work

Activity
All Comments History Work log

TQ
Add a comment...
Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Details

Assignee: **NT** Nguyễn Ngọc Thiên
[Assign to me](#)

Labels: None

Parent: ☒ SCRUM-5 Hoàn thành phần được gia

Sprint: S1 - Requirements

Story point estimate: None

Reporter: **QA** Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Spaces / My Software Team / SCRUM-5 / SCRUM-97

Viết Mục 4.2 – Use Case UC013 (Manage Settings)

+ ...

Description
Add a description...

Linked work items
Add linked work item

Agents Uses AI. Verify results.

Start work

Activity
All Comments History Work log

TQ
Add a comment...
Can I get more info...? Status update... Thanks...

Pro tip: press **M** to comment

Details

Assignee: **NT** Nguyễn Ngọc Thiên
[Assign to me](#)

Labels: None

Parent: ☒ SCRUM-5 Hoàn thành phần được gia

Sprint: S1 - Requirements

Story point estimate: None

Reporter: **QA** Quân Phạm Trần Anh

Development

More fields Atlassian Project

Automation Rule executions

Created 4 days ago

2

Problem Statement

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127250 – Phan Quang Tiến

Reviewed by: 24127226 – Phạm Trần Anh Quân

2.1 Context and Motivation

Electronic mail (email) remains one of the most important communication channels in both academic and professional environments. Students, lecturers, employees, and organizations rely on email for information exchange, notifications, task assignments, and formal communication. However, the increasing volume of incoming emails has created a significant information overload problem. Users often receive dozens of emails every day, including academic announcements, work-related notifications, promotional content, and spam messages.

Managing such a large amount of email data presents several challenges. First, users must spend considerable time manually reading and filtering emails to identify important information. Second, traditional email systems mainly rely on keyword-based search and rule-based filtering mechanisms, which are often unable to understand user intent or the semantic meaning of email content. Finally, existing email platforms provide limited support for integrating modern AI-driven workflows and intelligent automation.

To address these challenges, this project proposes **MeoArc**, an Email Intelligence Platform powered by Large Language Models (LLMs). MeoArc enables users to interact with their mailbox using natural language commands and provides intelligent features such as email summarization, email categorization, semantic search, and AI-assisted email management. In addition to serving human users through a web application, MeoArc also exposes selected email-management capabilities through APIs and a Model Context Protocol (MCP) server, allowing external AI agents to securely interact with the platform.

2.2 Operating Environment

MeoArc is designed as a cross-platform distributed web application consisting of the following layers:

Client Side

The user interface is implemented using React with TypeScript and is accessible through modern web browsers supporting HTML5, CSS3, and ECMAScript standards, including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

AI Gateway and Backend Layer

Backend services are implemented using FastAPI and operate as the central AI Gateway of the system. The gateway is responsible for request routing, authentication, communication with external services, AI workflow orchestration, and tool execution. The backend environment is containerized using Docker to ensure portability, dependency isolation, and consistent deployment across different environments.

Data Storage Layer

PostgreSQL is used as the primary relational database for storing user accounts, OAuth credentials, email metadata, system configurations, and application data. Redis is employed as an in-memory storage system for caching, session management, conversation context storage, and rate-limiting control.

External Services

The platform integrates with multiple mail providers — the Google Gmail API and the Microsoft Graph API (Outlook) — to access and manage users' email data after the user grants access via OAuth 2.0. Additionally, MeoArc communicates with the Google Gemini API to perform natural language understanding, email summarization, categorization, and function-calling operations.

Deployment Environment

During development, the system is executed locally using Docker Compose. For demonstration and production deployment, the frontend may be hosted on Vercel, the backend on Render, PostgreSQL on Neon, and Redis on Redis Cloud under their respective free-tier plans.

2.3 Design and Implementation Constraints

The development of MeoArc is subject to the following design and implementation constraints:

Technology Stack Constraint

React with TypeScript is selected for frontend development because it supports component-based architecture, code maintainability, and efficient user interface rendering. FastAPI is chosen for backend and AI Gateway development due to its high performance, asynchronous processing capabilities, and seamless integration with Python-based AI libraries. PostgreSQL is utilized as the primary database because it provides reliable relational data management, strong consistency, and support for complex queries. Redis is employed for caching and temporary session storage to improve application responsiveness and reduce unnecessary processing overhead. Docker is adopted to provide a consistent development and deployment environment across team members' machines.

Security Constraint

The system must not request or store users' raw passwords. Authentication and mailbox authorization are implemented using Google OAuth 2.0. Sensitive credentials, including API keys and OAuth secrets, are stored securely using environment variables and encrypted storage mechanisms. All communications between system components and external services must be protected using HTTPS.

Project Scope Constraint

Due to the six-week development schedule, the Minimum Viable Product (MVP) focuses on text-based email processing. Complex attachment analysis, such as advanced PDF understanding and multimedia content interpretation, is outside the project scope. Email deletion operations are implemented as soft-deletion actions, while AI interactions are protected through request rate limiting to prevent excessive API usage.

Documentation Constraint

All project artifacts, including requirements specifications, software design documents, implementation reports, and testing documents, must comply with the official templates and standards provided by the Faculty of Information Technology, VNU-HCMUS.

3

Requirements Overview

3.1 Stakeholders

Written by: 24127226 – Phạm Trần Anh Quân

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127545 – Nguyễn Ngọc Thiên

MeoArc is designed not merely as an end-user email client but as an *Email Intelligence Platform* that serves both human users and AI-driven agents. Its stakeholders therefore include the parties who interact directly with the system as well as those who hold an indirect interest in its operation. The following table identifies each stakeholder, classifies them as **Primary** (direct interaction with the system) or **Secondary** (indirect interest), and describes their role.

STT	Stakeholder	Type	Description
1	End User	Primary	Individuals who manage their Gmail mailbox through MeoArc's web interface using natural-language chat and the dynamic canvas. Representative personas include HR managers, office workers, and small-business owners who face daily email overload and need summarization, drafting, categorization, and bulk management.
2	Power User / External AI Agent	Primary	Technically proficient users who operate MeoArc through their own AI agent (e.g., Claude Desktop, Codex) via the MCP server, without opening the web application. The agent autonomously invokes MeoArc's fine-grained email tools (search, summarize, draft, send, label, bulk actions) under the user's authorization.

3	Development Team (Group 7)	Primary	The four-member student team responsible for analyzing requirements, designing, implementing, testing, and deploying the system within the academic schedule and free-tier resource constraints.
4	Google/ Microsoft (External Service Provider)	Secondary	External services: the Google Gmail API and the Microsoft Graph API (mail access & management), and the Google Gemini API (LLM reasoning/classification). MeoArc depends on Google for OAuth 2.0 authentication, mailbox data access, and generative-AI capabilities, and must comply with Google's authentication and security requirements.
5	Course Instructor & Teaching Assistants (Project Sponsor)	Secondary	Act as the academic sponsor of the project. They define deliverable requirements and templates, supervise progress through the chosen tools, and evaluate the quality of each artifact.

3.2 Requirements

3.2.1. Functional Requirements Specification

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127226 – Phạm Trần Anh Quân

Reviewed by: 24127529 – Nguyễn Chí Tài

3.2.1.1. Authentication and Account Management Requirements (FR-01)

- **FR-01.1 User Authentication:** The system shall allow users to sign in using OAuth 2.0
- **FR-01.2 Email Account Connection:** The system shall allow users to connect more than one email accounts after successful authentication
- **FR-01.3 Email Account Disconnection:** The system shall allow users to revoke access to connected email accounts.
- **FR-01.4 Session Management:** The system shall maintain authenticated user sessions after successful login

3.2.1.2. AI Conversation Requirements (FR-02)

- **FR-02.1 – Natural Language Command Input:** The system shall provide a chat interface where users can issue email management requests in natural language (e.g., "summarize my unread emails", "reply to John's email and say I'll be there")
- **FR-02.2 – Agent Planning Display:** The system shall display the agent's reasoning steps and tool calls as they are executed in real time
- **FR-02.3 – Clarification Request:** When a user's request is ambiguous or missing required information, the system shall ask the user a clarifying question instead of proceeding with assumptions
- **FR-02.4 – Confirmation Before Irreversible Actions:** Before executing any destructive or irreversible action (sending an email, deleting emails, bulk operations), the system shall pause and display a confirmation prompt showing the full details of the intended action. Execution shall proceed only upon explicit user approval
- **FR-02.5 – Task Completion Notification:** Upon completing an automated task, the system shall notify the user with a summary of what was executed, including any errors or items that could not be processed
- **FR-02.6 – Conversation History:** The system shall persist all chat sessions. Users shall be able to view (delete, rename, pin) past conversations and continue any prior session
- **FR-02.7 – AI Processing Scope Indicator:** The system shall display the current AI processing time window (per the user's subscription tier) as a persistent indicator near the chat input, and shall include an explicit notice in the agent's response when a user's request implies a scope exceeding that window.

3.2.1.3. Email Operation (via AI Chat Assistant or Direct UI) (FR-03)

- **FR-03.1 – Email List View:** The system shall display emails from a connected account as a list of cards, each showing sender, subject, timestamp, and a body preview snippet
- **FR-03.2 – Keyword & Criteria-Based Email Search:** The system shall allow users to search emails by structured criteria: sender, recipient, subject, date range, label, or read/unread status, displayed on the canvas as email cards.
- **FR-03.3 – Semantic Email Search:** The system shall allow users to search emails using natural language queries (e.g., "emails about the budget meeting last month"). Results shall reflect contextual meaning, not only exact keyword matches, and shall be displayed on the canvas in the same format as FR-03.1.
- **FR-03.4 – Email Summarization:** The system shall generate a concise natural language summary of one or more emails upon user request.
- **FR-03.5 – Task Extraction:** The system shall analyze a set of emails and extract actionable tasks, deadlines, and to-do items, presenting them to the user in a structured list on the canvas. (see NFR-SCO-02)

- **FR-03.6 – Email Digest:** The system shall produce a grouped summary report of emails received within a user-specified time period.
- **FR-03.7 – Email Drafting:** The system shall generate a draft email (subject and body) based on the user's instruction, intended recipient, and desired tone (formal, casual, assertive). The draft shall be displayed on the canvas for review before sending.
- **FR-03.8 – Email Reply Generation:** The system shall generate a contextually appropriate reply to an existing email thread based on the user's instruction. The reply shall be displayed on the canvas for review before sending.
- **FR-03.9 – Email Sending:** The system shall send a composed or generated email through a user-selected connected email account after user confirmation. (FR-02.4).
- **FR-03.10 – Inbox Triage:** The system shall classify a user's unread emails by priority, summarize the important ones, and suggest recommended next actions for each, presented as a report on the canvas, upon user request. This is a read-only operation and does not modify any email labels.
- **FR-03.11 – Meeting Preparation Brief:** The system shall analyze email threads related to an upcoming meeting and generate a brief containing extracted action items, deadlines, and key discussion points.
- **FR-03.12 – Bulk Email Operations:** The system shall allow users to apply a batch operation – delete, mark as read, mark as important, apply a label or remove label– to a set of emails matching specified criteria. The operation shall require user confirmation with an interactive checklist (FR-02.4) before execution
- **FR-03.13 – Background Email Synchronization:** The system shall continuously synchronize email data from all connected accounts in the background. New or updated emails shall be reflected in agent responses without requiring a manual refresh from the user.
- **FR-03.14 – Auto-Classification of New Emails:** The system shall automatically label newly synced emails. When enabled, each newly synced email shall automatically receive a Category label, and Priority/Status labels if the email is determined to be task-like, without requiring user confirmation. Users may review and manually correct any automatically applied label at any time.

3.2.1.4. Settings & Personalization (FR-04)

- **FR-04.1 – Language Selection:** The system shall allow users to select their preferred display language for the UI.
- **FR-04.2 – Theme Selection:** The system shall support at minimum a light and dark theme, switchable from the settings panel.

3.2.1.5. MCP Server Requirements (FR-05)

- **FR-05.1 – MCP Tool Invocation:** External AI agents (e.g., Claude Desktop) shall be able to connect to MeoArc and invoke email operations – search, summarize, draft, send, reply, bulk action, extract tasks – through the Model Context Protocol, without using the web interface.
- **FR-05.2 – User-Authorized MCP Access:** An external agent shall only be permitted to access or modify emails within the scope of permissions explicitly authorized by the authenticated MeoArc user.

3.2.2. Non-Functional Requirements Specification

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

Reviewed by: 24127250 – Phan Quang Tiến

3.2.2.1 Performance Requirements (NFR-01)

- **NFR-PER-01:** The system shall return API responses within 3 seconds for at least 95% of requests under normal operating conditions.
- **NFR-PER-02:** The system shall retrieve and display email lists within 5 seconds after receiving a user request.

3.2.2.2 Security Requirements (NFR-02)

- **NFR-SEC-01:** The system shall support user authentication through Google OAuth 2.0.
- **NFR-SEC-02:** The system shall not store users' raw passwords under any circumstances.
- **NFR-SEC-03:** All communications between clients, servers, and third-party services shall be protected using HTTPS/TLS encryption.
- **NFR-SEC-04:** Sensitive credentials, including API keys and OAuth secrets, shall be stored using environment variables and shall not be hard-coded in the source code.
- **NFR-SEC-05:** The system shall implement rate limiting to prevent excessive API requests from a single user or IP address.

3.2.2.3 Reliability Requirements (NFR-03)

- **NFR-REL-01:** The system shall gracefully handle temporary failures from Gmail API or Gemini API and provide meaningful error messages to users.
- **NFR-REL-02:** The system shall not lose conversation history or email metadata during unexpected system failures.

3.2.2.4 Usability Requirements (NFR-04)

- **NFR-USA-01:** The user interface shall be accessible through modern web browsers, including Google Chrome, Microsoft Edge, Mozilla Firefox, and Apple Safari.
- **NFR-USA-02:** Error messages and system notifications shall be clear, concise, and understandable to non-technical users.
- **NFR-USA-03:** The system shall provide a responsive user interface optimized for desktop devices.

3.2.2.5 Maintainability Requirements (NFR-05)

- **NFR-MAI-01:** The backend source code shall follow a modular architecture to facilitate future maintenance and extension.
- **NFR-MAI-02:** All backend APIs shall be documented using the OpenAPI/Swagger specification generated by FastAPI.
- **NFR-MAI-03:** The entire system shall be deployable through Docker Compose using a single configuration file.
- **NFR-MAI-04:** The system shall log all API requests (timestamp, endpoint, user ID, HTTP status) and all agent task executions (tool calls, errors, token usage) to a persistent log store for debugging purposes.

3.2.2.6 Scalability Requirements (NFR-06)

- **NFR-SCA-01:** The system architecture shall support future integration with additional Large Language Model (LLM) providers without requiring major architectural modifications.
- **NFR-SCA-02:** The system architecture shall support the addition of new email-management tools exposed through REST APIs or the Model Context Protocol (MCP).

3.2.2.7 Compatibility Requirements (NFR-07)

- **NFR-COM-01:** The system shall support multiple mail providers — Gmail (Google OAuth 2.0 + Gmail API) and Outlook (Microsoft OAuth 2.0 + Microsoft Graph API) — through a common provider-abstraction layer, so that email read/manage/send operations behave consistently regardless of provider.
- **NFR-COM-02:** The MCP server shall comply with the Model Context Protocol specification to ensure interoperability with compatible external AI agents

3.2.2.8 AI Processing Scope Constraints (NFR-08)

- **NFR-SCO-01:** On-demand AI operations that scan existing mailbox content (Semantic Search, Email Categorization, Summarization, Task Extraction, Reply context retrieval) shall limit the time range of emails processed according to the user's subscription tier: 90 days (Free), 180 days (Pro), 365 days (Pro Max).
- **NFR-SCO-02:** Task Extraction shall be capped at a maximum of 90 days regardless of tier, since extracted tasks beyond this range are unlikely to remain actionable.
- **NFR-SCO-03:** Email List View and Keyword & Criteria-Based Email Search (FR-03.1, FR-03.2) shall not be subject to this time window limitation.
- **NFR-SCO-04:** Auto-Classification of New Emails (FR-03.14) shall be exempt from this limitation, since it only processes emails received at or after the feature was enabled, and never scans historical mailbox content.

4

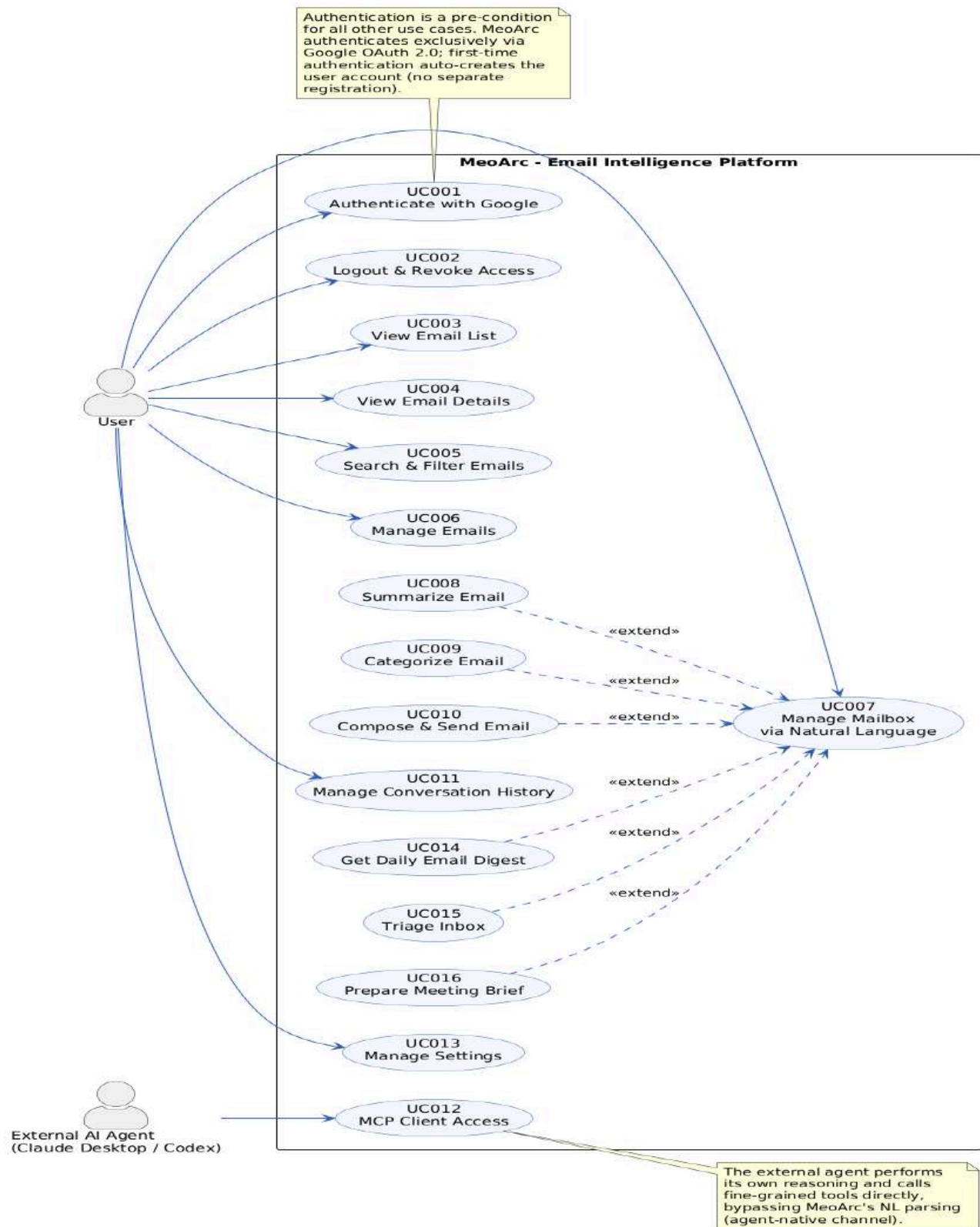
Requirements Analysis

4.1 *Use Case model*

Written by: 24127226 – Phạm Trần Anh Quân

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127250 – Phan Quang Tiến



The system has two primary actors. The **User** interacts with the application through the web interface and is associated with the base use cases. **Summarize Email (UC008)**, **Categorize Email (UC009)**, and **Compose & Send Email (UC010)** «extend» *Manage Mailbox via Natural Language (UC007)*, as they are AI-driven behaviors invoked conditionally based on user intent. The three reusable AI Skills – **Get Daily Email Digest (UC014)**, **Triage Inbox (UC015)**, and **Prepare Meeting Brief (UC016)** – also «extend» UC007, since each is an optional, predefined multi-step workflow built on top of the assistant's command loop to accomplish a common high-value email task in a single action. **Authenticate & Connect Mailbox (UC001)** is a common pre-condition for all other use cases; because MeoArc authenticates exclusively through Google OAuth 2.0, a first-time authentication automatically provisions the user account, and no separate registration use case is required. The **External AI Agent** accesses the system exclusively through **MCP Client Access (UC012)**; unlike UC007, this channel does not rely on MeoArc's natural-language interpretation – the external agent reasons independently and invokes the same fine-grained tools directly, the essence of the platform's agent-native design.

4.2 Use Case Specification

4.2.1 Authenticate & Connect Mailbox

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

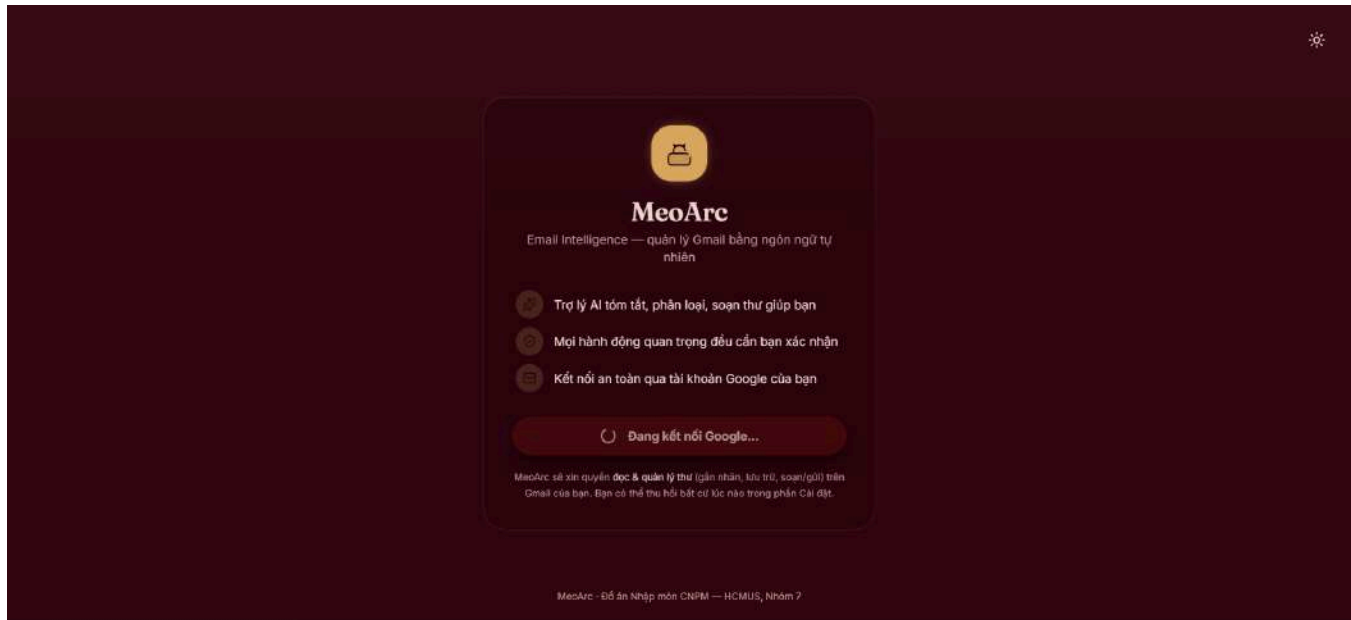
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC001
Use Case Name	Authenticate & Connect Mailbox
Brief Description	Describes how a user authenticates with MeoArc using OAuth 2.0 and grants the necessary permissions for accessing Google (Gmail) or Microsoft (Outlook) services. The process supports both first-time account creation and subsequent sign-in operations.
Actor	User
Pre-Conditions	1. The user accesses the MeoArc website using a supported web browser. 2. The user has a valid Google or Microsoft account.

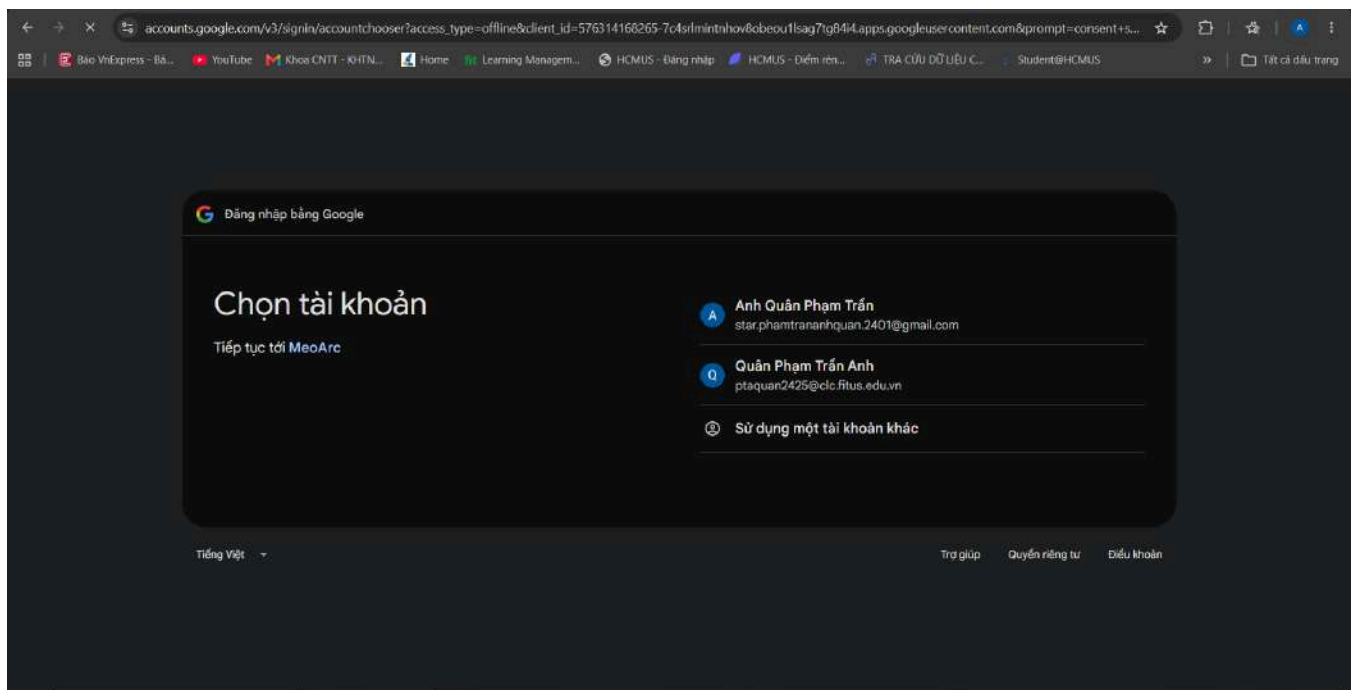
	3. The MeoArc authentication service is available.
Post-Conditions	1. The user is successfully authenticated. 2. Gmail or OutLook access permission has been granted to MeoArc. 3. An authenticated session is established. 4. The user is redirected to the main dashboard.
Main Scenario	1. The user clicks the "Đăng nhập với Google" or "Đăng nhập với Outlook" button on the MeoArc landing page. 2. The system redirects the user to Google's or Microsoft's authentication page. 3. The user signs in to their Google or Microsoft account if not already authenticated. 4. Google or Microsoft displays the permissions requested by MeoArc. 5. The user reviews and approves the requested permissions. 6. Google or Microsoft successfully verifies the user's identity and returns control to MeoArc. 7. The system processes the authorization response. If the user is accessing MeoArc for the first time, the system creates a new user profile; otherwise, the system retrieves the existing profile and establishes an authenticated session. 8. The system grants the user access to MeoArc services. 9. The user is redirected to the main dashboard.
Alternative Scenarios	A1. User denies permission (At Step 5) 5a. The user selects "Cancel" or denies the requested permissions. 5b. Google returns an authorization failure response.

	5c. The system displays the message: "Login failed because access permission was denied." 5d. The user remains on the login page. A2. Authentication service unavailable (At Step 6-7) 6a. Google authentication service is temporarily unavailable or a network error occurs. 6b. The system cannot complete the authentication process. 6c. The system displays the message: "Google authentication service is currently unavailable. Please try again later."
Non-Functional Constraints	NFR-SEC-01: The system shall authenticate users exclusively through Google OAuth 2.0. NFR-SEC-02: The system shall not request or store users' Google passwords under any circumstances. NFR-SEC-03: All authentication communications shall be protected using HTTPS/TLS encryption. NFR-PER-01: The login process should be completed within 5 seconds under normal operating conditions.

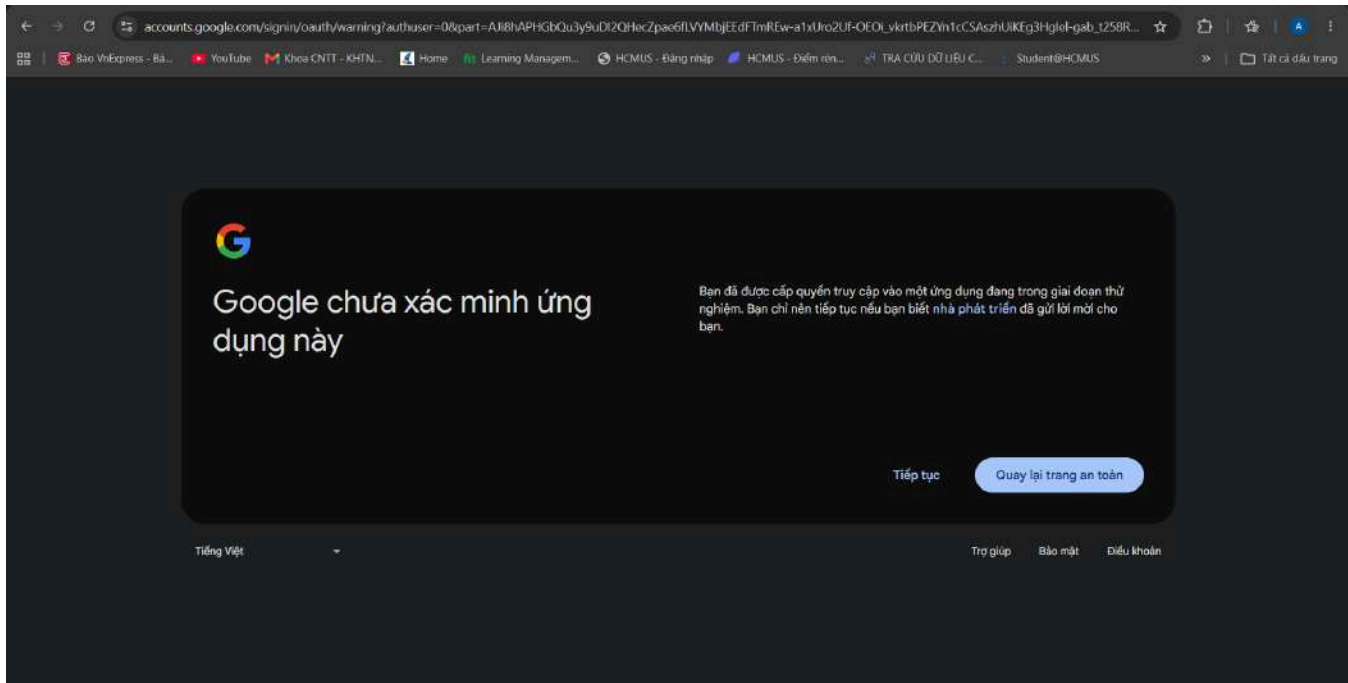
Khi vào web, ta nhấn “Đăng nhập với Google”:



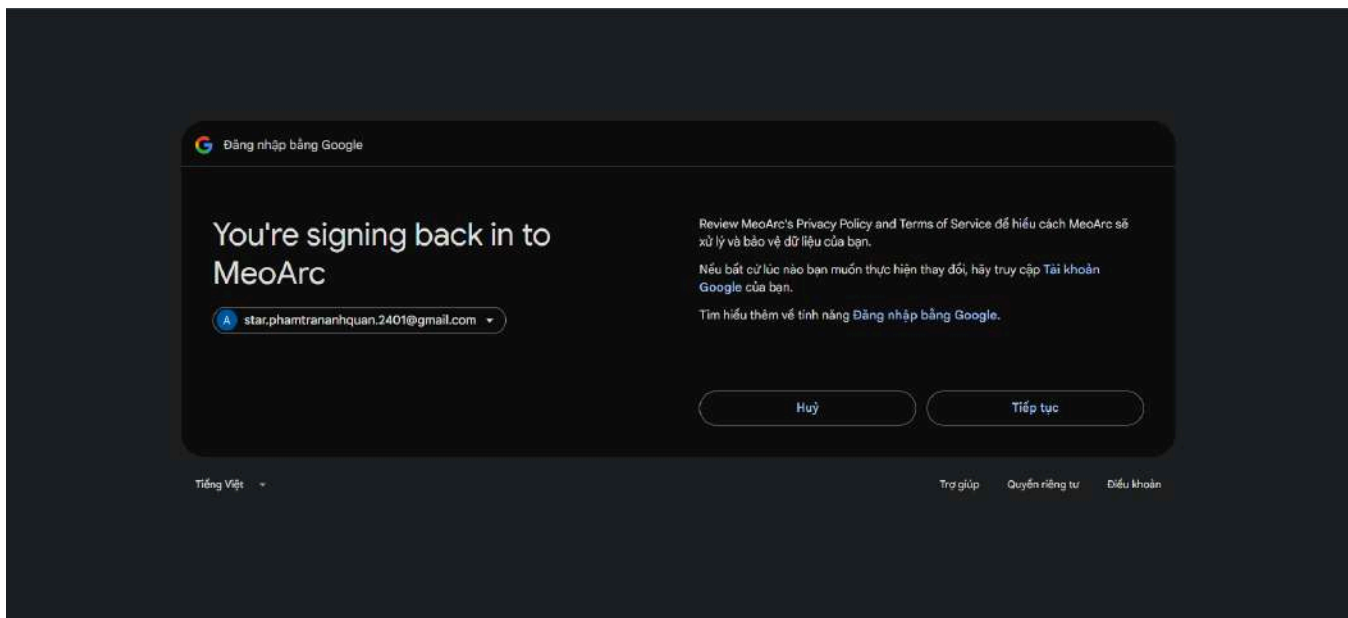
Người dùng sẽ được dẫn sang trang web để đăng nhập Google:



Sau khi chọn một tài khoản, Google sẽ hỏi lại người dùng để xác nhận một số điều về bảo mật:

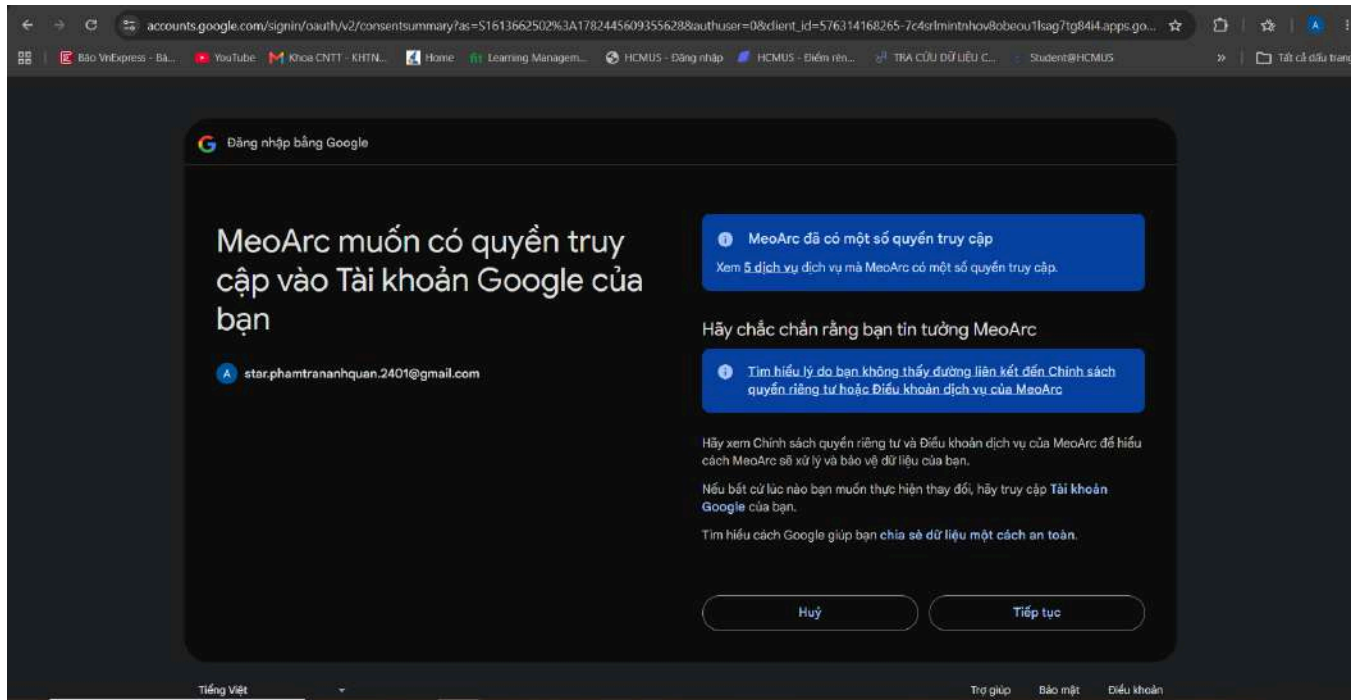


Sau khi nhấn “Tiếp tục”, người dùng sẽ được hỏi lại một lần cuối về website MeoArc:

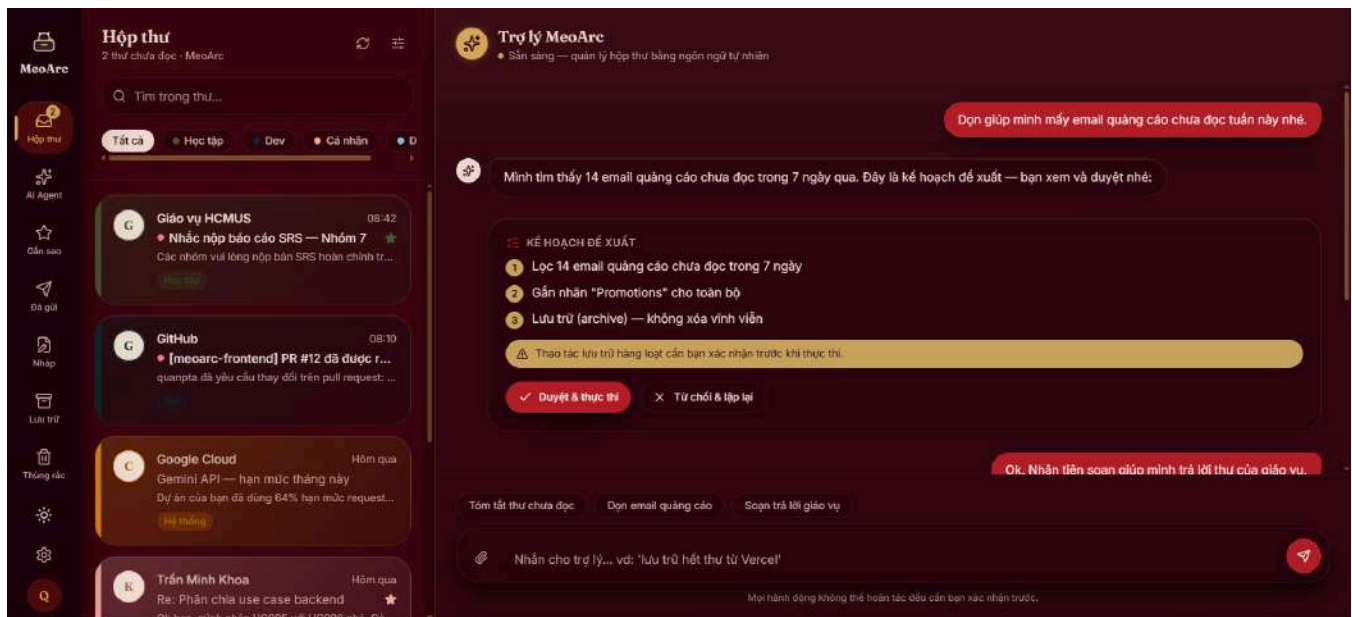


Sau khi nhấn “Tiếp tục”, Google sẽ thông báo với người dùng về các quyền mà MeoArc có thể làm với Gmail của người dùng và đề nghị người dùng xem lại chính sách về quyền riêng tư và

bảo mật:



Khi nhấn “Tiếp tục” thì người dùng vào trong web MeoArc:



4.2.2 Logout & Revoke Access

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

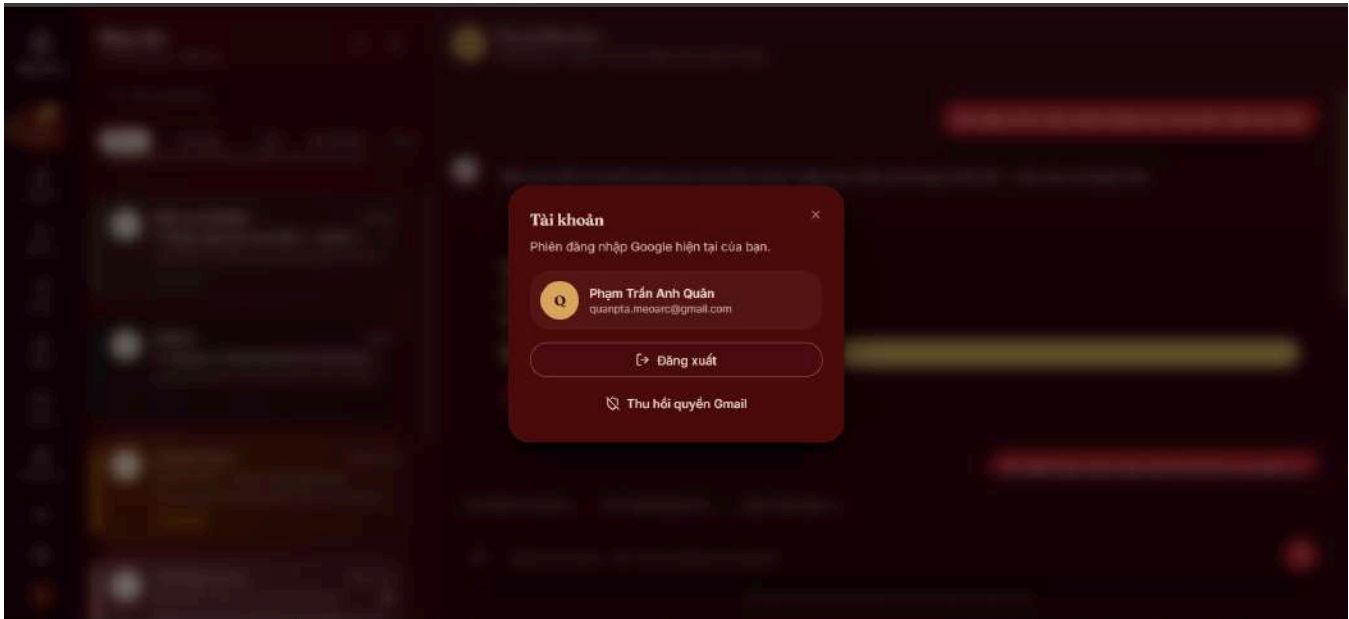
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC002
Use Case Name	Logout & Revoke Access
Brief Description	Describes how a user securely signs out of MeoArc and revokes previously granted Gmail access permissions.
Actor	User
Pre-Conditions	1. The user has successfully authenticated through UC001 – Authenticate & Connect Mailbox. 2. The user currently has an active session in MeoArc.
Post-Conditions	1. The user session is terminated. 2. MeoArc no longer has permission to access the user's Gmail account. 3. All locally stored authentication information is invalidated. 4. The user is redirected to the login page.
Main Scenario	1. The user opens the account menu from the application interface. 2. The system displays available account actions, including "Logout". 3. The user selects "Logout". 4. The system displays a confirmation dialog informing the user that Gmail access permissions will be revoked. 5. The user confirms the action. 6. The system requests revocation of the previously granted Gmail access permissions.

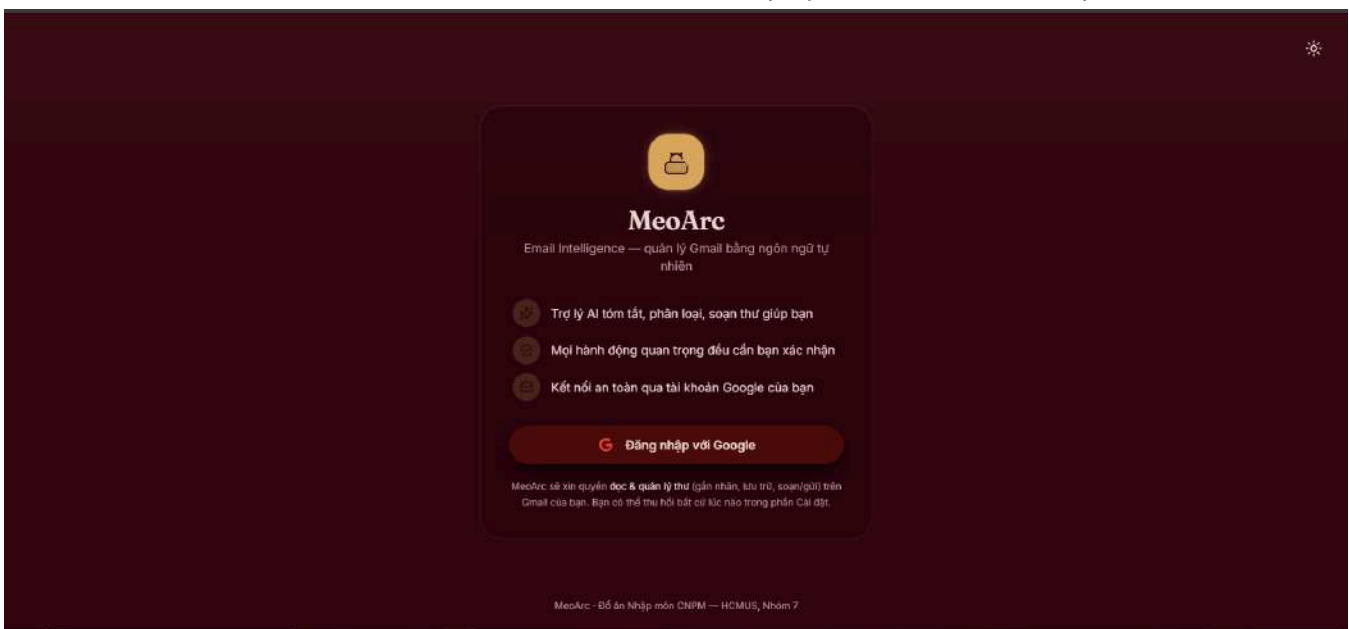
	7. The system terminates the current authenticated session. 8. The system removes locally stored authentication information. 9. The system redirects the user to the login page. 10. The system displays the message: "Logout and access revocation completed successfully."
Alternative Scenarios	A1. User cancels the operation (At Step 5) 5a. The user selects "Cancel" in the confirmation dialog. 5b. The system closes the dialog. 5c. The current session remains active. 5d. The user continues using MeoArc without any changes. A2. Permission revocation service unavailable (At Step 6) 6a. The system is unable to complete the permission revocation request due to a temporary service or network issue. 6b. The system continues terminating the local session to protect user security. 6c. The system records the failure for administrative monitoring. 6d. The system redirects the user to the login page and displays the message: "You have been signed out successfully. Gmail access revocation may not have completed. Please review your Google Account permissions if necessary."
Non-Functional Constraints	NFR-SEC-03: All communication related to authentication and permission revocation shall be protected using HTTPS/TLS encryption. NFR-SEC-05: Logout requests shall be protected by rate-limiting mechanisms to prevent abuse.

	NFR-PER-01: The logout process should be completed within 5 seconds under normal operating conditions.
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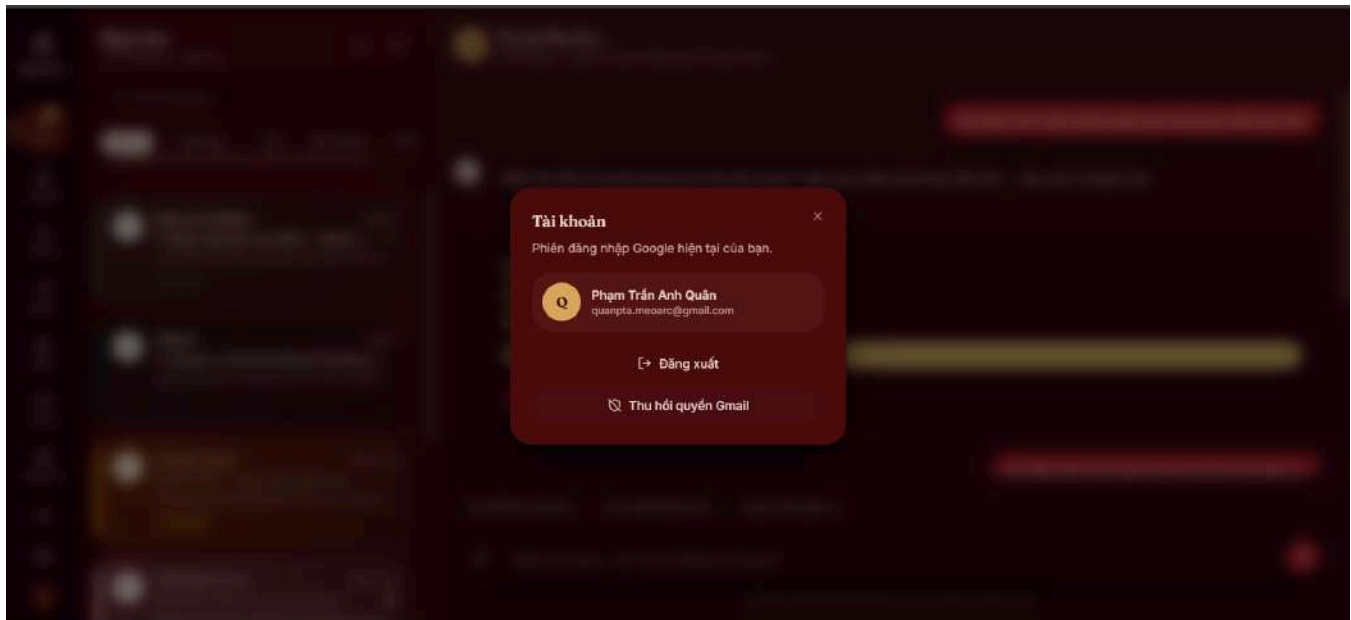
Khi nhấn hình avatar góc dưới cùng bên trái, web sẽ hiện lên mục “Đăng xuất”:



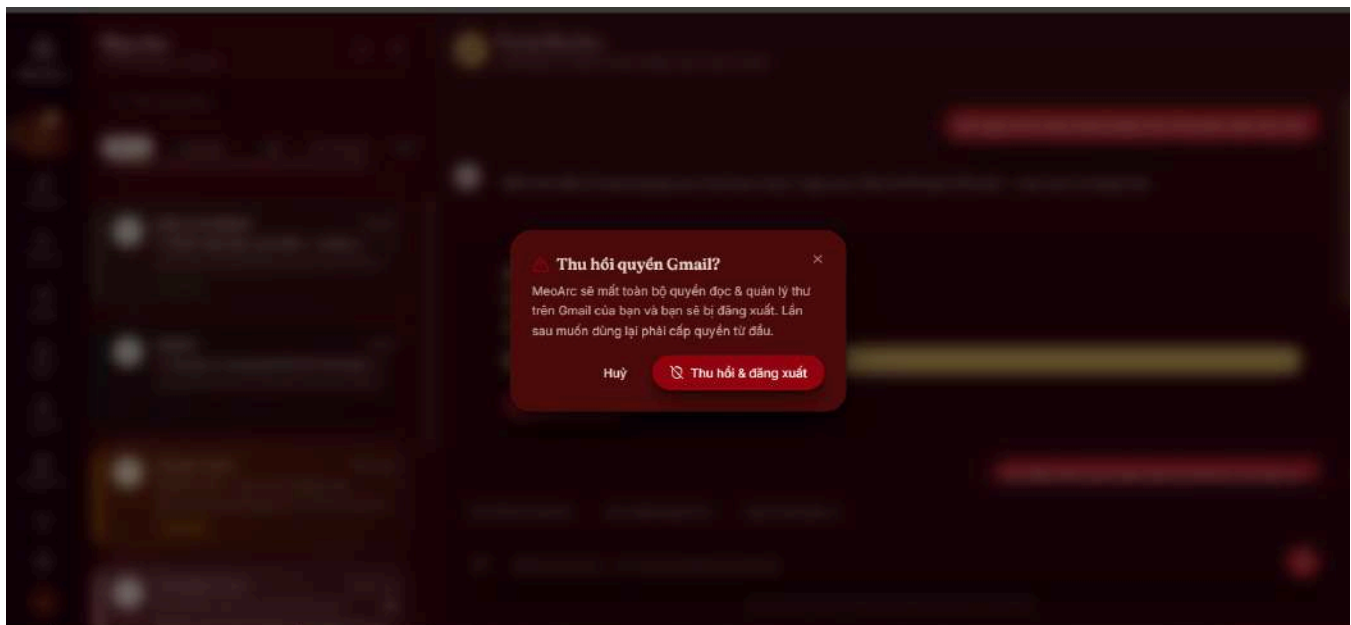
Sau khi nhấn nút “Đăng xuất”, web sẽ đưa người dùng quay lại trang đăng nhập ban đầu:



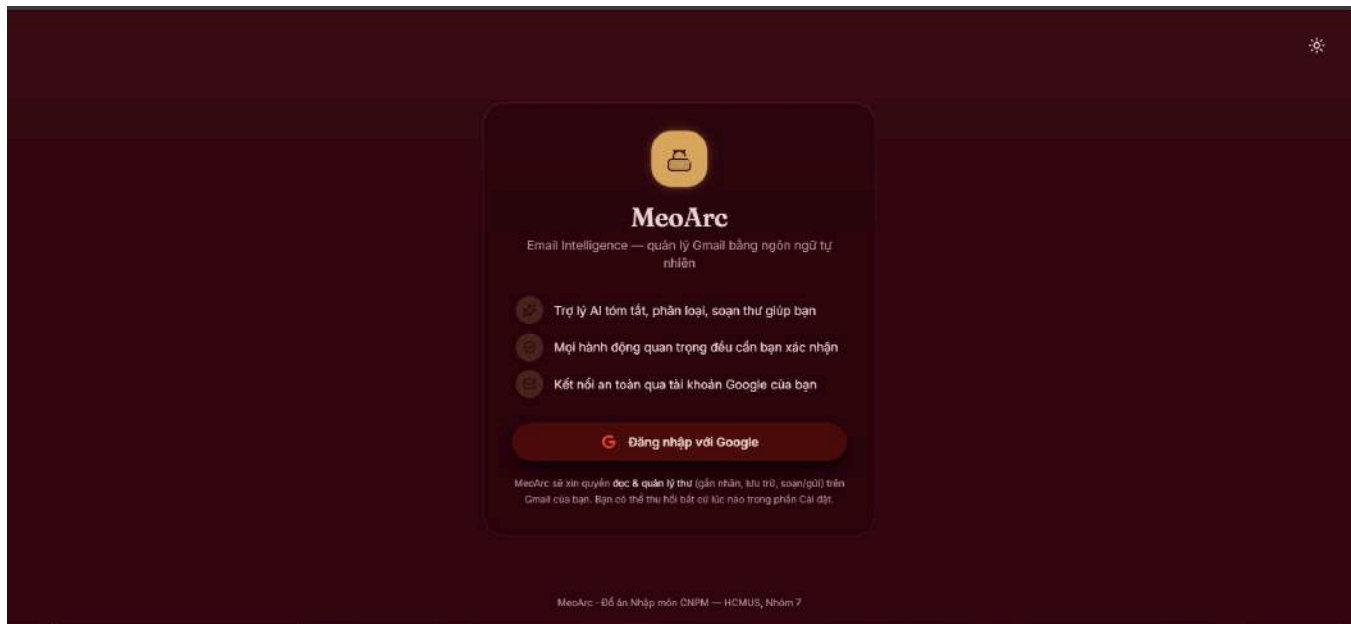
Tương tự với “Đăng xuất”, nhưng sẽ đối với “Thu hồi quyền Gmail”:



Khi nhấn “Thu hồi quyền Gmail”, web sẽ hỏi để xác nhận lại người dùng:



Khi người dùng nhấn “Thu hồi & Đăng xuất” thì web sẽ đưa người dùng quay về trang đăng nhập ban đầu:



4.2.3 View Email List

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127250 – Phan Quang Tiến

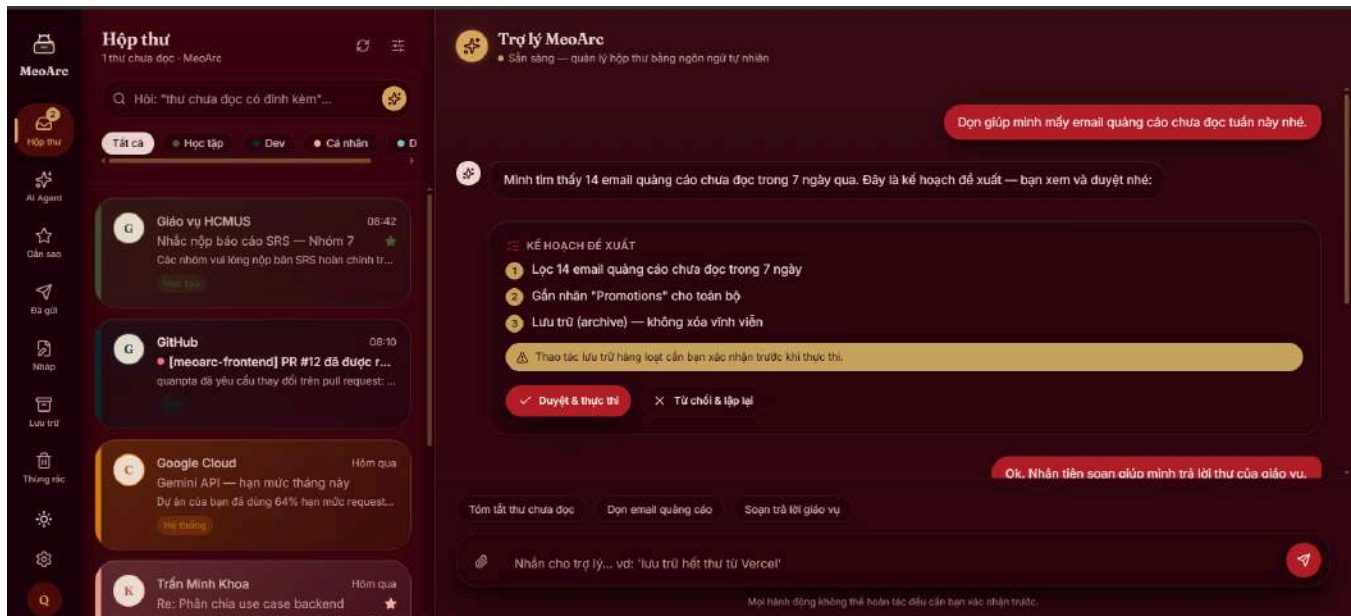
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC003
Use Case Name	View Email List
Brief Description	Describes how a user views the list of emails available in their Gmail mailbox through the MeoArc dashboard.
Actor	User
Pre-Conditions	1. The user has successfully authenticated through UC001 – Authenticate & Connect Mailbox. 2. The user has an active session in MeoArc.

	3. The user has granted Gmail access permissions to MeoArc.
Post-Conditions	1. The email list is displayed successfully. 2. The user can view basic information for each email, including sender, subject, timestamp, and status. 3. The displayed data reflects the most recent information available to the system.
Main Scenario	1. The user accesses the MeoArc dashboard. 2. The system retrieves the user's email data. 3. The system synchronizes email information if newer data is available. 4. The system organizes the emails by recency and prepares them for display. 5. The system displays the email list on the dashboard. 6. The user reviews the displayed emails, including sender, subject, timestamp, and read/unread status.
Alternative Scenarios	A1. Gmail access token has expired (At Step 2–3) 2a. The system detects that the current Gmail authorization is no longer valid. 2b. The system attempts to refresh the authorization automatically. 2c. If the refresh operation succeeds, the use case continues at Step 3. 2d. If the refresh operation fails, the system requests the user to log in again through UC001. A2. Gmail service temporarily unavailable (At Step 3)

	3a. The system is unable to retrieve the latest email data due to a temporary service or network issue. 3b. The system loads the most recently available email data. 3c. The system displays a warning message: "Offline data is being displayed. Some emails may not be up to date." 3d. The use case continues at Step 5.
Non-Functional Constraints	NFR-PER-01: The system shall respond to email retrieval requests within 3 seconds for at least 95% of requests under normal operating conditions. NFR-PER-02: The email list shall be displayed within 5 seconds after the user accesses the dashboard. NFR-PER-03: The system shall support at least 50 concurrent users without significant performance degradation. NFR-USA-04: The email list interface shall provide a responsive and user-friendly layout optimized for desktop web browsers.

Khi vào bên trong web sau khi đã đăng nhập xong thì web sẽ hiện danh sách cái emails trong khung “Hộp thư”:



4.2.4 View Email Details

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127250 – Phan Quang Tiến

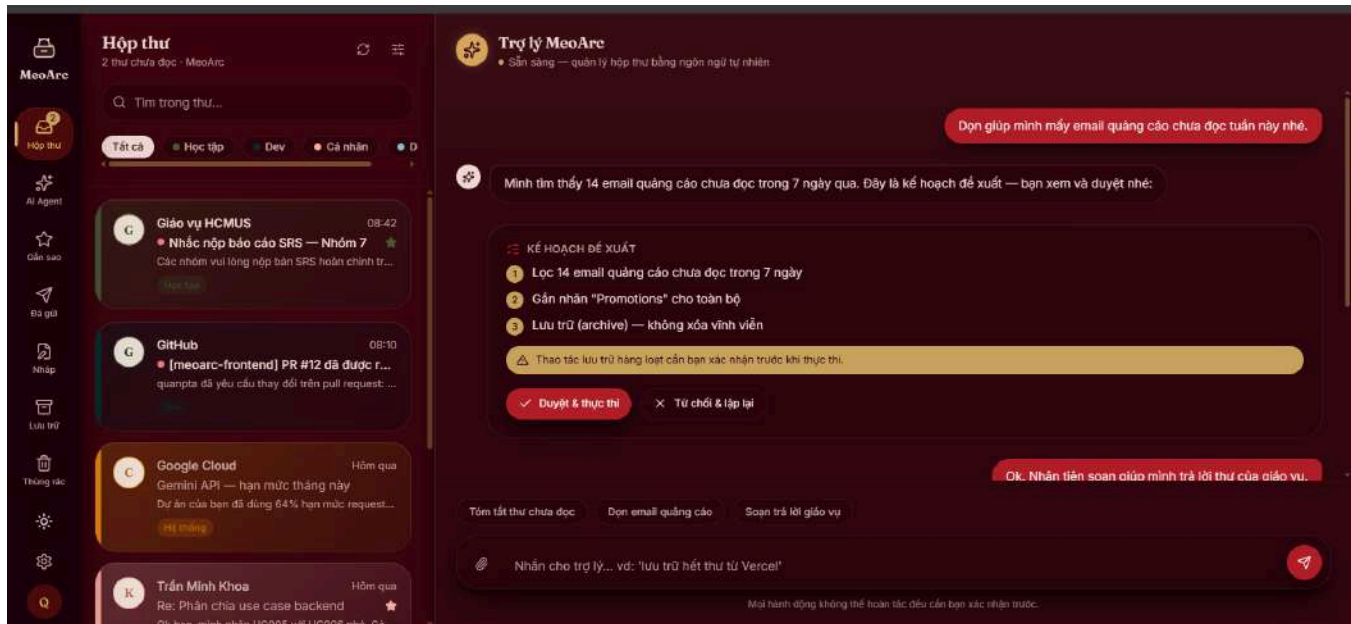
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC004
Use Case Name	View Email Details
Brief Description	Describes how a user opens and views the complete content of a selected email through the MeoArc interface.
Actor	User
Pre-Conditions	1. The user has successfully authenticated through UC001 – Authenticate & Connect Mailbox.

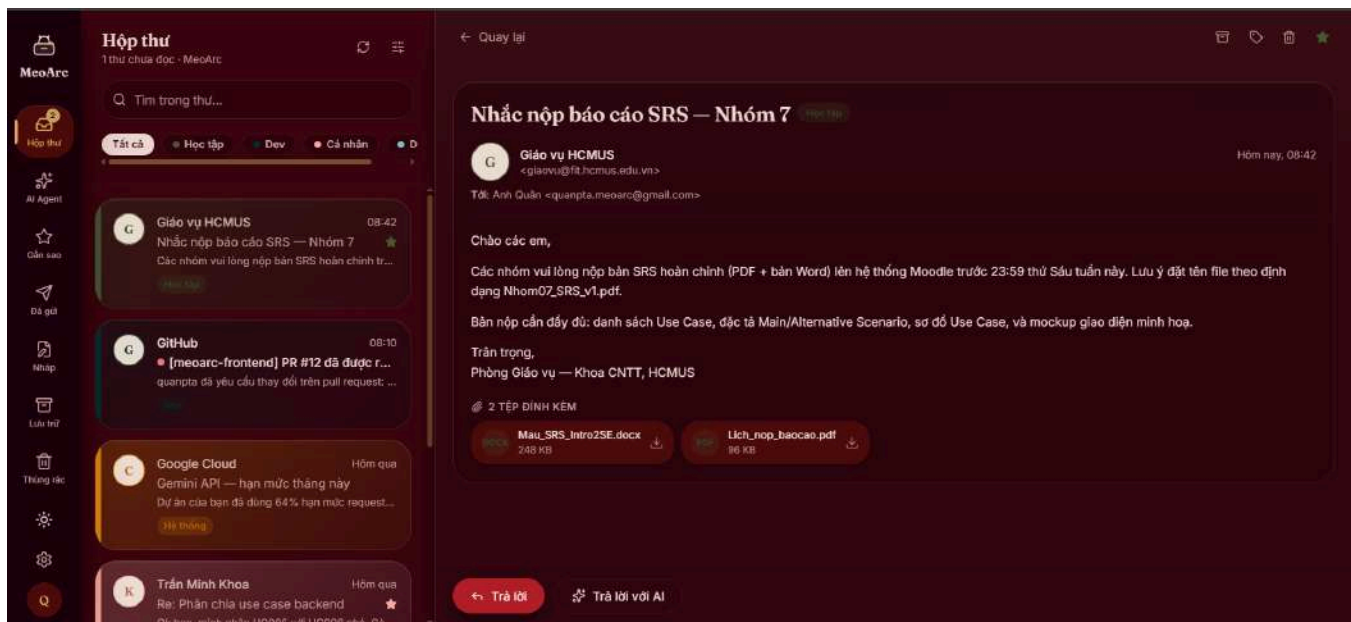
	2. The user has an active session in MeoArc. 3. The email list is currently displayed (UC003).
Post-Conditions	1. The selected email is displayed successfully. 2. The user can view the sender, recipients, subject, timestamp, email body, and available attachments. 3. The email status may be updated to "Read" if applicable.
Main Scenario	1. The user selects an email from the displayed email list. 2. The system retrieves the complete content of the selected email. 3. The system processes and prepares the email content for display. 4. The system displays the email details, including sender information, subject, timestamp, message content, and attachments (if any). 5. The system updates the email status to "Read" when applicable. 6. The user reviews the email content.
Alternative Scenarios	A1. Selected email is no longer available (At Step 2) 2a. The system cannot retrieve the selected email because it has been deleted or is no longer accessible. 2b. The system displays the message: <i>"This email is no longer available."</i> 2c. The system removes the unavailable email from the displayed list if necessary. A2. Email content cannot be fully retrieved (At Step 2-3)

	2a. The system is unable to retrieve the complete email content due to a temporary service or network issue. 2b. The system displays the most recently available preview data if available. 2c. The system displays the warning message: <i>"Full email content is currently unavailable. A preview version is being displayed."</i>
Non-Functional Constraints	NFR-PER-01: The system shall display email details within 3 seconds for at least 95% of requests under normal operating conditions. NFR-SEC-03: All email content transmitted between the client and server shall be protected using HTTPS/TLS encryption. NFR-SEC-06: Email content containing HTML elements shall be sanitized before rendering to prevent malicious script execution. NFR-USA-03: Email details shall be presented in a clear and readable layout suitable for desktop web browsers.

Ở phần Hộp thư, ta thấy cái emails hiện lên, ta chọn một mail cụ thể:



Ở đây ta nhấn mail Giáo vụ HCMUS, ta sẽ thấy nội dung mail ở khung bên phải:



4.2.5 Search & Filter Emails

Written by: 24127250 – Phan Quang Tiến

Edited by: 24127529 – Nguyễn Chí Tài

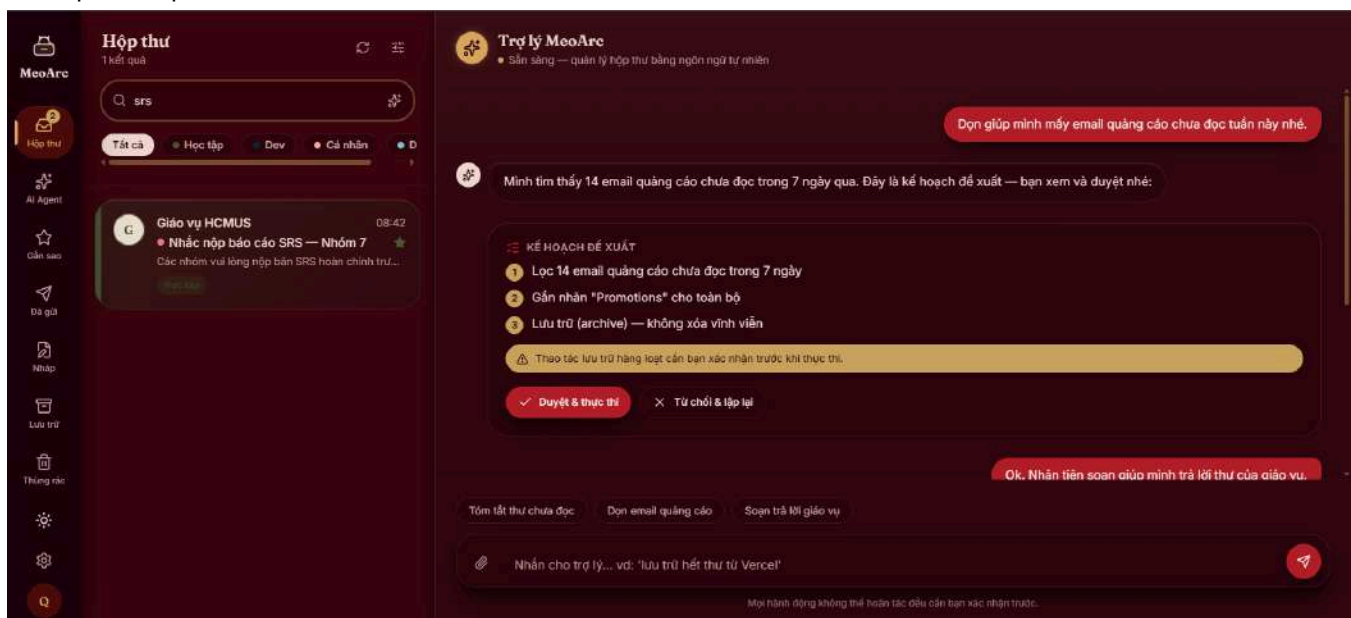
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC005
Use Case Name	Search & Filter Emails
Brief Description	Describes how a user searches for specific emails or filters the email list based on structured criteria or natural language queries through the MeoArc interface.
Actor	User
Pre-Conditions	1. The user has successfully authenticated through UC001 - Authenticate & Connect Mailbox. 2. The user has an active session in MeoArc. 3. The email list is currently displayed (UC003).
Post-Conditions	1. The system displays a filtered list of emails matching the specified search criteria. 2. The user can view and interact with the filtered results. 3. The original email list remains intact and restorable.
Main Scenario	1. The user enters a keyword or selects filter criteria (sender, recipient, subject, date range, label or read/unread status) in the search interface. 2. The system validates the input and constructs a query based on the provided criteria. 3. The system sends the query to the Gmail API and retrieves matching emails. 4. The system processes and ranks the results by relevance or recency.

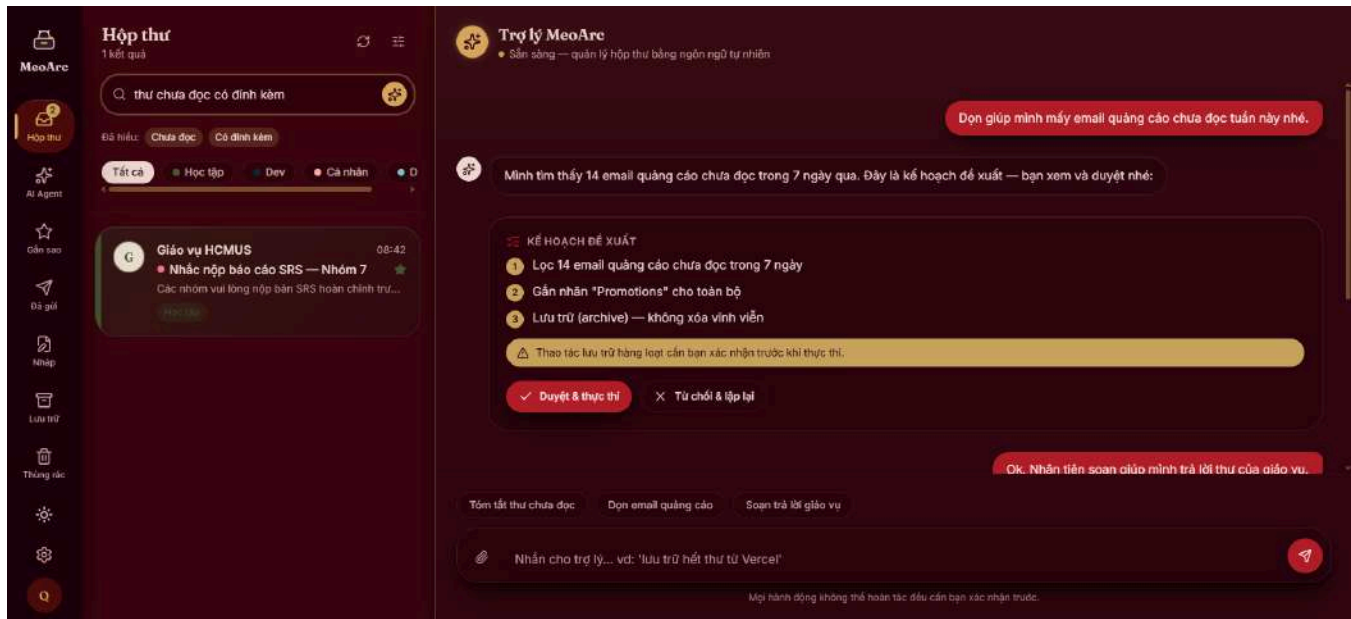
	5. The system displays the filtered email list to the user. 6. The user reviews the search results
Alternative Scenarios	<i>A1. No matching emails found (At step 3-4)</i> 3a. The system receives an empty result set from the query. 3b. The system displays the message: “No emails match your search criteria.” 3c. The user may modify the search criteria and retry. <i>A2. User performs a semantic/natural language search (At step 1)</i> 1a. The user enters a natural language query (e.g., “emails about the budget meeting last month”). 1b. The system forwards the query to the AI layer for semantic interpretation. 1c. The AI layer extracts structured search parameters from the natural language input. 1d. The use case continues from Step 3 using the extracted parameters. <i>A3. Gmail service temporarily unavailable (At step 3)</i> 3a. The system is unable to query Gmail due to a temporary service or network issue. 3b. The system displays the message: “Search is temporarily unavailable. Please try again later.”

	3c. The user remains on the current email list view.
Non-Functional Constraints	NFR-PER-01: The system shall return search results within 3 seconds for at least 95% of requests under normal operating conditions. NFR-SEC-03: All search queries and responses shall be transmitted using HTTPS/TLS encryption. NFR-USA-02: Search error messages shall be clear and understandable to non-technical users.

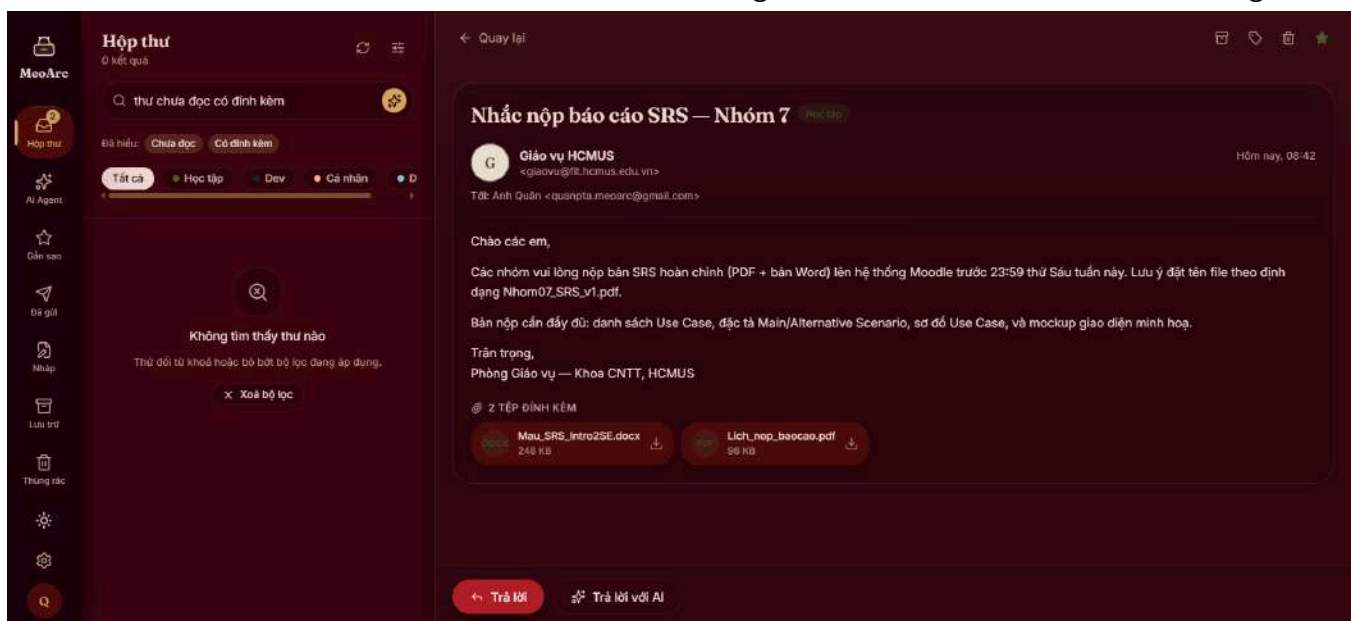
Tìm theo nội dung (thật): gõ vào ô tìm → lọc trực tiếp theo người gửi, email, tiêu đề, nội dung, nhân — không phân biệt dấu/hoa-thường (gõ "srs" vẫn ra email của Giáo vụ có keyword đấy). Số "kết quả" cập nhật theo.



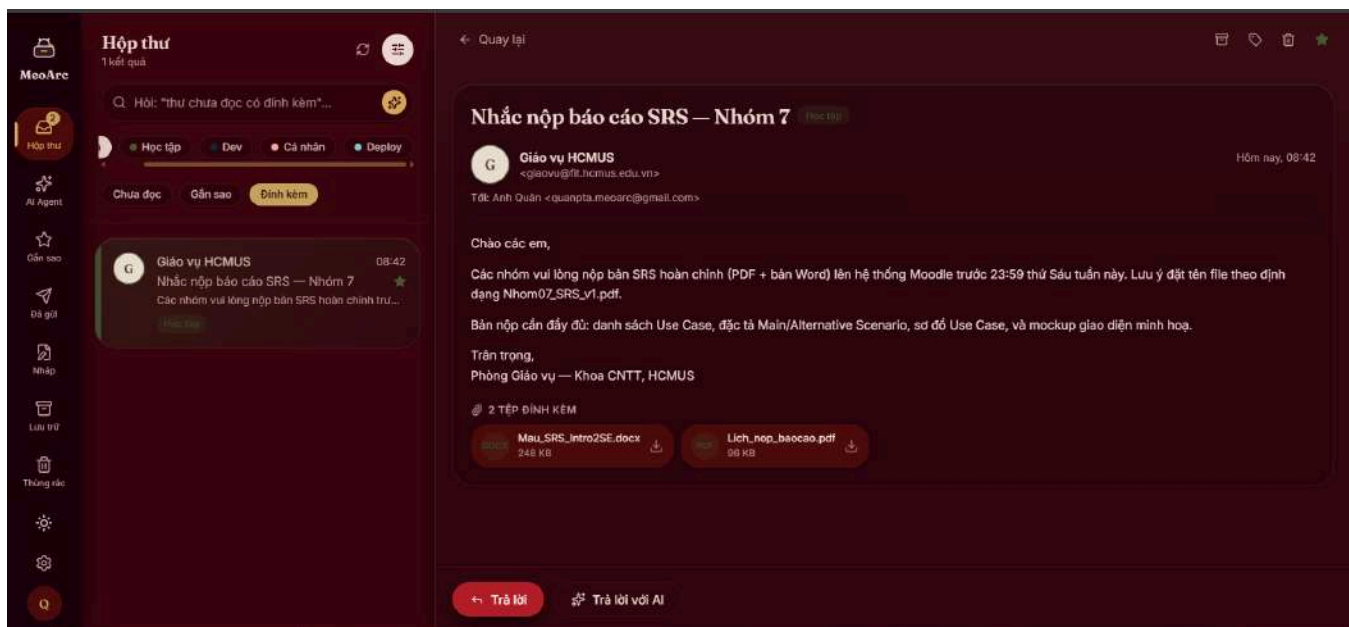
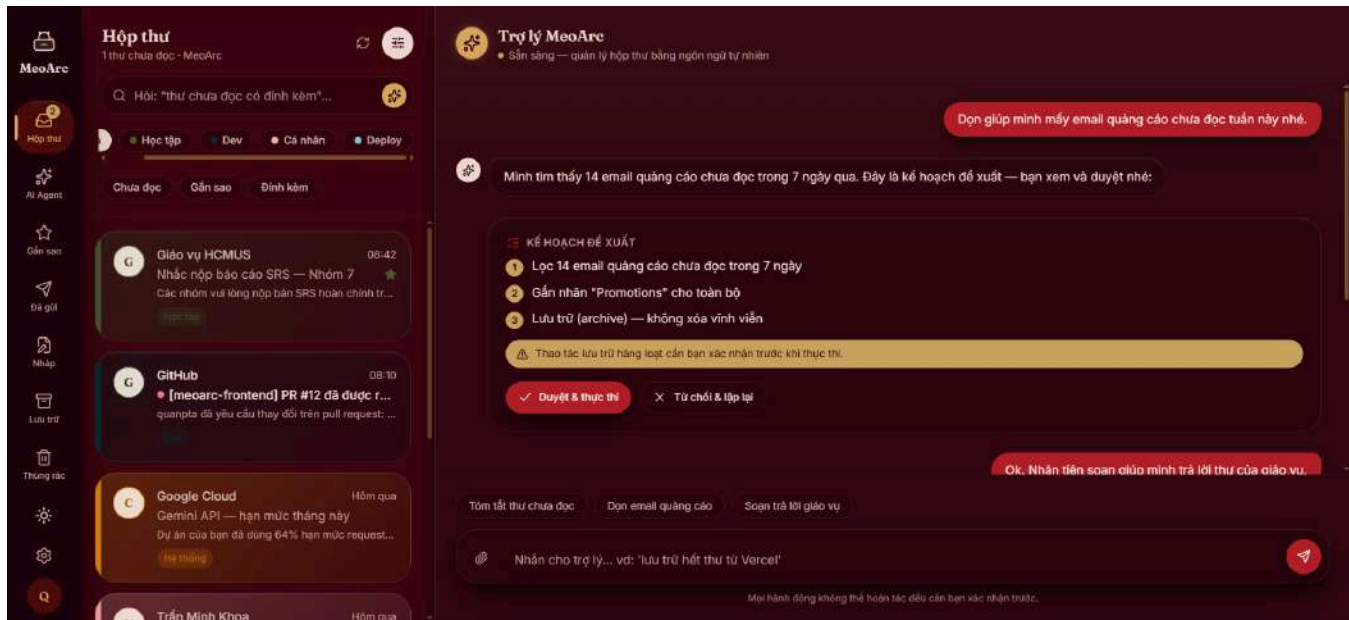
Tìm bằng ngôn ngữ tự nhiên (semantic, mock): bấm nút ✨ trong ô tìm để bật chế độ Natural Language → gõ ví dụ "thư chưa đọc có đính kèm" → hệ thống "hiểu" và hiện chip tiêu chí (Chưa đọc, Có đính kèm) rồi lọc đúng.



Sau khi nhận mail Giáo vụ HCMUS thì thư đó đã được gán “Đã đọc” nên ô tìm kiếm sẽ trống:



Lọc theo tiêu chí: bấm nút (SlidersHorizontal) → hiện bộ lọc nhanh Chưa đọc / Gắn sao / Đính kèm; cộng với chip category (Học tập/Dev/Cá nhân/Deploy) — tất cả kết hợp kiểu AND.



4.2.6 Manage Emails

Written by: 24127250 – Phan Quang Tiến

Edited by: 24127529 – Nguyễn Chí Tài

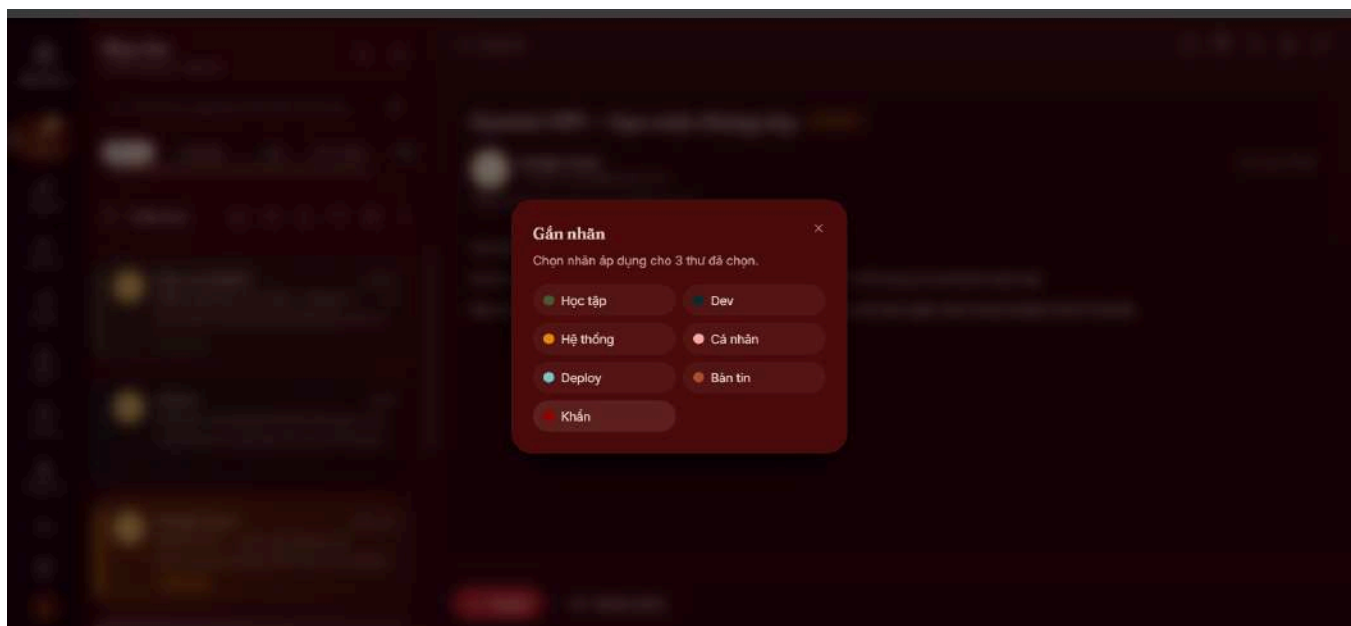
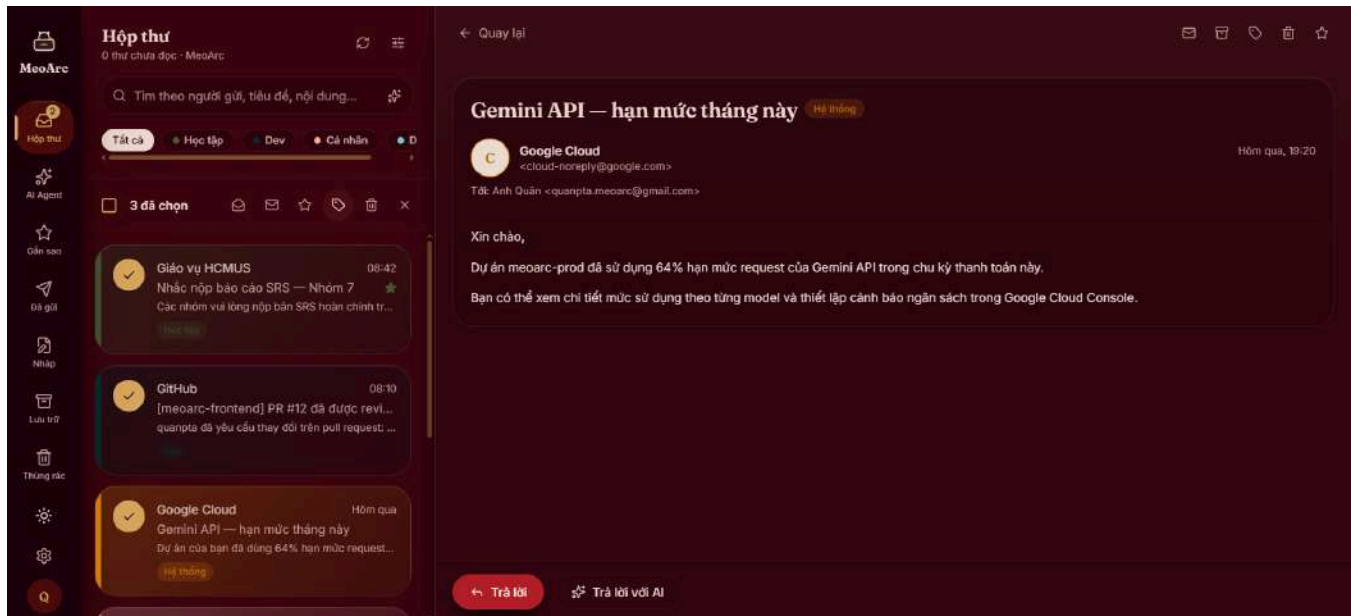
Reviewed by: 24127226 – Phạm Trần Anh Quân

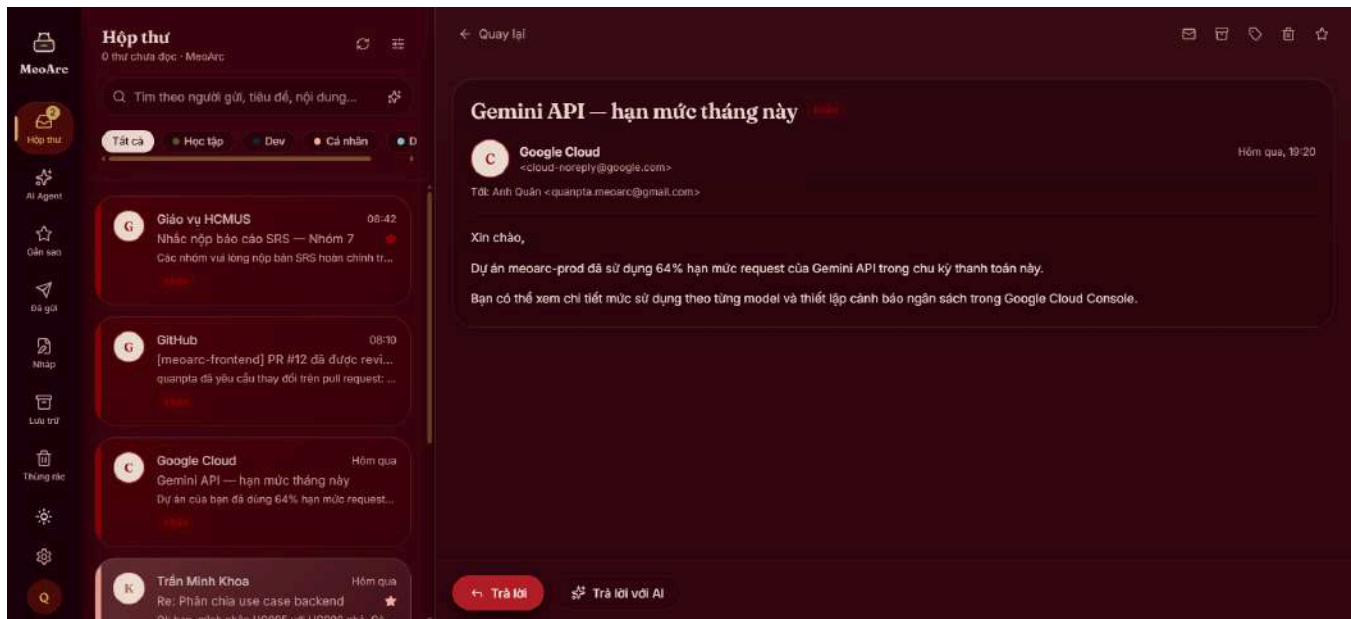
Use case ID	UC006
Use Case	Manage Emails
Brief Description	Describes how a user performs management actions on one or multiple emails including marking as read/unread, marking as important, applying labels and deleting emails through the MeoArc interface.
Actor	User
Pre-Conditions	1. The user has successfully authenticated through UC001 - Authenticate & Connect Mailbox. 2. The user has an active session in MeoArc. 3. The email list is currently displayed (UC003).
Post-Conditions	1. The selected action has been successfully applied to the target email(s). 2. The email list is updated to reflect the changes. 3. The user receives a confirmation notification of the completed action.
Main Scenario	1. The user selects 1 or more emails from the email list. 2. The system highlights the selected emails and displays the available management actions (mark as read, mark as unread, mark as important, apply label, delete). 3. The user selects the desired action.

	4. If the action is irreversible (e.g., delete), the system displays a confirmation dialog showing the details of the intended action. 5. The user confirms the action. 6. The system sends the request to the Gmail API to execute the action. 7. The system updates the email list to reflect the applied changes. 8. The system notifies the user with a success message (e.g. "2 emails have been deleted successfully.").
Alternative Scenarios	<i>A1. User cancels the confirmation dialog (At step 5)</i> 5a. The user select "Cancel" in the confirmation dialog. 5b. The system closes the dialog. 5c. No changes are applied to the selected emails. 5d. The user returns to the email list view. <i>A2. Bulk operation on emails matching criteria (At step 1)</i> 1a. The user specifies a set of criteria (e.g. all unread emails from a specific sender) instead of manually selecting individual emails. 1b. The system retrieves all emails matching the criteria. 1c. The system displays the matched email count and the intended action for confirmation. 1d. The use case continues from Step 4. <i>A3. Gmail service temporarily unavailable (At step 6)</i>

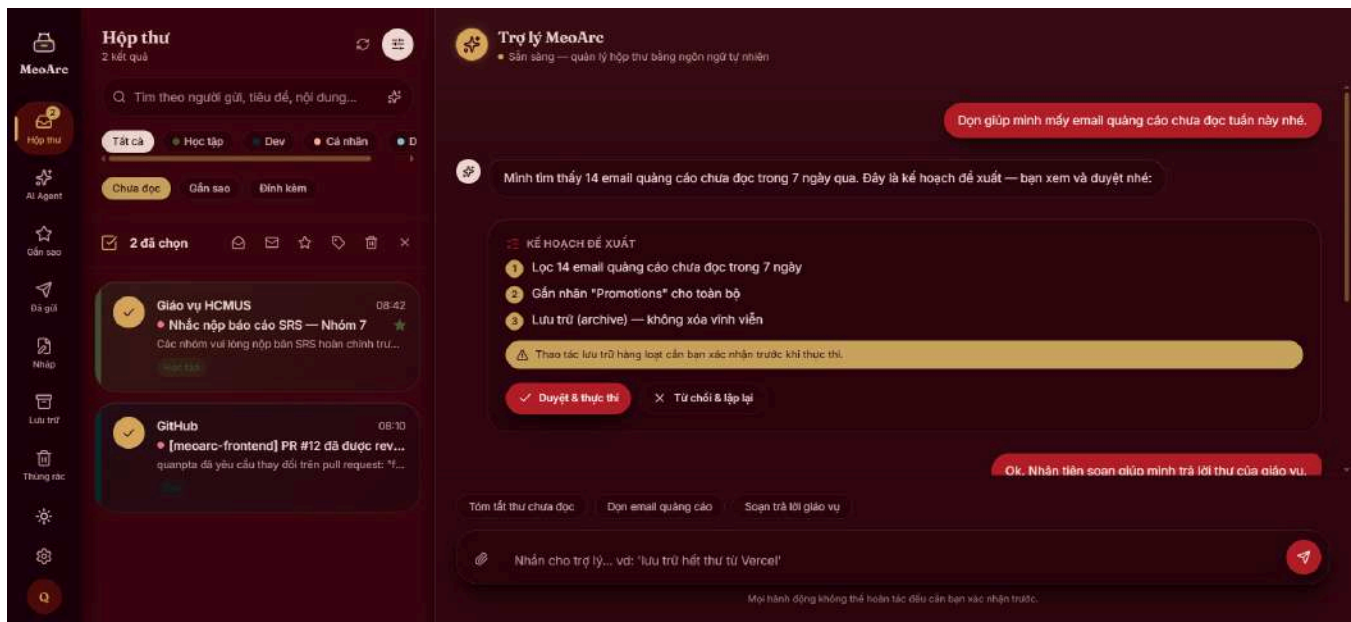
	6a. The system is unable to execute the action due to a temporary service or network issue. 6b. The system displays a message: "The action could not be completed. Please try again later." 6c. No changes are applied to the selected emails. 6d. The user remains on the current email list view. <i>A4. Partial failure in bulk operation (At step 6-7)</i> 6a. The system successfully applies the action to some emails but fails on others. 6b. The system updates the email list for successfully processed emails. 6c. The system displays a partial success message: "Action applied to X of Y emails. Some emails could not be processed."
Non-Functional Constraints	NFR-PER-01: The system shall complete email management actions within 3 seconds for at least 95% of requests under normal operating conditions. NFR-SEC-03: All email management requests shall be transmitted using HTTPS/TLS encryption. NFR-USA-02: Confirmation dialogs and result notifications shall be clear and understandable to non-technical users. NFR-REL-01: The system shall gracefully handle temporary Gmail API failures and provide meaningful error messages to users.

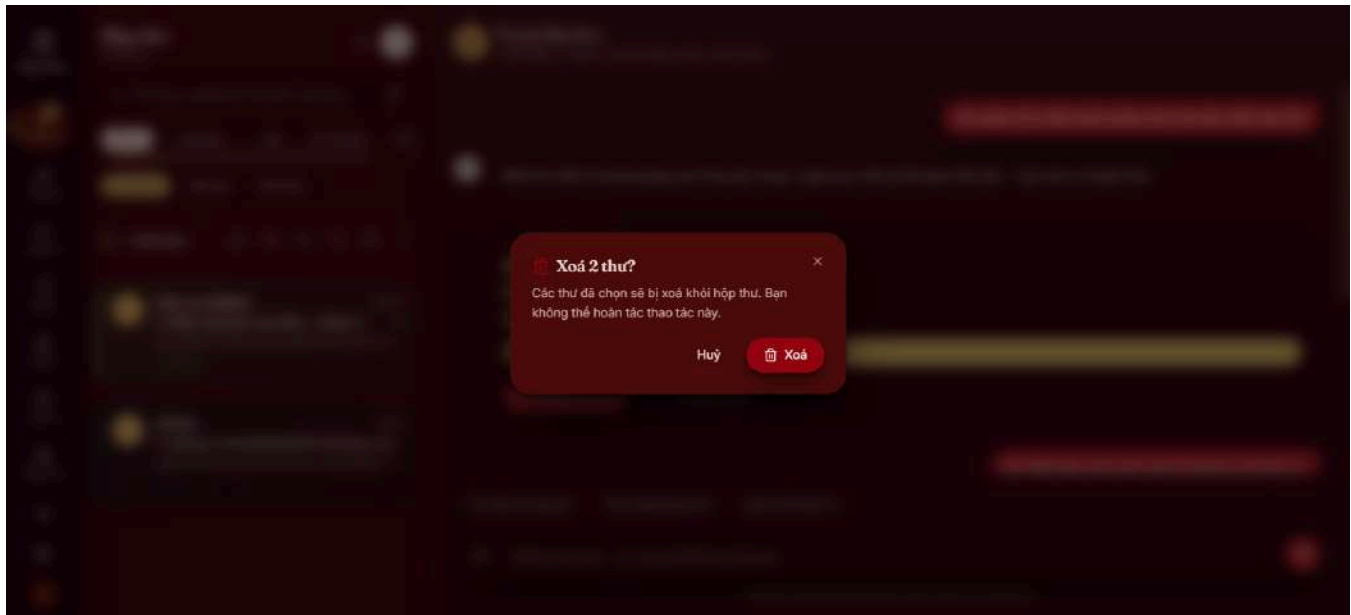
Rê chuột lên 2-3 thư → tick chọn → bấm  gắn nhãn "Khẩn" → nhãn đổi ngay.



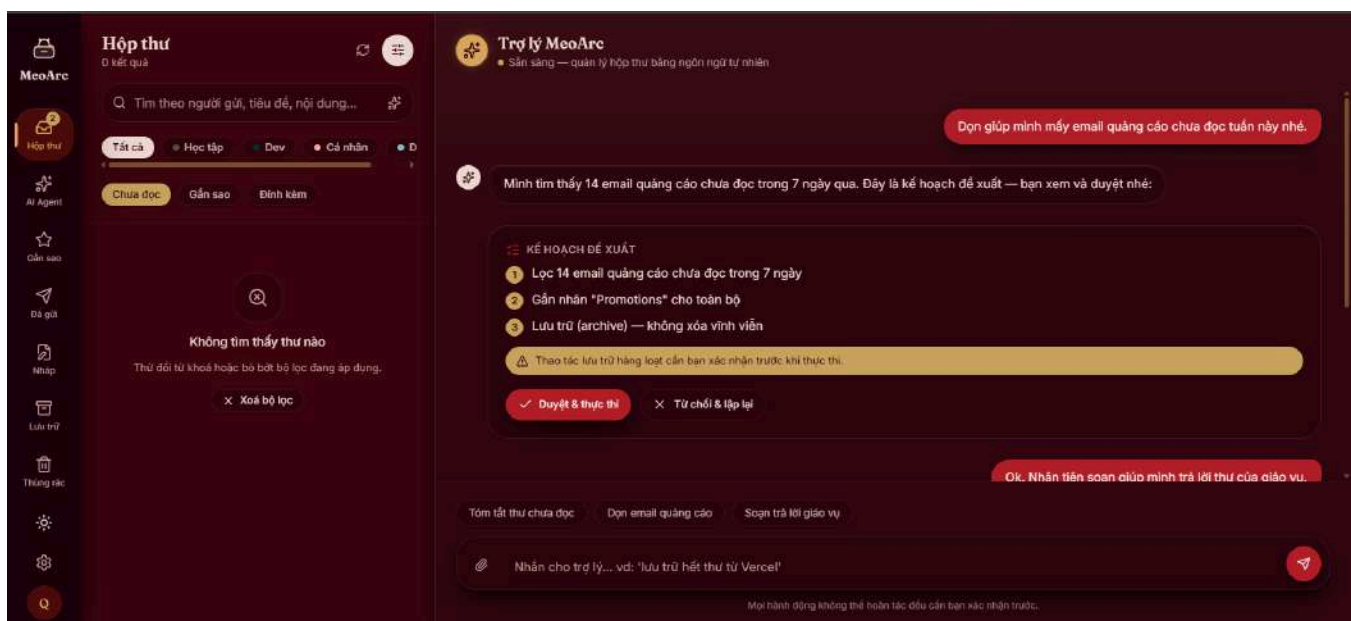


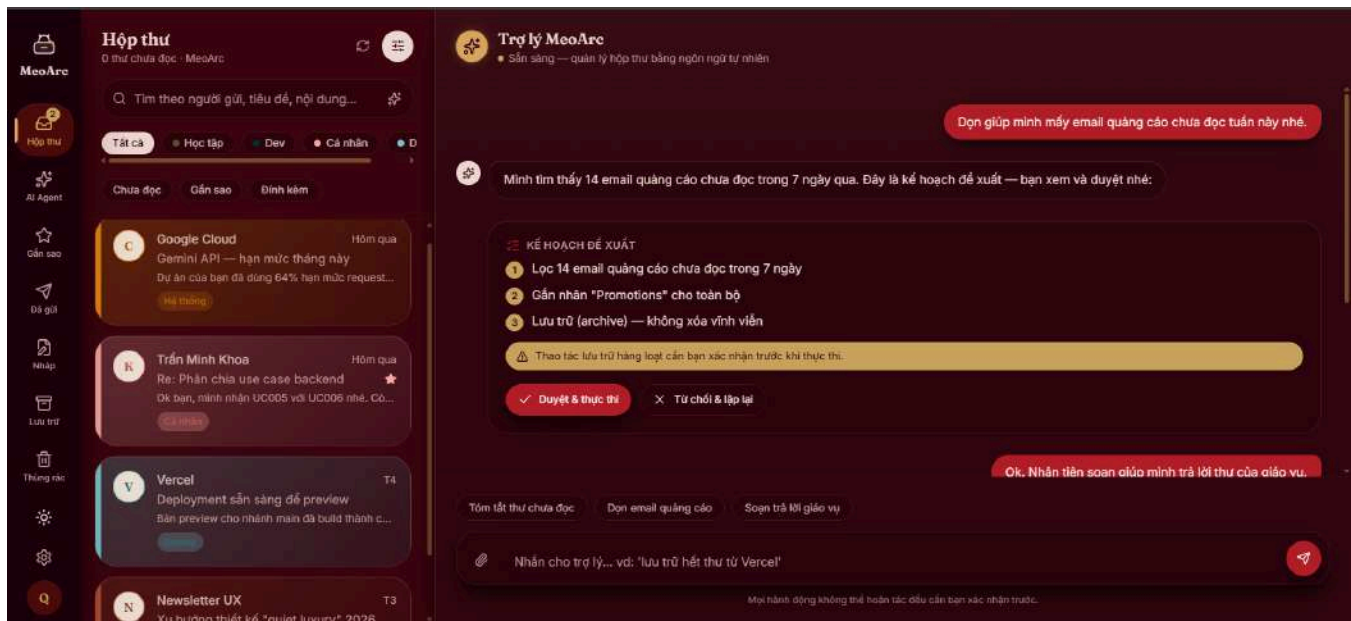
Bật bộ lọc Chưa đọc → bấm chọn tất cả trong thanh bulk → → xác nhận → xóa cả nhóm.



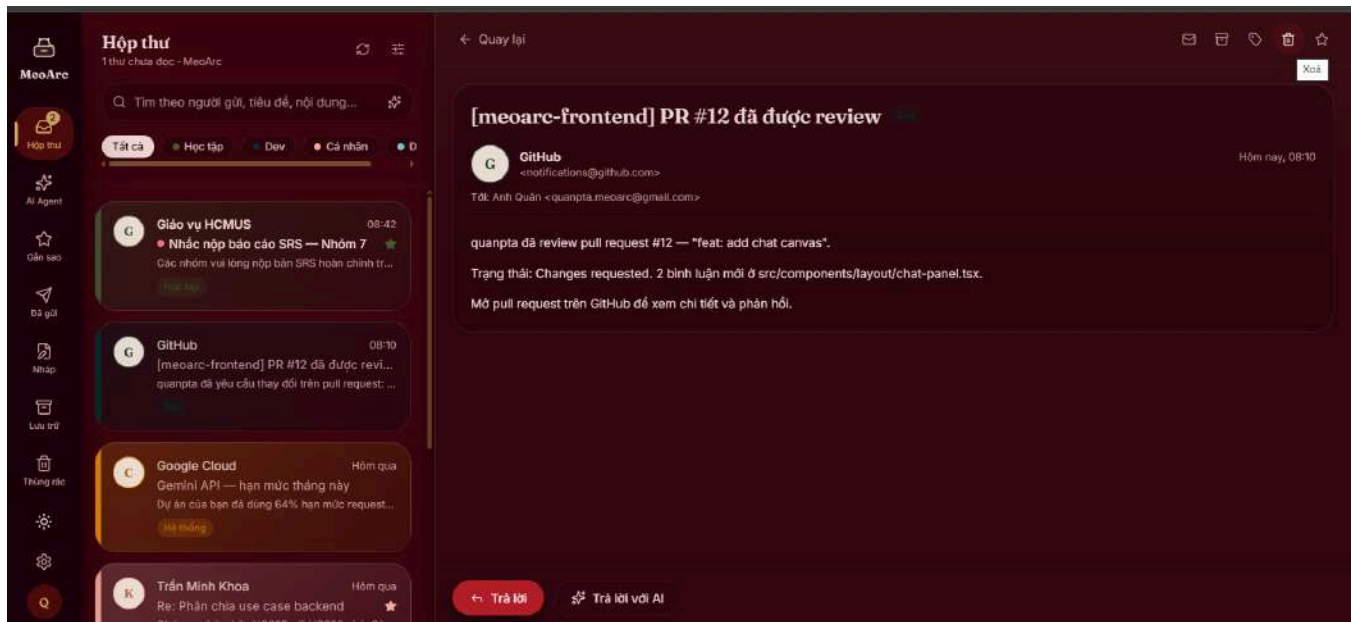


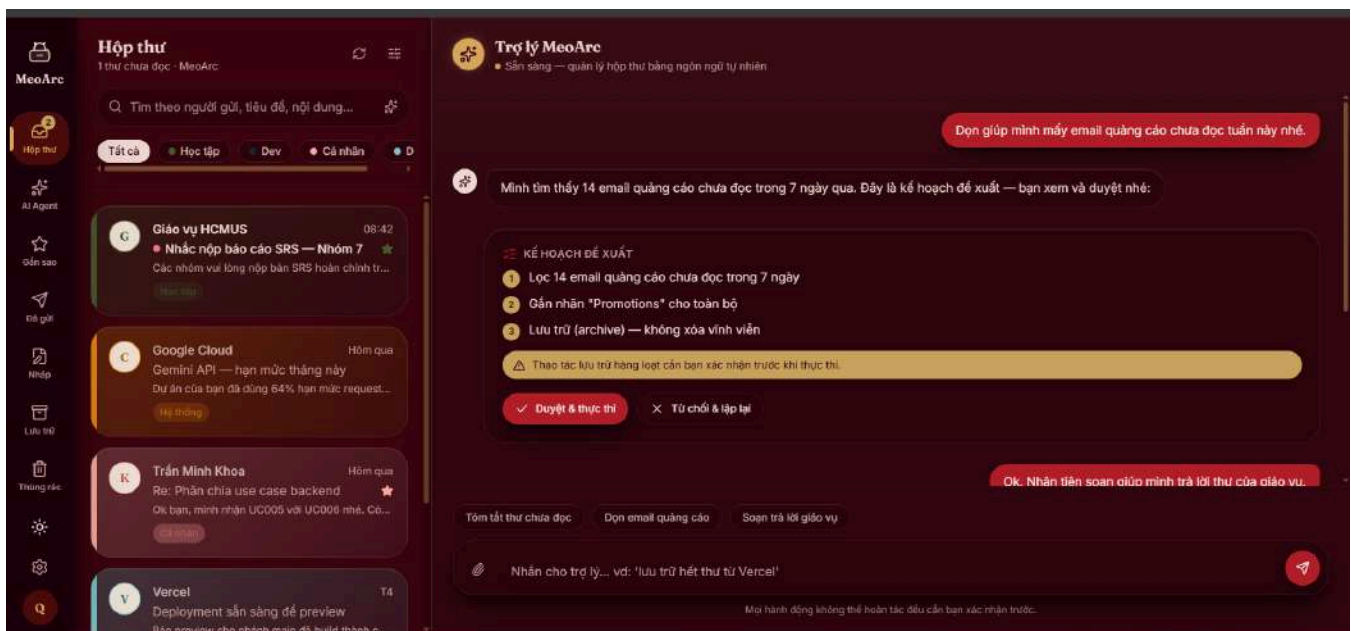
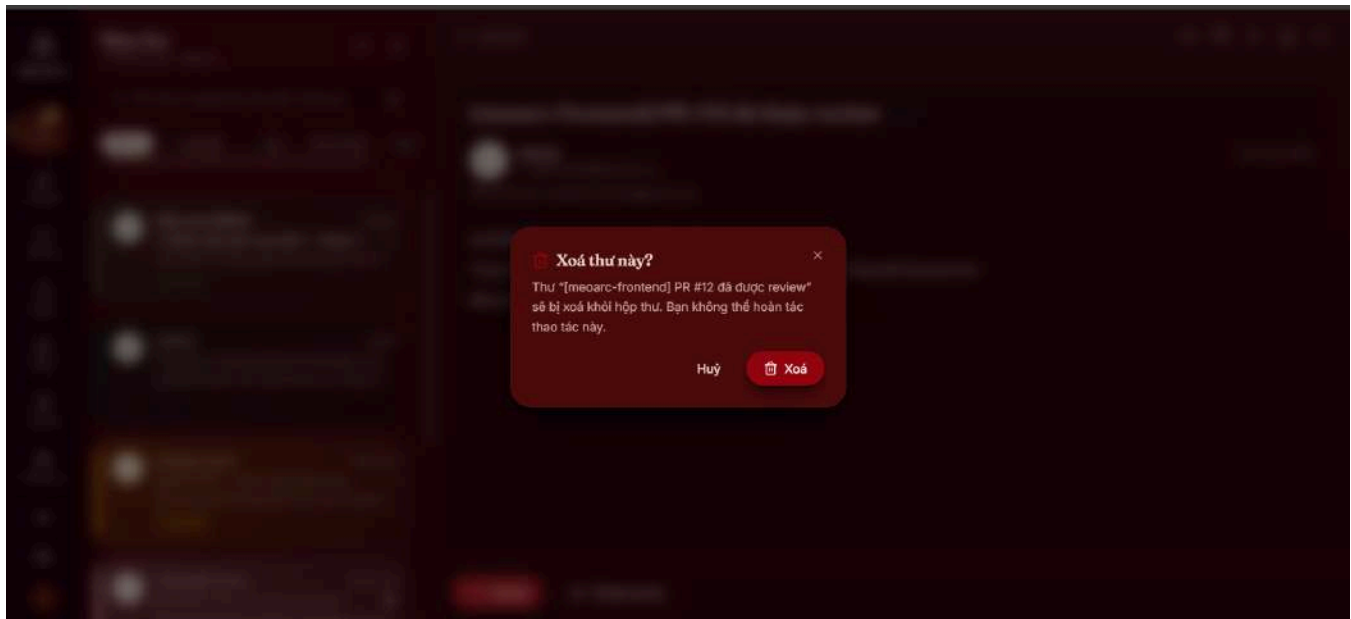
Đã bị xóa





Mở một thư → bấm 🗑 ở header → dialog xác nhận.





4.2.7 Manage Mailbox via Natural Language

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID

UC007

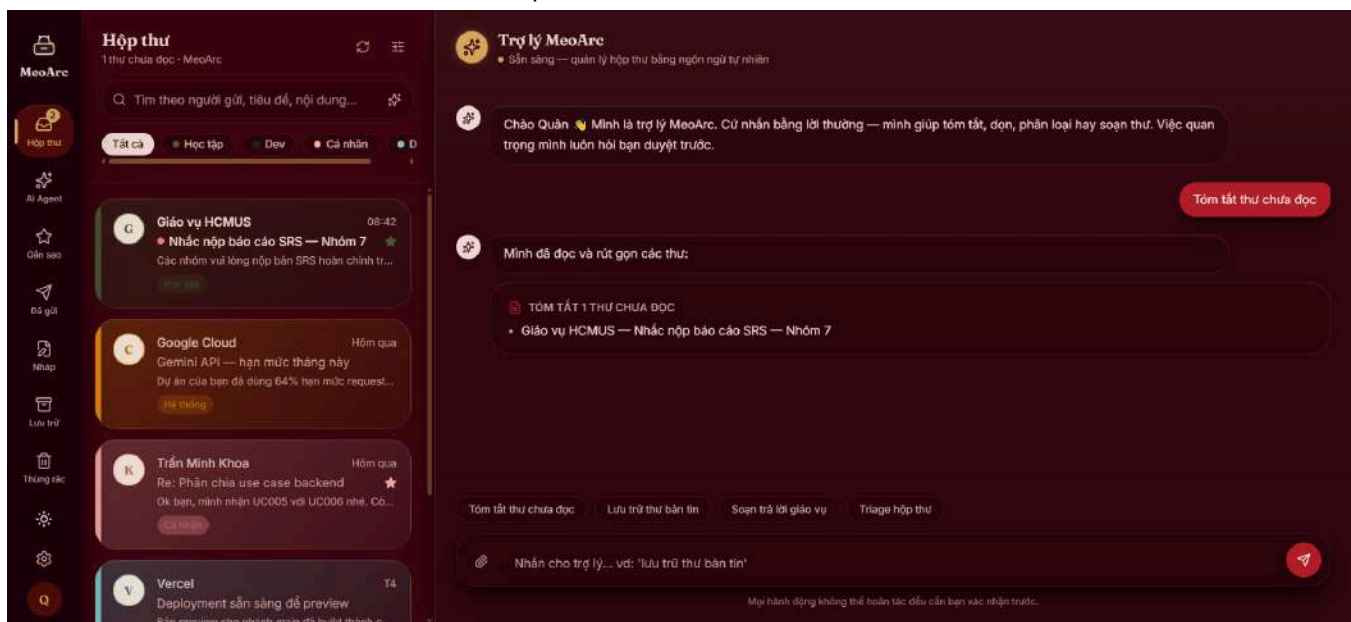
Use Case	Manage Mailbox via Natural Language
Brief Description	This is the central use case of MeoArc. The user interacts with the AI Agent through a natural language chat interface to manage their mailbox. The LLM acts as the controller: it receives the request, forms a plan, selects the appropriate tools, executes each step, and returns the result. This use case serves as the entry point for all AI-driven actions including UC008, UC009, UC010, UC014, UC015, UC016
Actor	Primary: User Secondary: Gemini API (Agent in MeoArc), Gmail API
Pre-Condition	1. User has successfully authenticated (UC001) 2. At least one email account has been connected (FR-01.2) 3. User opens the chat interface - either a new session or a continued session from history (the session must be not full context)
Post-Condition	1. The requested task has been executed successfully or cancelled if the user denied 2. The system sends a completion notification summarizing what executed including any errors 3. All conversation turns are persisted in conversation history
Main Scenario	1. The user enters a natural language request (can add attachments) 2. Input guardrails validate the request for safety/policy compliance. 3. Memory Manager loads relevant conversation history; Skill Loader retrieves relevant skill(s) from Skill Repository via semantic search 4. Agent reasoning loop begins: LLM thinks, and selects a tool from the Tool Registry (or decides to call request_confirmation tool if the action is risky). 5. LangGraph runtime executes the selected tool call. If the tool involves mailbox operations, the call is routed through the MCP Client Adapter to the external mail provider. 6. Agent observes the tool result, evaluates whether the loop should continue (more tools needed) or end

	7. Steps 4–7 repeat as needed (the reasoning loop). 8. Output guardrails check the final response and any proposed tool actions before returning to user. 9. Response is delivered via the canvas; conversation state persisted to PostgreSQL.
Alternative Scenarios	A1 – Input Rejected by Guardrails (at step 2) 2a. Input guardrails detect unsafe/malicious content (e.g. prompt injection) 2b. System rejects the request with an explanatory message A2 – Clarification Needed (at step 4 within loop) 4a. The agent determines the request is ambiguous or missing required parameters. Instead of selecting a tool, it generates a clarifying question 4b. System returns this to the user and pauses the loop, awaiting reply 4c. Flow resumes at Step 1 with the user's clarification appended to context. A3 – Confirmation Required (at step 4 within loop) 4a. Agent determines the next action is risky/irreversible (send, delete, bulk operation) and calls the <code>request_confirmation</code> tool instead of the action tool directly 4b. The system displays the confirmation prompt with full action details. Loop pauses until the user responds. A3.1 – User Approves 4c. User confirms 4d. Loop resumes at Step 4; agent proceeds to call the actual action tool. A3.2 – User Denies 4c'. User rejects 4d'. Agent records the cancellation, generates a response acknowledging it, and the loop ends without performing the action. A4 – Tool Execution Failure (at step 5)

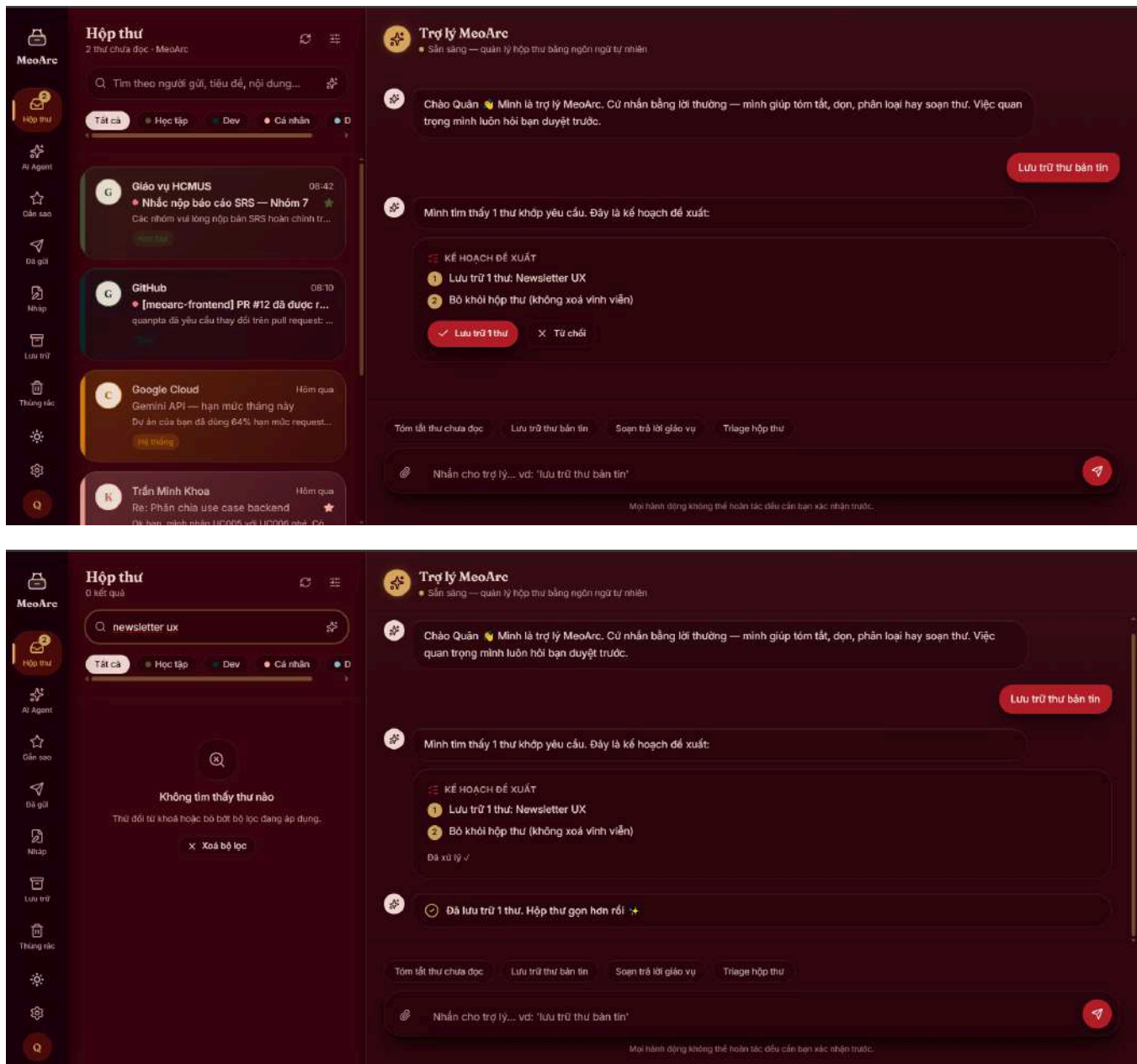
	5a. External provider (Gmail API) returns an error 5b. Gemini API decides how to proceed (retry, skip or report) A5 – External LLM Service Unavailable (at step 4 or 6) a. Gemini API times out or returns an error. b. System displays a user-friendly error message and prompts the user to retry -> use case end A6 – Output Rejected by Guardrails (at step 8) 8a. Output guardrails detect the generated response or proposed tool action violates policy (e.g. leaking another user's data). 8b. The system blocks the response and returns a generic fallback message instead
Non-Functional Constraints	NFR-SEC-03: All communications between clients, servers, and third-party services shall be protected using HTTPS/TLS encryption. NFR-SEC-05: The system shall implement rate limiting to prevent excessive API requests from a single user or IP address. NFR-REL-01: The system shall gracefully handle temporary failures from Gmail API or Gemini API and provide meaningful error messages to users. NFR-REL-02: The system shall not lose conversation history or email metadata during unexpected system failures. NFR-MAI-04: The system shall log all API requests (timestamp, endpoint, user ID, HTTP status) and all agent task executions (tool calls, errors, token usage) to a persistent log store for debugging purposes. NFR-SCA-01: The system architecture shall support future integration with additional Large Language Model (LLM) providers without requiring major architectural modifications. NFR-SCA-02: The system architecture shall support the addition of new email-management tools exposed through REST APIs or the Model Context Protocol (MCP).

NFR-SCO-01: On-demand AI operations that scan existing mailbox content (Semantic Search, Email Categorization, Summarization, Task Extraction, Reply context retrieval) shall limit the time range of emails processed according to the user's subscription tier: 90 days (Free), 180 days (Pro), 365 days (Pro Max).

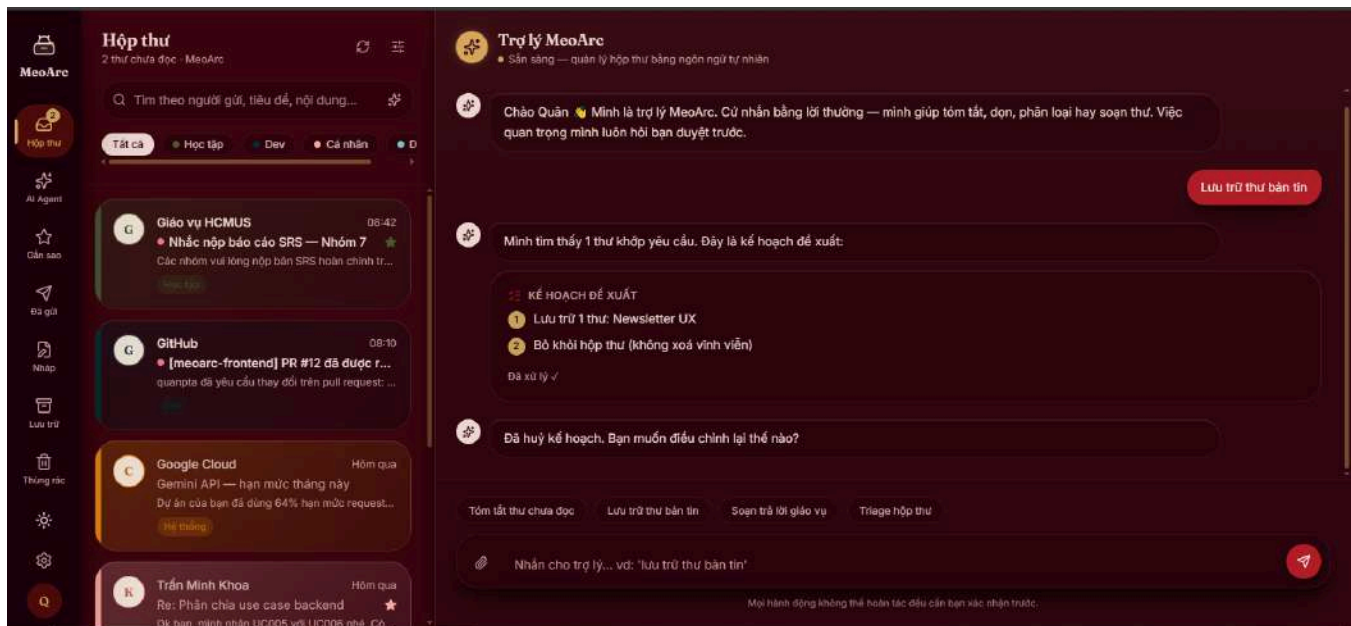
"Tóm tắt thư chưa đọc" (hoặc bấm chip) → result card liệt kê thư



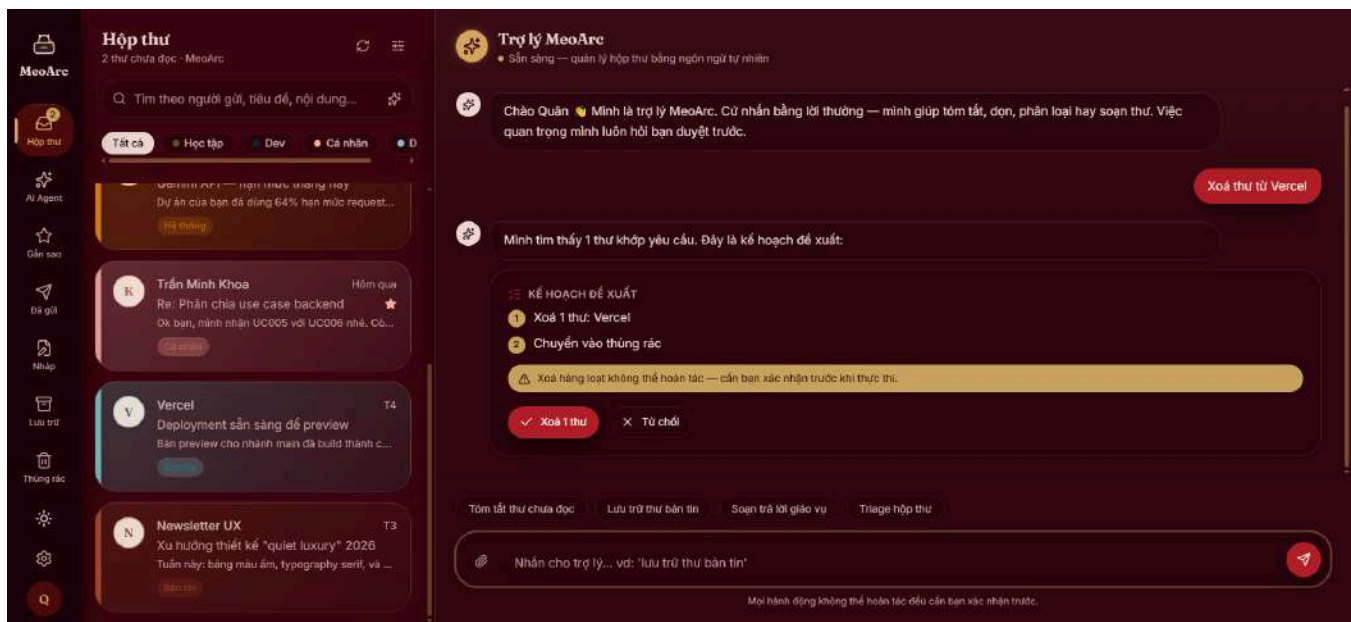
"Lưu trữ thư bản tin" → plan card (các bước) → Duyệt → Newsletter biến mất khỏi list giữa + completion summary ✓.

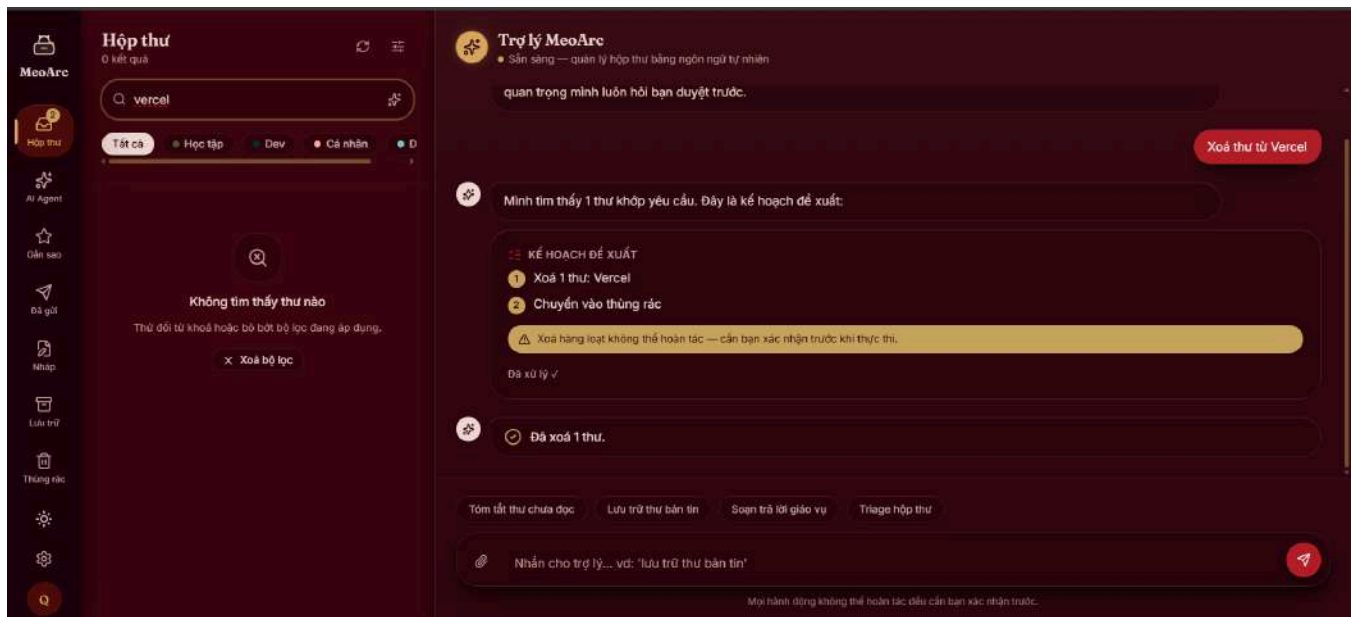


Bấm Từ chối → agent huỷ & hỏi lại.

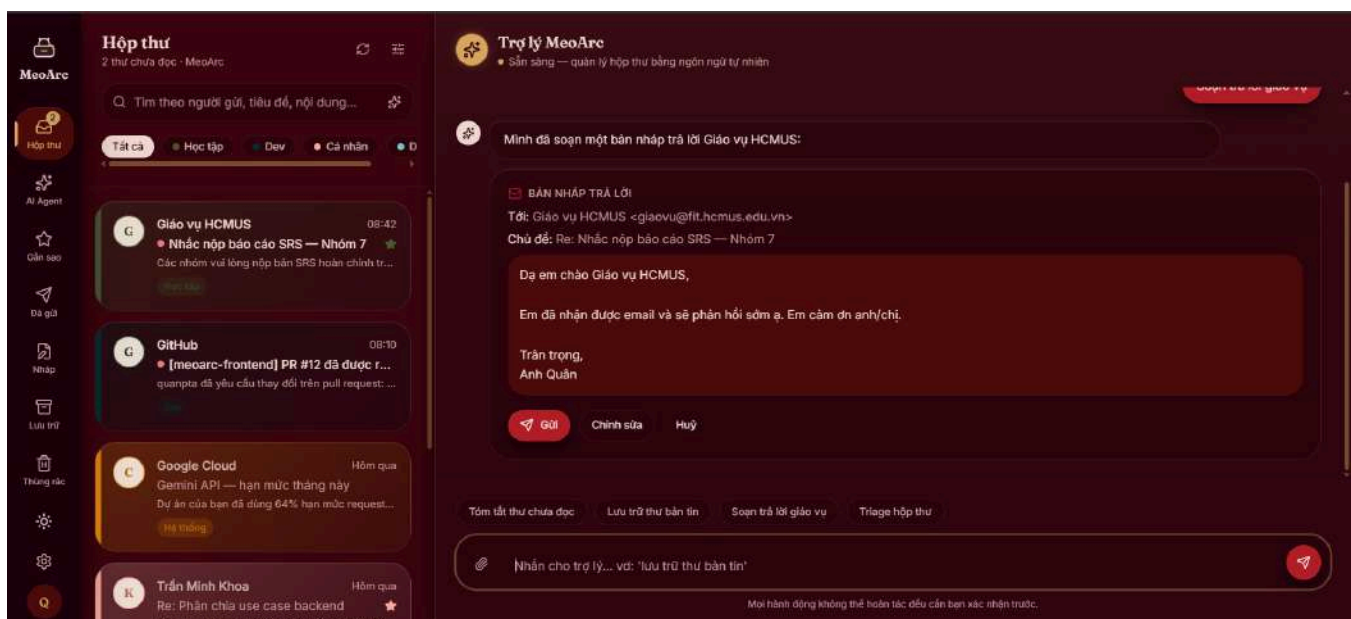


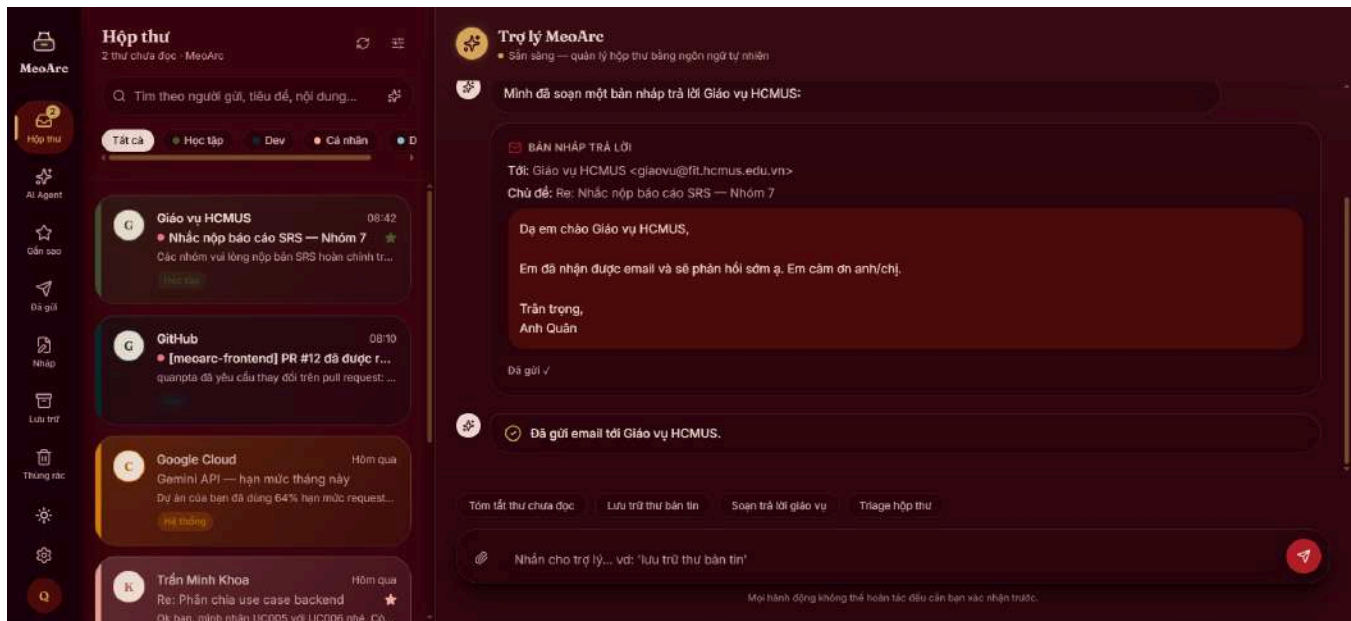
"Xoá thư từ Vercel" → plan có cảnh báo không hoàn tác → Duyệt mới xoá (UC006 qua NL).



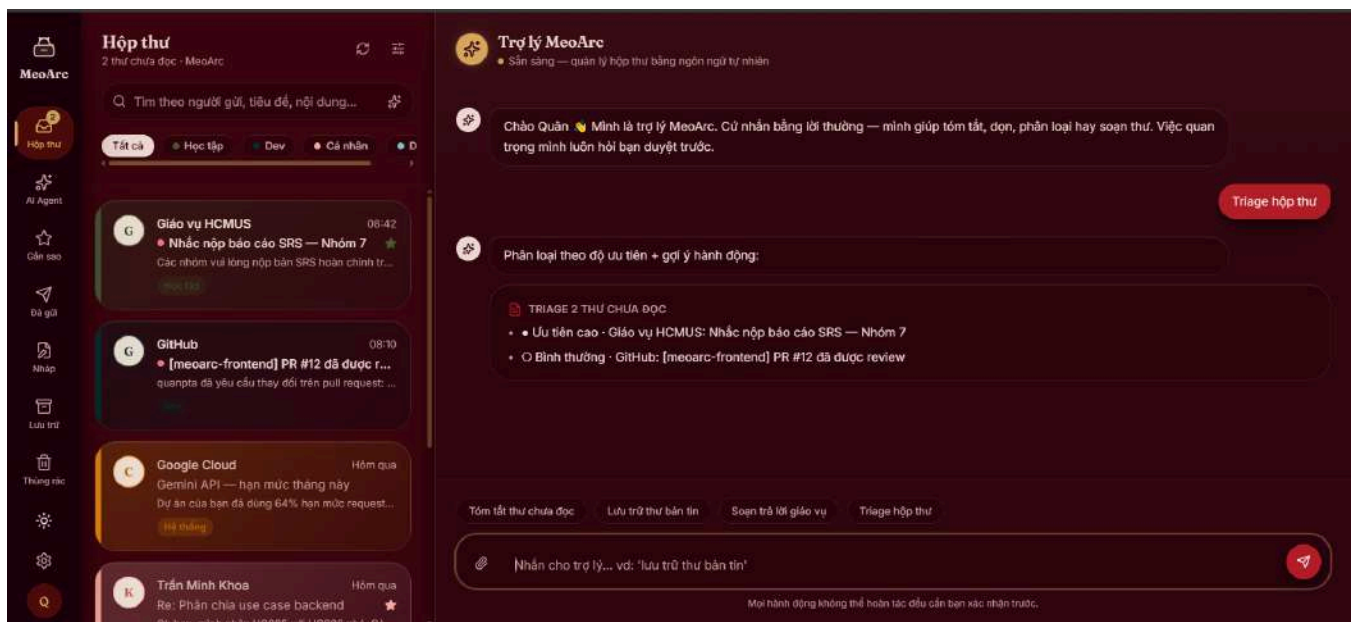


"Soạn trả lời giáo vụ" → draft preview → Gửi → "đã gửi" (UC010).

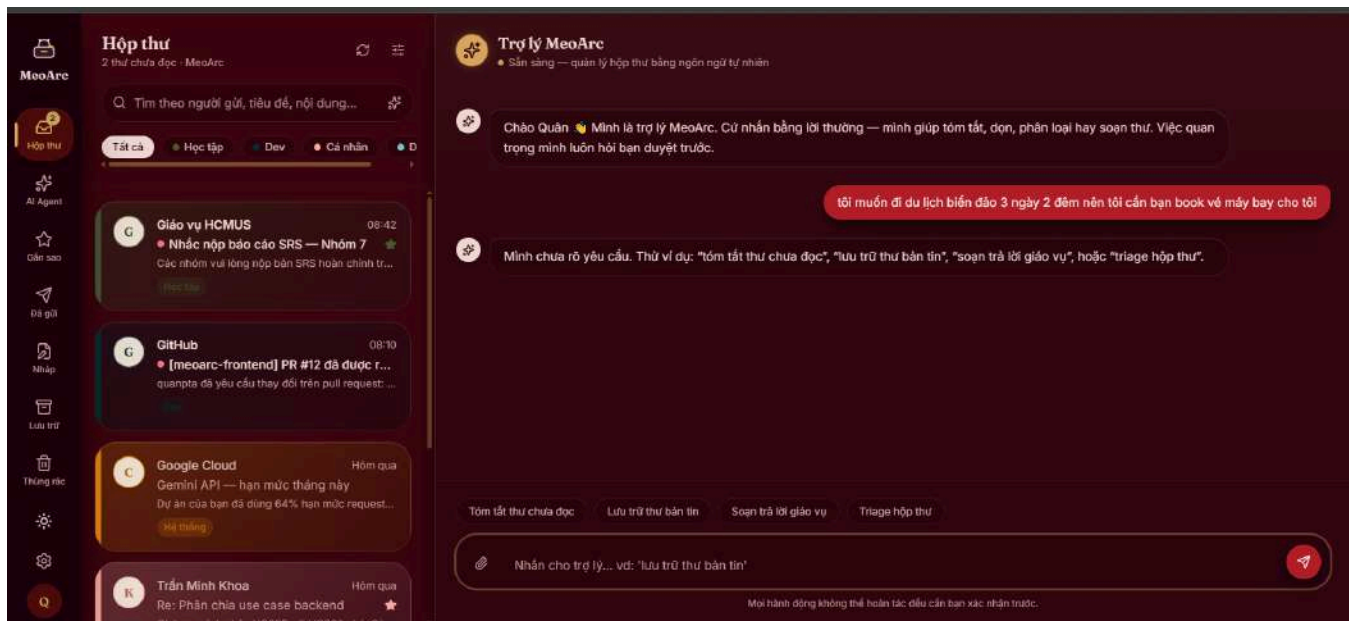




"Triage hộp thư" → result card phân loại ưu tiên + gợi ý (UC015).



Cô linh tinh → agent hỏi lại (clarify).



4.2.8 Summarize Email

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127226 – Phạm Trần Anh Quân

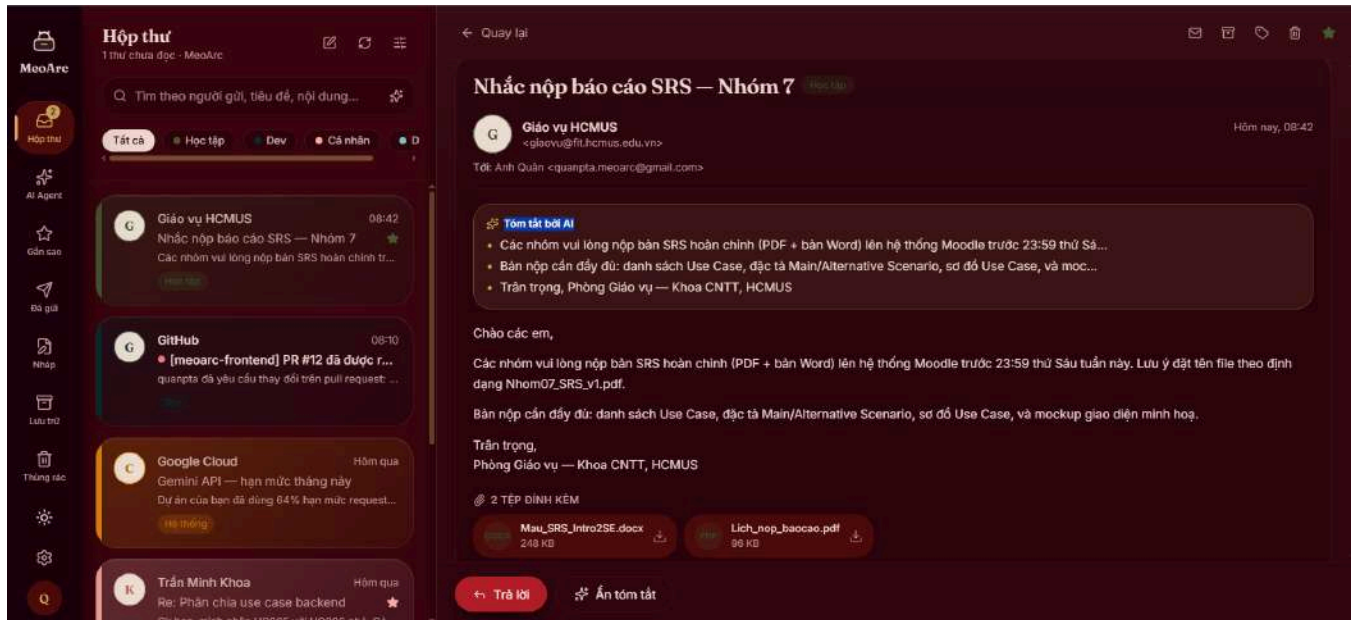
Use case ID	UC008
Use Case	Summarize Email
Brief Description	The user requests a concise natural language summary of one or more emails through the chat interface. This is a sub-use-case of UC007 (Manage Mailbox via Natural Language) – it represents a specific instance of the agent reasoning loop where the Skill Loader injects the relevant summarization skill (if applicable) and the agent selects summarization-related tools to produce the result.
Actor	Primary: User Secondary: Gemini API (agent), Gmail API
Pre-Condition	1. User has successfully authenticated (UC001)

	- At least one email account has been connected (FR-01.2) - The target email exist in the system and have been synchronized - User opens the chat interface - either a new session or a continued session from history (the session must be not full context)
Post-Condition	- A natural language summary of the requested email has been generated and displayed in the canvas - The summary, along with the tool calls and reasoning steps that produced it, is persisted to conversation history. - No change occurs to the email data (read-only operation).
Main Scenario	- The user enters a summarization request in natural language (e.g. "Summarize today's emails from my manager") - Input guardrails validate the request. - Memory Manager loads relevant conversation history; Skill Loader searches for skill but for a simple single-email summary, no skill is needed and the agent proceeds directly. - Agent reasoning loop: parses intent, extracts parameters, and decides the next tool call (e.g. <code>search_email</code> with current criteria). - LangGraph runtime executes the selected tool call. - Provider returns matching email content (or summarization result, if <code>summarize_email</code> was called) via the MCP Client Adapter / Gemini API. - Agent observes the result and evaluates (to repeat Steps 4-7 as needed) - Output guardrails check the response - System sends a completion notification - The response is delivered to the user via the canvas; conversation state is persisted to PostgreSQL.
Alternative Scenarios	A1 – Input Rejected by Guardrails (at step 2) - Input guardrails detect unsafe/malicious content (e.g. prompt injection) - System rejects the request with an explanatory message

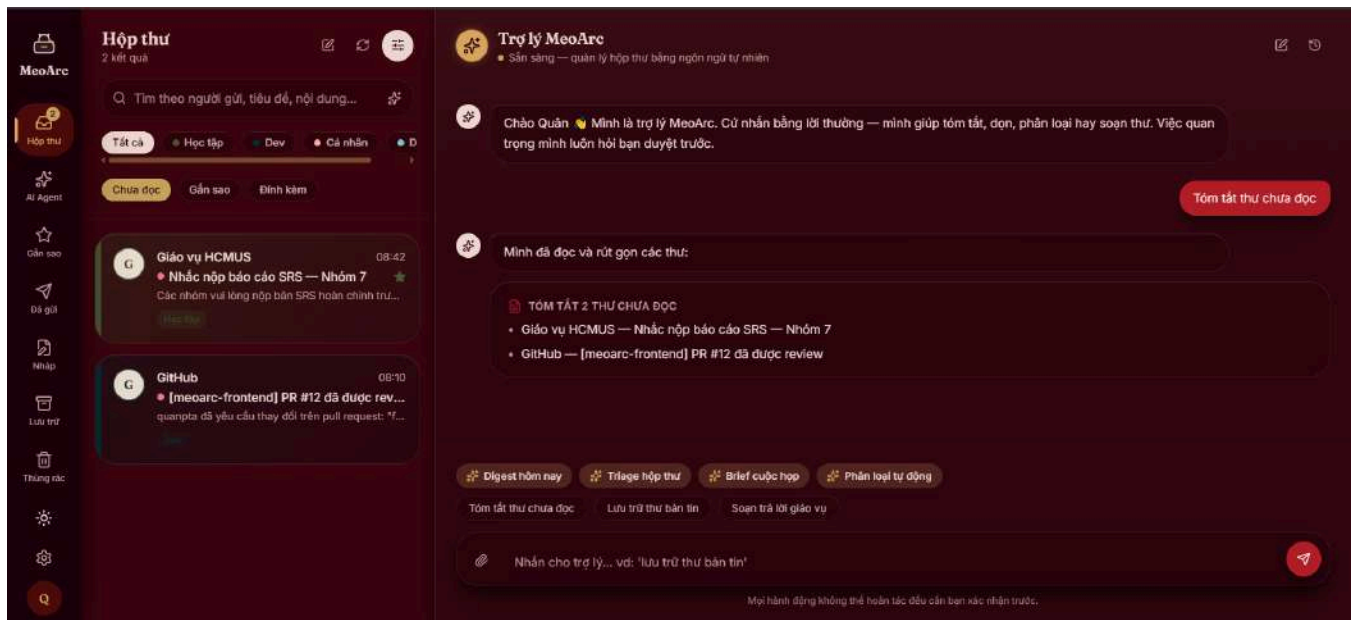
	A2 – Clarification Needed (at step 4 within loop) 4a. Agent determines the request is ambiguous or missing required parameters. Instead of selecting a tool, it generates a clarifying question 4b. System returns this to the user and pauses the loop, awaiting reply 4c. Flow resumes at Step 1 with the user's clarification appended to context. A4 – Tool Execution Failure (at step 5) 5a. External provider (Gmail API) returns an error 5b. Gemini API decides how to proceed (retry, skip or report) A5 – No Matching Emails (at step 7) 7a. Agent observes an empty search result and determines the task is complete with nothing to summarize 8b. Proceeds directly to the final response informing the user no matching emails were found. A6 – External LLM Service Unavailable (at step 4 or 6 or 7) a. Gemini API times out or returns an error. b. System displays a user-friendly error message and prompts the user to retry -> use case end A7 – Output Rejected by Guardrails (at step 8) 8a. Output guardrails detect the generated response or proposed tool action violates policy (e.g. leaking another user's data). 8b. The system blocks the response and returns a generic fallback message instead
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Non-Functional Constraints	NFR-SEC-03: All communications between clients, servers, and third-party services shall be protected using HTTPS/TLS encryption. NFR-SEC-05: The system shall implement rate limiting to prevent excessive API requests from a single user or IP address. NFR-REL-01: The system shall gracefully handle temporary failures from Gmail API or Gemini API and provide meaningful error messages to users. NFR-REL-02: The system shall not lose conversation history or email metadata during unexpected system failures. NFR-MAI-04: The system shall log all API requests (timestamp, endpoint, user ID, HTTP status) and all agent task executions (tool calls, errors, token usage) to a persistent log store for debugging purposes. NFR-SCA-01: The system architecture shall support future integration with additional Large Language Model (LLM) providers without requiring major architectural modifications. NFR-SCA-02: The system architecture shall support the addition of new email-management tools exposed through REST APIs or the Model Context Protocol (MCP). NFR-SCO-01: On-demand AI operations that scan existing mailbox content (Semantic Search, Email Categorization, Summarization, Task Extraction, Reply context retrieval) shall limit the time range of emails processed according to the user's subscription tier: 90 days (Free), 180 days (Pro), 365 days (Pro Max).
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Tóm tắt một thư cụ thể



Tóm tắt nhiều thư



4.2.9 Categorize Email

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC009
Use Case	Categorize Email
Brief Description	The user requests the AI Agent, via the chat interface, to classify one or more emails using the system's fixed multi-dimensional label taxonomy: Category – always assigned, exactly one of seven fixed values (School, Work, Finance, Social, Shopping, System, Personal) – plus Priority (High, Medium, Low) and Status (Todo, Waiting, Done), which are assigned only for task-like emails. (Spam is not a Category value; unsolicited mail is handled separately by the provider's spam filter.). Since label assignment is a reversible action, the agent applies the determined labels directly – no confirmation popup is required. The result is then displayed as a checklist for the user to review and adjust afterward. This use case only operates with the system's predefined label set; if the user requests a label name outside this taxonomy, the system informs them it is not supported (User do it manually). Processing scope is bounded by the user's subscription tier (AI Processing Scope Limitation). This is a sub-use-case of UC007.
Actor	Primary: User Secondary: Gemini API (Agent). Gmail API
Pre-Condition	1. User has successfully authenticated (UC001) 2. At least one email account has been connected (FR-01.2) 3. User opens the chat interface – either a new session or a continued session from history (the session must be not full context) 4. Target email(s) exist in the system and have been synchronized
Post-Condition	1. For each targeted email, the agent has applied a Category label, and – if the email was determined to be task-like – a Priority and Status label, directly on the mail provider 2. The system sends a completion notification summarizing the result

	3. The conversation turn, including the applied checklist, is persisted to conversation history for later review/adjustment
Main Scenario	1. The user enters a categorization request in natural language via the chat interface, e.g. <i>"Categorize my emails from the last 30 days"</i>. 2. Input guardrails validate the request for safety/policy compliance. 3. Memory Manager loads relevant conversation history; Skill Loader retrieves a categorization skill from Skill Repository via semantic search 4. Agent reasoning loop begins: LLM determines the target scope from the request and selects the next tool – search_emails to retrieve candidates (default last 30 days) 5. The system applies the user's tier-based AI Processing Scope to the requested date range as a hard upper bound – if the request exceeds it, the system silently proceeds using the clamped range (the maximum allowed under the user's tier) rather than failing or asking for clarification. This always happens, regardless of whether the original request fits within or exceeds the limit. 6. LangGraph runtime executes the selected tool call, routed through the MCP Client Adapter to Gmail API as needed. 7. Agent observes the retrieved emails. 8. Agent evaluates whether the loop should continue (e.g. more emails to fetch within the bounded scope) or proceed to classification. 9. Steps 4–8 repeat as needed (the reasoning loop). 10. For each retrieved email, agent determines: Category (always assigned) – exactly one of the seven fixed values: School, Work, Finance, Social, Shopping, System, Personal. (Spam is not a Category; it is filtered upstream by the mail provider.). Whether the email is task-like (i.e. requires an action, response, or follow-up from the user)

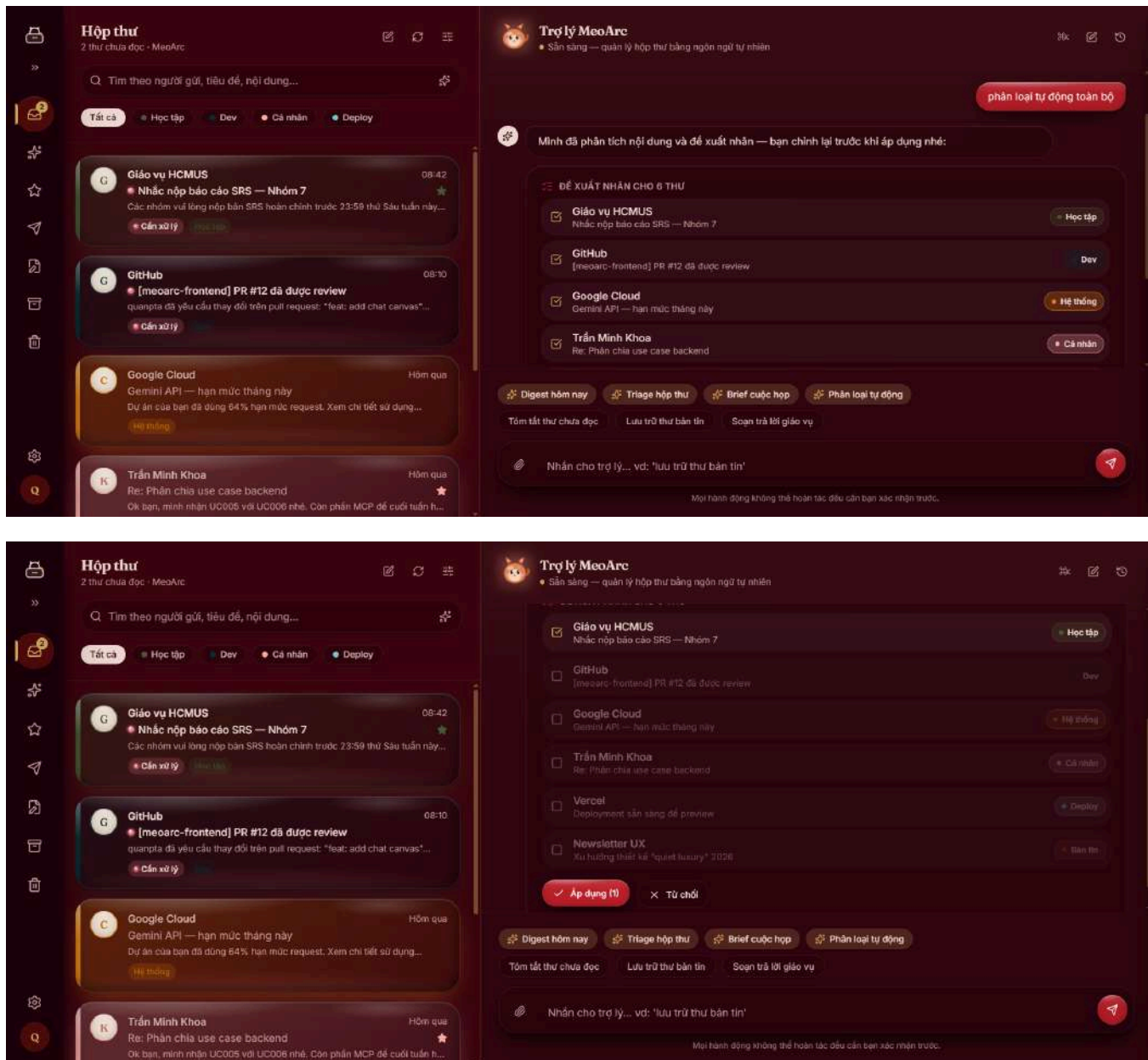
	a. If task-like: Priority (mandatory, High/Medium/Low) and Status (mandatory, Todo/Waiting/Done) are both determined. b. If not task-like (e.g. promotional, spam, automated notification): Priority and Status are left unassigned. 11. Output guardrails check the determined label set for all emails 12. System invokes apply_labels directly for the determined mapping – no request_confirmation call is made, since this is a reversible action. 13. Gmail API applies the corresponding labels to each email. 14. System displays the result as a checklist (grouped by email, showing applied Category, and Priority/Status if task-like), allowing the user to uncheck or change any assignment after the fact. 15. System sends a completion notification (e.g. <i>"18 emails categorized: 12 as task-like with Priority/Status assigned, 6 as non-task-like (Category only)."</i>). 16. Response is shown in the canvas and conversation state is persisted to PostgreSQL.
Alternative Scenarios	A1 – Input Rejected by Guardrails (at step 2) 2a. Input guardrails detect unsafe/malicious content (e.g. prompt injection) 2b. System rejects the request with an explanatory message -> use case end A2 – Unsupported Label Requested (at step 4 or 10) a. The user's request explicitly names a label outside the fixed taxonomy (e.g. a Category or Priority value that doesn't exist in the system). b. Agent does not attempt to apply it. System informs the user this label is not supported and lists the valid fixed values. c. Agent proceeds with the request using only the valid/supported parts, or asks for clarification (via request_clarification) if the entire request depended on the unsupported label.

	A3 — No Target Emails Found (at step 7) 7a. <code>search_emails</code> returns no matching emails within the bounded scope. 7b. Agent informs the user no emails were found to categorize -> use case ends. A4 — Notifying When Scope Was Limited (at step 15) 15a. If the date range was clamped in Step 5 (i.e. the user's original request implied a broader scope than their tier allows), the completion notification includes an explicit notice: "Note: your current plan processes emails from the last XX days. Some emails you mentioned may be outside this range. Upgrade to extend coverage." 15b. If the request was already within the tier's limit, no such notice is shown — the notification proceeds normally. A5 — Mandatory Dimension Cannot Be Confidently Determined (at step 10) 10a. Agent cannot confidently determine Category for a given email, OR cannot confidently determine whether the email is task-like (which governs whether Priority/Status apply). 10b. For Category: agent makes a best-effort assignment using the closest match — Category is never left blank. 10c. For task-like determination: if ambiguous, agent defaults to treating the email as non-task-like (leaves Priority/Status unassigned) rather than forcing an arbitrary Status value, and flags this email as "uncertain" in the checklist for user attention. A6 — User Adjusts Already-Applied Labels (at step 14)
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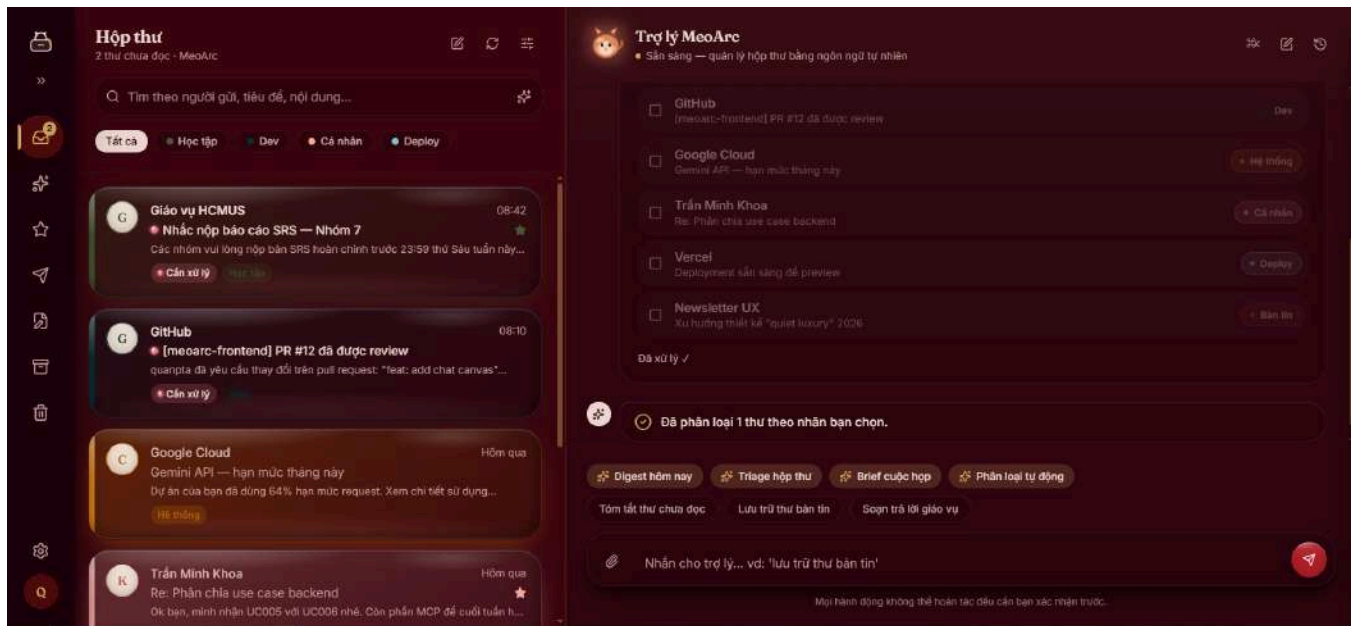
	14a. User unchecks an applied label, changes a Category/Priority/Status value via the checklist, or manually marks a previously non-task-like email as task-like (or vice versa). 14b. System invokes apply_labels again with the correction – directly, no confirmation required, since this remains a reversible action. 14c. System updates the checklist to reflect the correction. A7 – Partial Label Application Failure (at step 12) 12a. Gmail API returns an error for some emails (e.g. rate limit, email no longer exists). 12b. Agent records which items succeeded and which failed. 12c. System notifies the user with a breakdown (e.g. <i>"15 of 18 emails labeled successfully. 3 failed due to a temporary error – you can retry."</i>) and reflects the failed items in the checklist as unapplied. A8 – Output Rejected by Guardrails (at step 11) 11a. Output guardrails detect the determined labeling violates policy (e.g. signs of manipulated input affecting classification). 11b. The affected label is withheld from apply_labels; agent either regenerates a corrected determination (returns to Step 10) or excludes that email from the result with a note. 11c. System logs the rejection reason for audit purposes. A9 – External Service Unavailable (at step 4, 7, 10, or 13) a. Gemini API or Gmail API is unresponsive or times out. b. System displays a user-friendly error message and prompts the user to retry. Use case ends.
Non-Functional Constraints	NFR-SCO-01: On-demand AI operations that scan existing mailbox content (Semantic Search, Email Categorization, Summarization, Task Extraction, Reply context retrieval) shall limit the time range of emails processed according to the user's

	subscription tier: 90 days (Free), 180 days (Pro), 365 days (Pro Max). NFR-SEC-03: All communications between clients, servers, and third-party services shall be protected using HTTPS/TLS encryption. NFR-SEC-05: The system shall implement rate limiting to prevent excessive API requests from a single user or IP address. NFR-REL-01: The system shall gracefully handle temporary failures from Gmail API or Gemini API and provide meaningful error messages to users. NFR-REL-02: The system shall not lose conversation history or email metadata during unexpected system failures. NFR-MAI-04: The system shall log all API requests (timestamp, endpoint, user ID, HTTP status) and all agent task executions (tool calls, errors, token usage) to a persistent log store for debugging purposes. NFR-SCA-01: The system architecture shall support future integration with additional Large Language Model (LLM) providers without requiring major architectural modifications. NFR-SCA-02: The system architecture shall support the addition of new email-management tools exposed through REST APIs or the Model Context Protocol (MCP).
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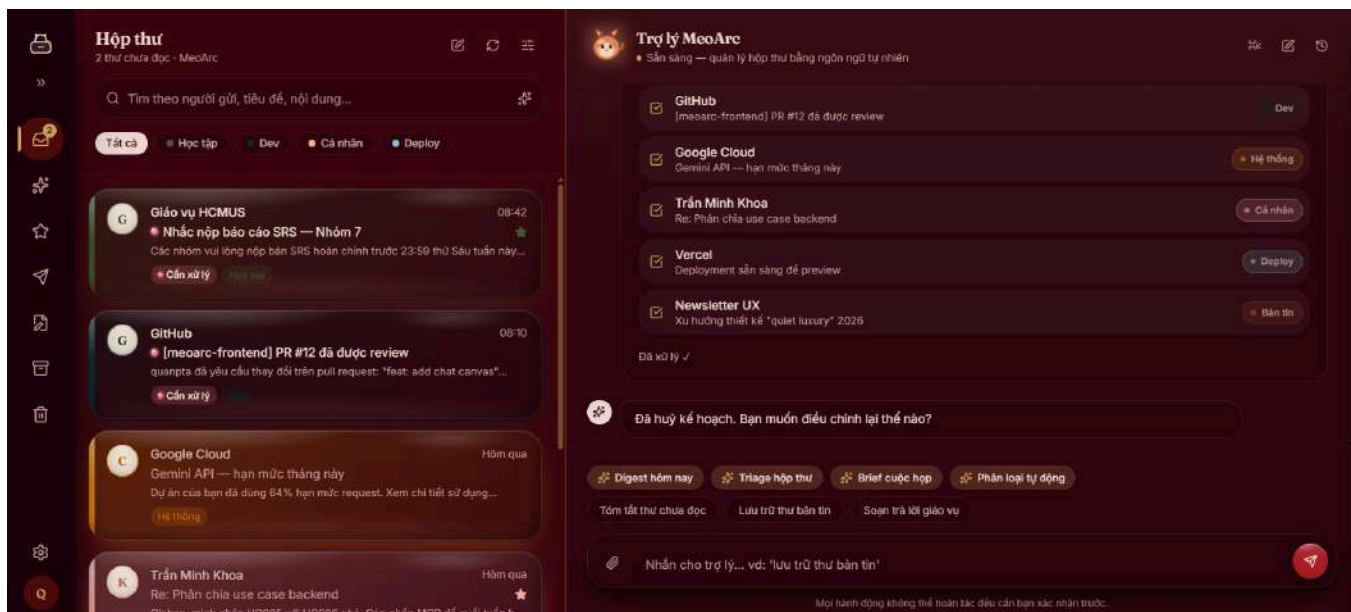
Panel AI → gõ phân loại tự động toàn bộ → ra CategorizeWidget.



Nếu đồng ý thì:



Còn nếu hủy thì:



4.2.10 Compose & Send Email

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC010
Use Case	Compose and Send Email
Brief Description	The user requests the AI Agent to compose a new email or reply to an existing thread via natural language, optionally attaching a file from the current chat. The agent resolves the recipient (for new compositions) or inherits it from the original thread (for replies), determines tone, and generates a draft together with a confirmation request in the same reasoning session. The draft is presented with structured actions – Confirm, Edit, Rewrite, Cancel – ensuring the system never loses track of which draft is being acted upon, regardless of how much time passes before the user responds. This is a sub-use-case of UC007
Actor	Primary: User Secondary: Gemini API (Agent), Gmail API
Pre-Condition	1. User has successfully authenticated (UC001) 2. At least one email account has been connected (FR-01.2) 3. User opens the chat interface – either a new session or a continued session from history (the session must be not full context) 4. For Reply: the source email thread exists and has been synchronized
Post-Condition	1. The email has been sent through the user-selected connected account, or the action was cancelled by the user 2. The system sends a completion notification 3. The conversation turn is persisted to conversation history 4. For Reply: the new message is correctly threaded with the original email on the mail provider.
Main Scenario	1. The user enters a request in natural language (optionally attaching a file) 2. Input guardrails validate the request for safety/policy compliance

	3. Memory Manager loads relevant conversation history; Skill Loader retrieves an email-writing skill from Skill Repository via semantic search (optional – e.g. tone/format guidance). 4. Agent reasoning loop begins: LLM determines whether this is Compose or Reply, and selects the next tool – recipient lookup (Compose) or thread retrieval (Reply) – from the Tool Registry. 5. LangGraph runtime executes the selected tool call, routed through the MCP Client Adapter to Gmail API as needed. 6. Agent observes the result. If an attachment was provided in the chat, agent reads its content as part of the available context. Agent determines the tone: uses the tone explicitly stated in the user's prompt if present, otherwise applies the system default (formal) 7. Agent evaluates whether the loop should continue (e.g. recipient still unresolved) or proceed to generate the draft 8. Steps 4–7 repeat as needed (the reasoning loop) 9. Agent generates the draft (recipient, subject, body, attachment reference) using the resolved information and selected tone. 10. Output guardrails check the draft 11. Within the same loop iteration, agent immediately calls the <code>request_confirmation</code> tool, attaching the generated draft – the loop pauses here, it does not end. The draft and its pending confirmation state are kept together as a single unit, so no separate turn is needed to "remember" the draft later. 12. UI displays the draft together with four structured actions bound to this specific draft instance: Confirm, Edit, Rewrite, Cancel. 13. The user clicks Confirm 14. LangGraph runtime executes the <code>send_email</code> tool call directly – no re-interpretation of intent is needed, since Confirm is a structured signal tied to the exact draft. 15. System sends a completion notification 16. Response is showed in the canvas and conversation state are persisted to PostgreSQL
Alternative Scenarios	A1 – Input Rejected by Guardrails (at step 2)

	2a. Input guardrails detect unsafe/malicious content (e.g. prompt injection). 2b. System rejects the request with an explanatory message. Use case ends. A2 – Reply Source Not Found (at step 5, Reply only) 5a. The referenced email/thread cannot be found with the given description. 5b. Agent performs a broader <code>search_emails</code> call (relaxed criteria, e.g. sender name only) to surface candidate emails. 5c. If candidates are found, agent calls <code>request_clarification</code>, presenting the candidate list (sender, subject, date) for the user to pick from – not an open-ended question. If no candidates are found at all, agent informs the user clearly and suggests alternatives: provide more detail (exact sender email, date range), or open the target email directly from the email list and use the in-context reply action there. 5e. User selects a candidate (or provides more detail); loop continues at Step 4. A3 – Recipient Cannot Be Resolved (at step 7, Compose only) 7a. The named recipient doesn't match any known contact, or matches multiple people. 7b. Agent calls <code>request_clarification</code> tool instead of proceeding (recipient, tone, original instruction, additional notes) 7c. System displays the question to the user and pauses the loop. 7d. User confirms the contact (or provides the exact email address) and/or adjusts the tone. 7e. Loop continues at Step 4 with the new information appended to context – not a restart of the use case.
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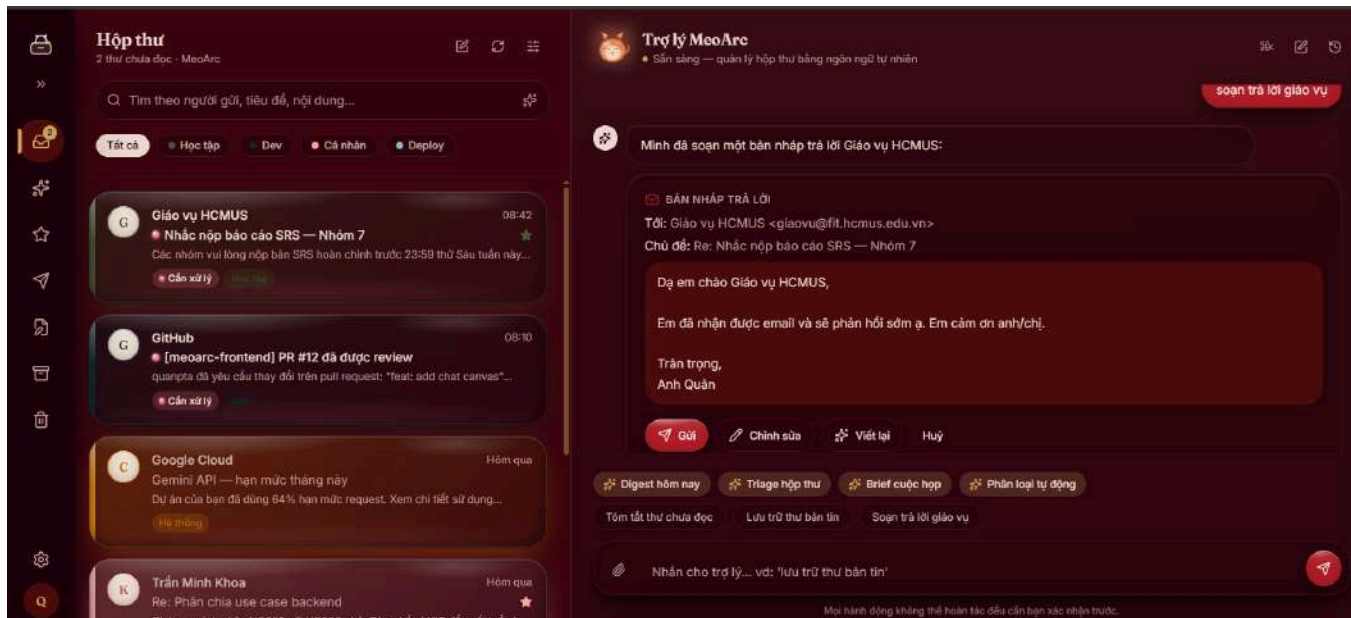
	A4 – User Edits the Draft (at step 12) 12a. User clicks Edit on the presented draft. 12b. UI switches the draft fields (recipient, subject, body) into a directly editable state, and enables attachment management (remove an existing attachment, or add a new one from the current chat). 12c. User modifies the content and/or attachment(s) manually and clicks Confirm. 12d. If an attachment was added or changed, system re-validates it (file type, size limit) before allowing confirmation. 12e. LangGraph runtime executes the <code>send_email</code> tool call using the user-edited content and final attachment set – the agent does not regenerate or re-interpret the draft, since the edit is a direct user action on the existing draft instance. 12f. Flow continues at Step 13. A4.1 – Attachment Validation Failure (at step 12d) 12d-1. The added attachment exceeds the size limit or has an unsupported file type. 12d-2. System displays an inline error and keeps the draft in editable state, without invoking <code>send_email</code>. User may remove/replace the attachment or proceed without it. A5 – User Requests Rewrite (at step 12) 12a. User clicks Rewrite and optionally provides additional instruction (e.g. "make it shorter", "more formal"). 12b. The pending <code>request_confirmation</code> call is withdrawn; the loop resumes with the new instruction appended to context. 12c. Flow returns to Step 9 – agent regenerates the draft (same resolved recipient/thread/tone unless the new instruction explicitly changes them), then re-attaches it to a new <code>request_confirmation</code> call. 12d. UI displays the updated draft with the same four structured actions, replacing the previous draft instance.
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	A6 – User Cancels (at step 12) 12a. User clicks Cancel. 12b. Agent withdraws the pending <code>request_confirmation</code> call without invoking <code>send_email</code>. 12c. System notifies the user the action was cancelled. 12d. The cancelled draft and the conversation turn are persisted to conversation history. Use case ends. A7 – Send Failure (at step 13) 13a. Gmail API returns an error while executing <code>send_email</code> (e.g. rate limit, invalid recipient address, attachment too large). 13b. Agent observes the failure and generates a response explaining what went wrong. 13c. System notifies the user the email failed to send and offers to retry (returns to the same draft, re-attached to a new <code>request_confirmation</code> call) or cancel. A8 – External Service Unavailable (at step 4, 6, 9, or 13) a. Gemini API or Gmail API is unresponsive or times out. b. System displays a user-friendly error message and prompts the user to retry. Use case ends A9 – Draft Rejected by Output Guardrails (at step 10) 10a. Output guardrails detect the generated draft violates policy (e.g. contains inappropriate content, leaks unrelated email data, or shows signs of prompt-injected instructions from an attachment). 10b. The draft is not attached to a <code>request_confirmation</code> call and is not shown to the user as-is. 10c. Agent discards the rejected draft and either: • regenerates a corrected draft (returns to Step 9), or • if regeneration also fails, returns a generic fallback message informing the user the draft could not be completed safely.
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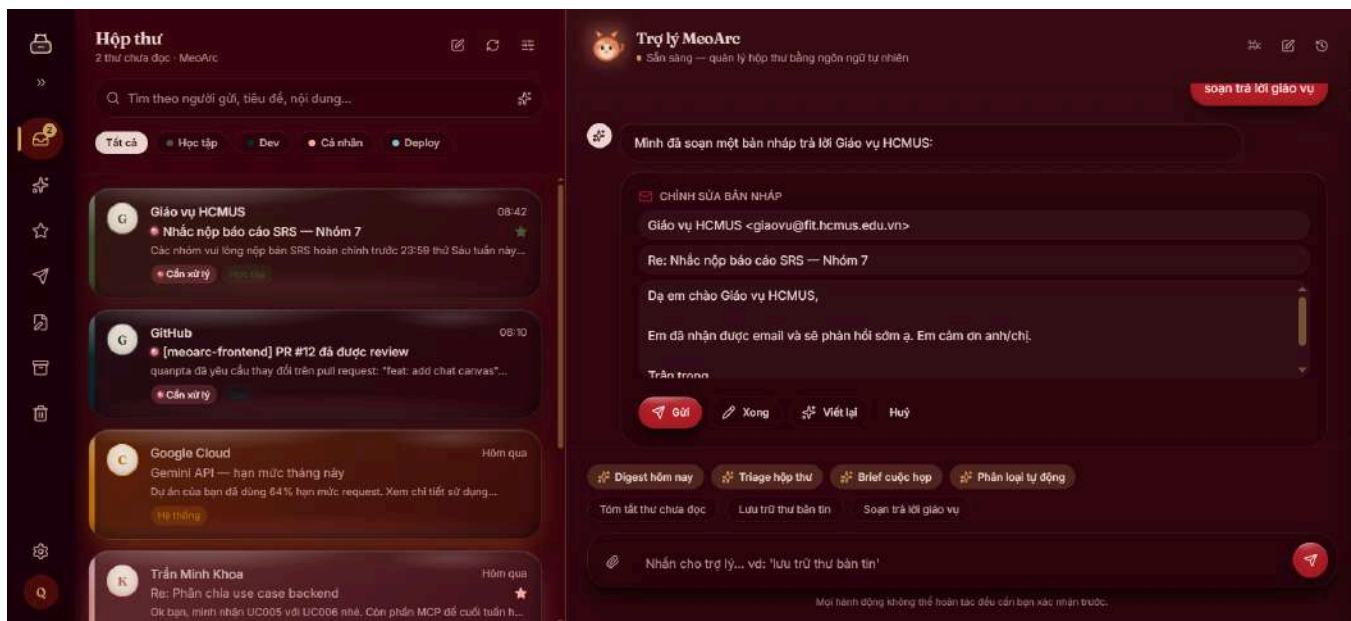
	10d. System logs the rejection reason for audit purposes (NFR-MAI-04). Use case ends if fallback is returned; otherwise continues at Step 10 with the new draft.
Non-Functional Constraints	NFR-SEC-03: All communications between clients, servers, and third-party services shall be protected using HTTPS/TLS encryption. NFR-SEC-05: The system shall implement rate limiting to prevent excessive API requests from a single user or IP address. NFR-REL-01: The system shall gracefully handle temporary failures from Gmail API or Gemini API and provide meaningful error messages to users. NFR-REL-02: The system shall not lose conversation history or email metadata during unexpected system failures. NFR-MAI-04: The system shall log all API requests (timestamp, endpoint, user ID, HTTP status) and all agent task executions (tool calls, errors, token usage) to a persistent log store for debugging purposes. NFR-SCA-01: The system architecture shall support future integration with additional Large Language Model (LLM) providers without requiring major architectural modifications. NFR-SCA-02: The system architecture shall support the addition of new email-management tools exposed through REST APIs or the Model Context Protocol (MCP). NFR-SCO-01: On-demand AI operations that scan existing mailbox content (Semantic Search, Email Categorization, Summarization, Task Extraction, Reply context retrieval) shall limit the time range of emails processed according to the user's subscription tier: 90 days (Free), 180 days (Pro), 365 days (Pro Max).

Với bước soạn gửi mail, người gửi nhập vào Panel AI → gõ soạn trả lời giáo vụ → ra DraftCard:

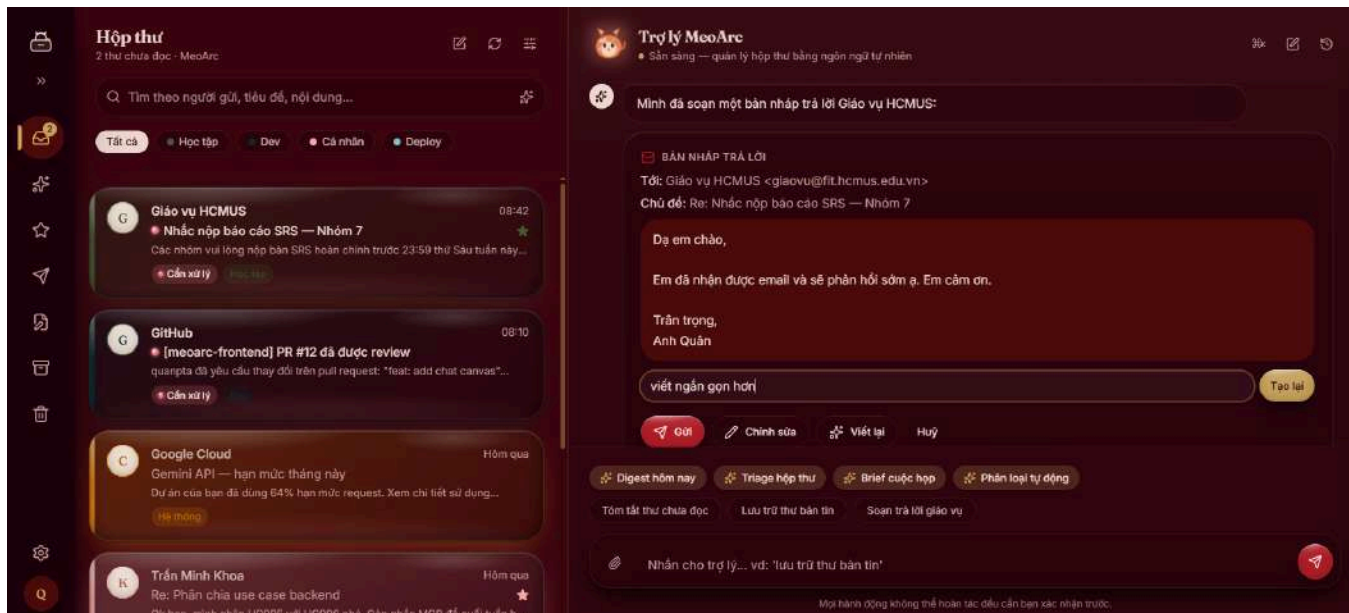
Ra draft + 4 nút Gửi · Chỉnh sửa · Viết lại · Huỷ.



Khi bấm Chỉnh sửa → 3 field (Tới/Chủ đề/Nội dung) sửa inline:



Khi bấm Viết lại → ô gợi ý "ngắn gọn hơn...":



4.2.11 Manage Conversation History

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127226 – Phạm Trần Anh Quân

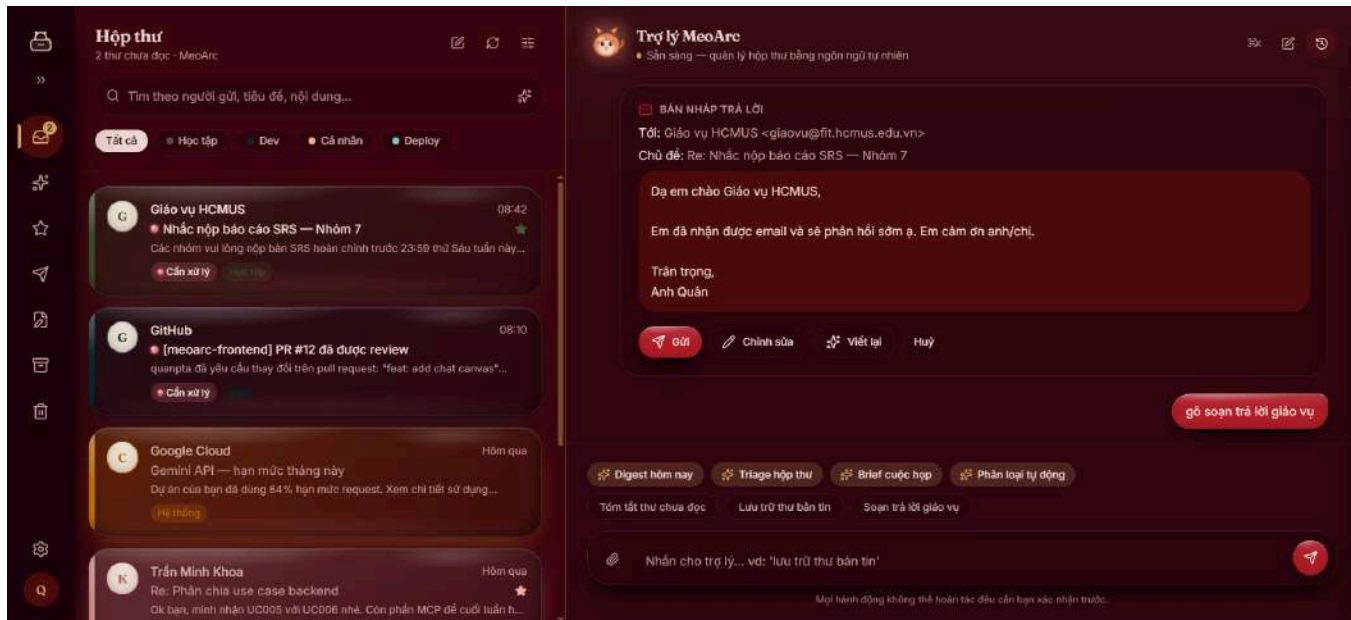
Use case ID	UC011
Use Case	Manage Conversation History
Brief Description	The user manages their past chat sessions with the AI assistant – browsing the list of conversations, resuming a previous session to continue where they left off, renaming a session for easier identification, pinning important sessions for quick access and deleting sessions no longer needed. (View, Continue, Delete, Rename, Pin (Star))
Actor	Primary: User
Pre-Condition	1. User has successfully authenticated (UC001)

	2. At least one prior conversation session exists (i.e. user has used UC007 at least once)
Post-Condition	1. View: The list of conversation sessions (with metadata: title, the last accessed time, pinned status) is displayed to the user 2. Continue: The selected session's full message history is loaded into the chat interface and the agent's context/memory is restored, allowing the conversation to continue 3. Rename: The session's display title is updated and persisted 4. Pin: The session's pinned status is toggled and persisted; pinned sessions are sorted to the top of the list 5. Delete: The session and its associated messages are permanently removed from the system
Main Scenario	1. The user opens the conversation history panel 2. System retrieves and displays the list of the user's conversation sessions (pinned first then recency) each showing title, the last accessed time, pinned status 3. The user selects one or more sessions from the list 4. Highlights the selected session(s) and displays the available actions (Resume – single selection only, Rename – single selection only, Pin/Unpin, Delete). 5. The user selects the desired action 6. If the action is irreversible (i.e. Delete), system displays a confirmation dialog showing the session(s) to be affected 7. The user confirms the action / input (for Rename) 8. Executes the action against the persisted session data (update title, toggle pinned flag, or remove session and its messages). 9. Updates the conversation history list to reflect the applied changes.
Alternative Scenarios	A1 – Empty History (at step 2) 2a. System displays an empty state message instead of a list -> use case end A2 – Continue Session (at step 5) 5a. User selects Continue on a single session

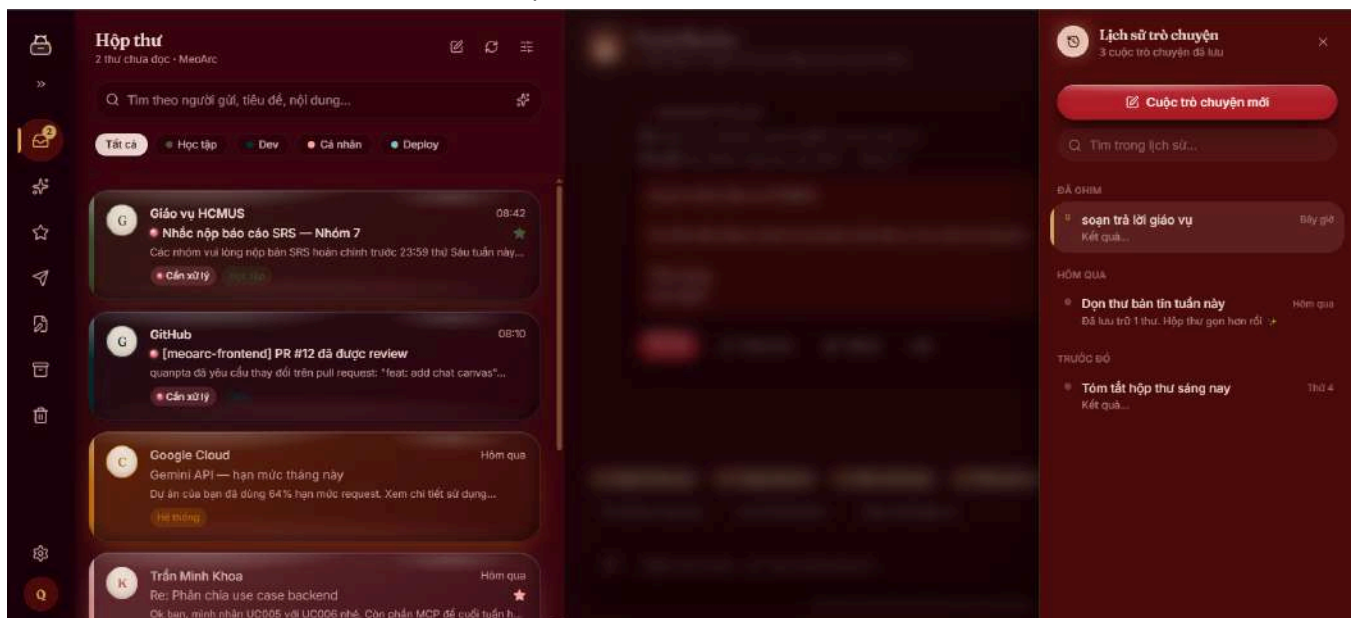
	5b. System loads its full message history and memory context, then displays it in the chat interface then restores context -> user case end A3 – Action Cancelled (at step 7) 7a. User declines the confirmation dialog (Delete only) 7b. No changes are made; system returns to Step 3 A4 – Rename Validation Failure (at step 7) 7a. User enters a new title is empty or exceeds the length limit 7b. System displays an inline error and keeps the dialog open for correction A5 – Action Execution Failure (at step 8) 8a. Database error occurs while applying the action then return an error 8b. System displays an error message and leaves the session list unchanged. A6 – Multi-Select with Single-Only Action (at step 4) 4a. User has multiple sessions selected but chooses Continue or Rename (which only support one session at a time) 4b. System disables these options when more than one session is selected
Non-Functional Constraints	NFR-PER-01: The system shall return API responses within 3 seconds for at least 95% of requests under normal operating conditions. NFR-SEC-03: All communications between clients, servers, and third-party services shall be protected using HTTPS/TLS encryption. NFR-REL-02: The system shall not lose conversation history or email metadata during unexpected system failures. NFR-USA-02: Error messages and system notifications shall be clear, concise, and understandable to non-technical users. NFR-MAI-04: The system shall log all API requests (timestamp, endpoint, user ID, HTTP status) and all agent task executions (tool

calls, errors, token usage) to a persistent log store for debugging purposes.

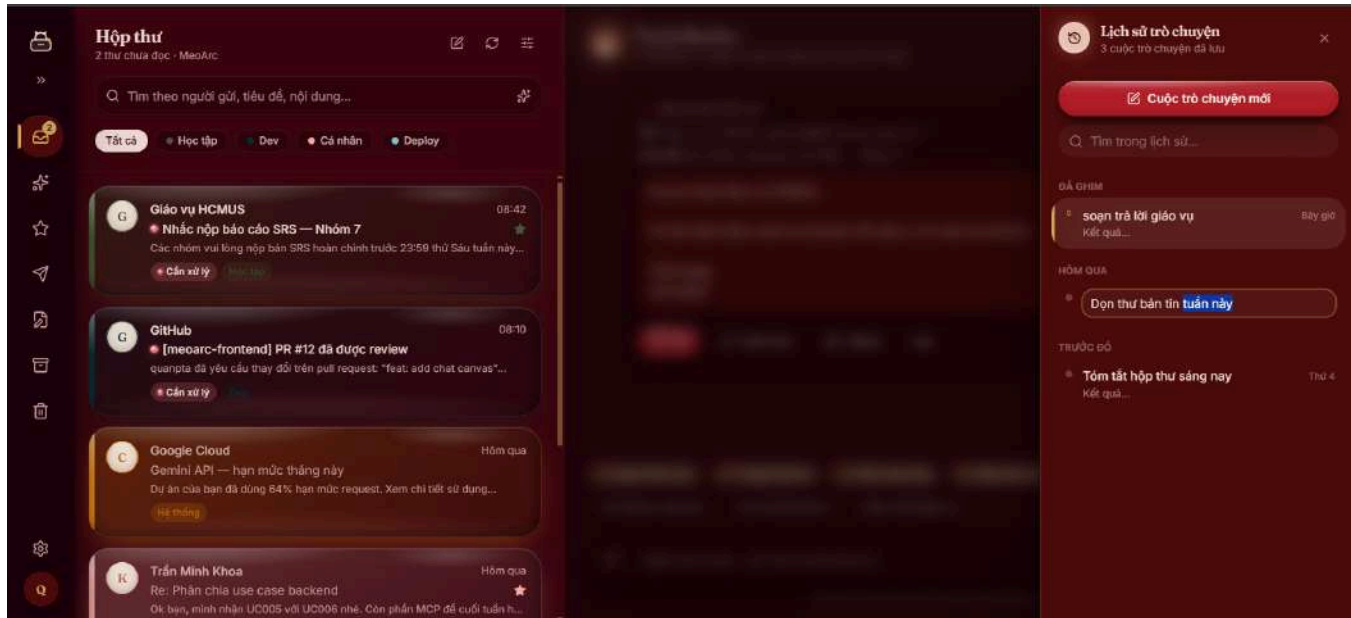
Mở History (icon đồng hồ/ lịch sử ở góc phải header panel AI).



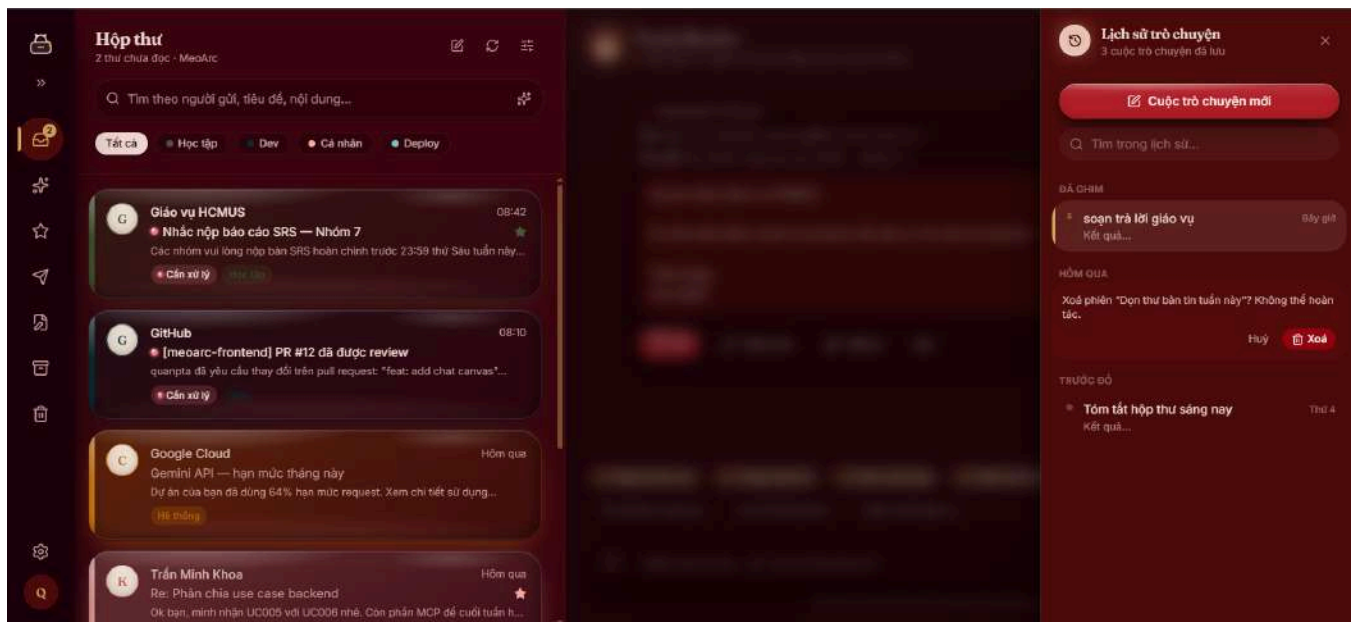
Hiện danh sách + nhóm thời gian, có 1 phiên đã ghim nằm trên cùng



Lúc đang Đổi tên (ô input hiện trong dòng)



Khi xóa: xuất hiện dải xác nhận Xóa "không thể hoàn tác" (bấm icon thùng rác).



4.2.12 MCP Client Access

Written by: 24127250 – Phan Quang Tiến

Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127226 – Phạm Trần Anh Quân

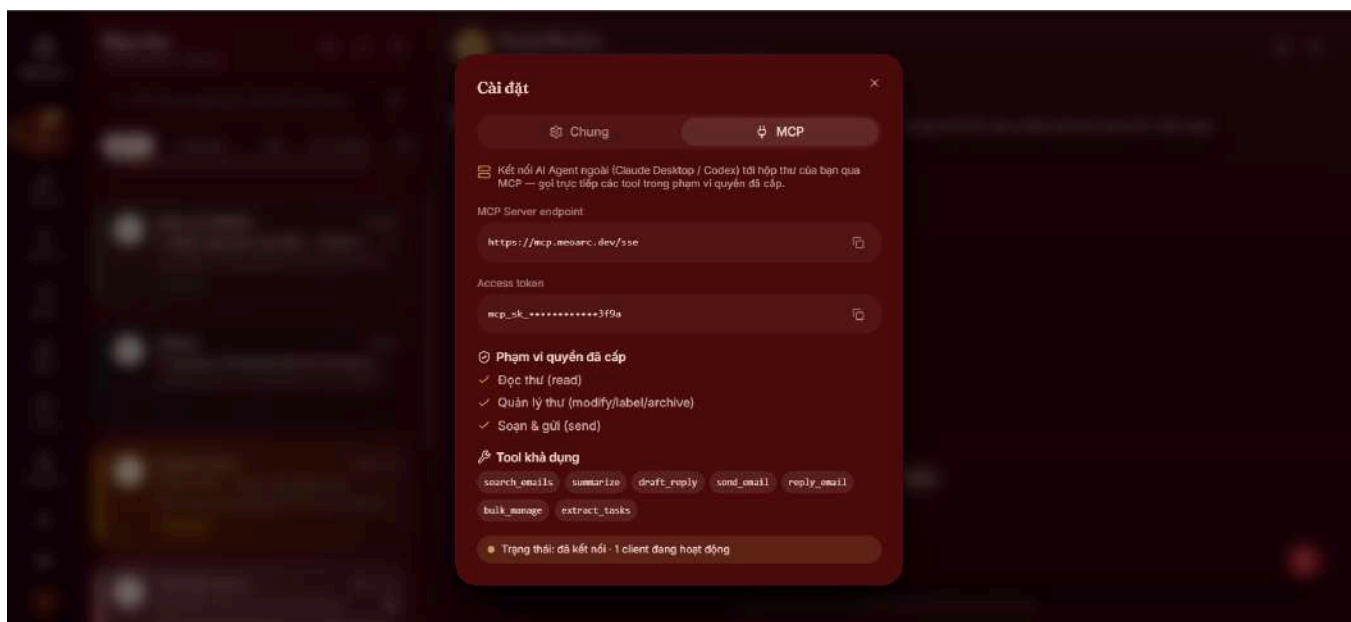
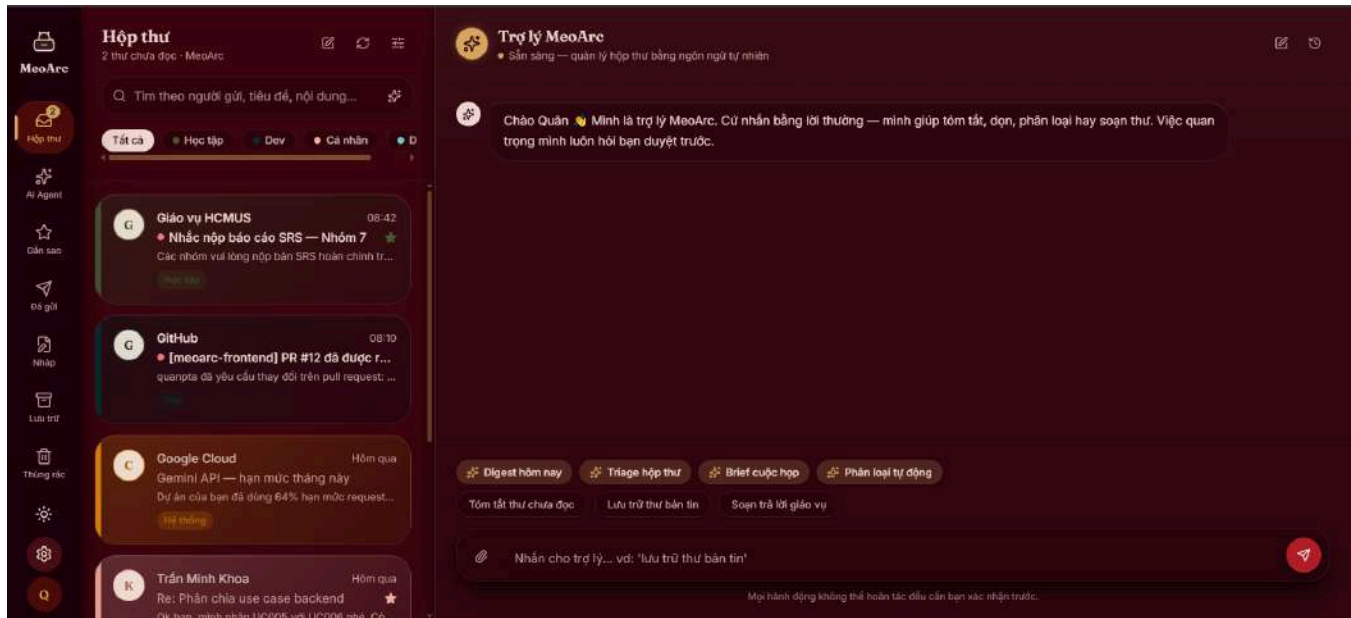
Use case ID	UC012
Use Case Name	MCP Client Access
Brief Description	Describes how an external AI agent (e.g. Claude Desktop, Codex) connects to MeoArc via the Model Context Protocol (MCP) server and invokes fine-grained email management tools directly, bypassing MeoArc's natural language parsing layer, under the scope of permissions authorized by the authenticated user.
Actor	External AI Agent
Pre-Conditions	1. The user has successfully authenticated through UC001 – Login with Google and has an active MeoArc account. 2. The user has explicitly authorized the external AI agent to access MeoArc on their behalf. 3. The MeoArc MCP server is running and accessible. 4. The external AI agent supports the Model Context Protocol specification.
Post-Conditions	1. The external AI agent has successfully invoked the requested email tool(s). 2. The requested operation has been executed within the scope of the user's authorized permissions. 3. The result of the tool invocation is returned to the external AI agent.

	4. All tool executions are logged for auditing and debugging purposes.
Main Scenario	1. The external AI agent establishes a connection to the MeoArc MCP server using the user's authorization credentials. 2. The MCP server validates the agent's credentials and verifies the scope of the user's granted permissions. 3. The external AI agent sends a tool invocation request, specifying the target tool (e.g. search, summarize, draft, send, reply, bulk action, extract tasks) and the required parameters. 4. The MCP server receives the request and maps it to the corresponding internal email management function. 5. The system executes the requested tool operation against the user's Gmail account via the Gmail API. 6. The system returns the operation result to the external AI agent through the MCP protocol. 7. The external AI agent processes the result and continues its own reasoning workflow independently.
Alternative Scenarios	<i>A1. Invalid or expired authorization credentials (At step 2)</i> 2a. The MCP server detects that the provided credentials are invalid or have expired. 2b. The MCP server returns an authentication error response to the external AI agent. 2c. The system logs the failed access attempt. 2d. The external AI agent must prompt the user to re-authenticate through UC001 before retrying.

	<i>A2. Requested tool is outside the user's authorized permission scope (At step 2-3)</i> 2a. The MCP server verifies that the requested tool or target resource exceeds the scope of permissions granted by the user. 2b. The MCP server returns an authorization error response: "The requested operation is not permitted under the current authorization scope." 2c. No email data is accessed or modified. 2d. The system logs the unauthorized access attempt for auditing. <i>A3. Invalid tool invocation request (At step 3-4)</i> 3a. The external AI agent sends a malformed or incomplete tool invocation request. 3b. The MCP server validates the request and detects missing or invalid parameters. 3c. The MCP server returns an error response specifying the validation failure details. 3d. The external AI agent may correct the request and retry. <i>A4. Gmail API temporarily unavailable (At step 5)</i> 5a. The system is unable to execute the tool operation due to a temporary Gmail API or network issue. 5b. The MCP server returns an error response to the external AI agent: "The requested operation could not be completed due to a temporary service unavailability. Please retry later." 5c. No partial changes are applied to the user's email data.
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Non-Functional Constraints	NFR-SEC-03: All communications between the external AI agent and the MeoArc MCP server shall be protected using HTTPS/TLS encryption. NFR-SEC-05: The MCP server shall implement rate limiting to prevent excessive tool invocation requests from a single agent or user. NFR-PER-01: The MCP server shall return tool execution results within 3 seconds for at least 95% of requests under normal operating conditions. NFR-REL-01: The system shall gracefully handle temporary Gmail API failures and return meaningful error responses to the external AI agent. NFR-MAI-04: All MCP tool invocations, including tool name, parameters, execution result, errors, and token usage, shall be logged to a persistent log store for auditing and debugging purposes. NFR-COM-02: The MCP server shall comply with the Model Context Protocol specification to ensure interoperability with compatible external AI agents.
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Nhấn hình răng cưa bên thanh dọc bên trái, ta sẽ ra mục cài đặt có phần MCP:



4.2.13 Manage Settings

Written by: 24127250 – Phan Quang Tiến

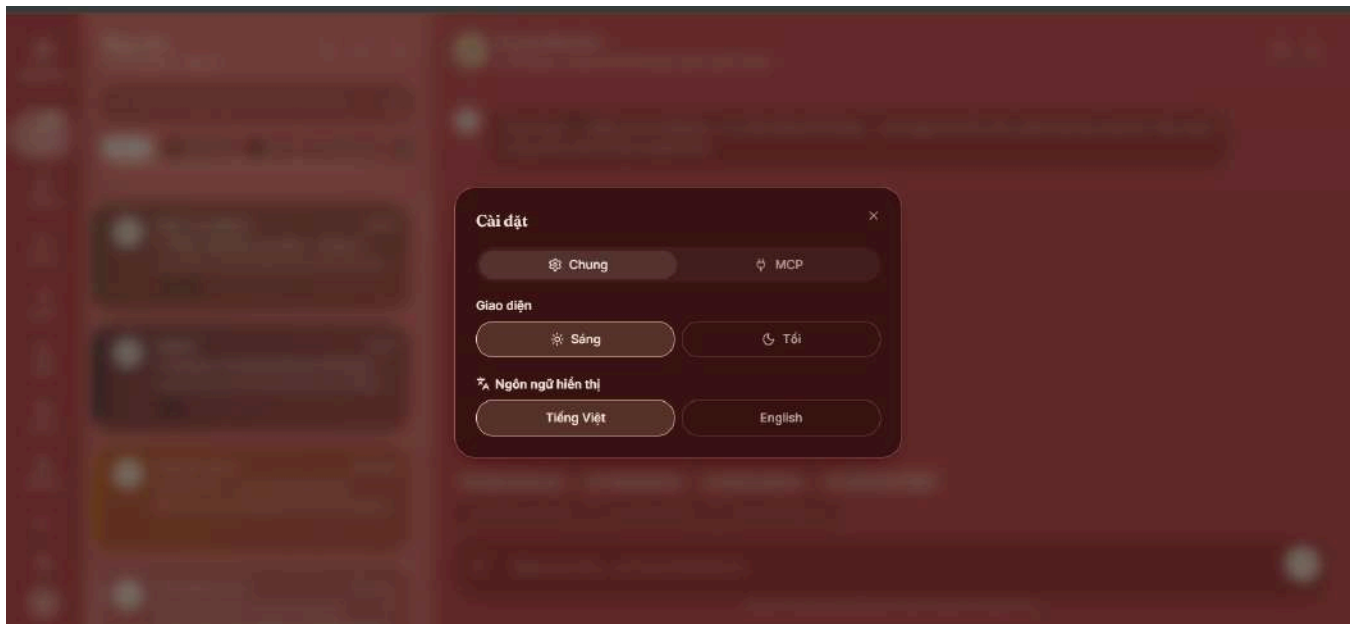
Edited by: 24127529 – Nguyễn Chí Tài

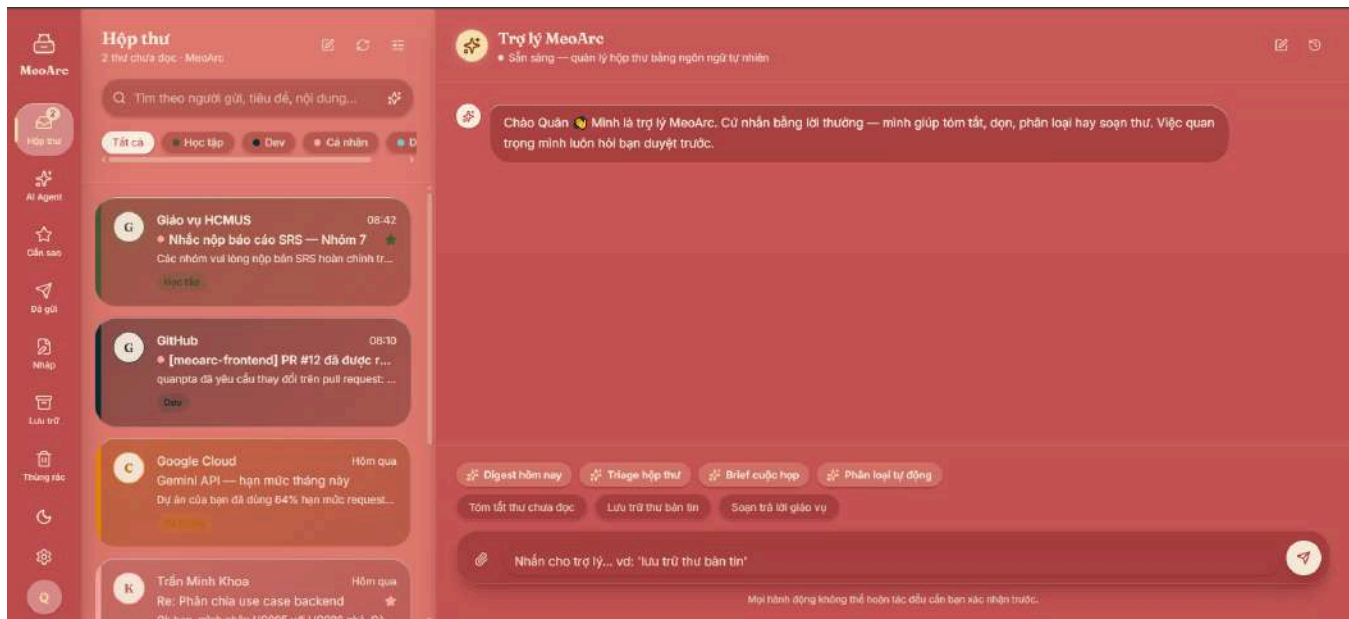
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC013
Use Case Name	Manage Settings
Brief Description	Describes how a user customizes personal preferences within the application, including changing the display language and toggling between light and dark themes
Actor	User
Pre-Conditions	1. The user has successfully authenticated through UC001 - Authenticate & Connect Mailbox. 2. The user has an active session in MeoArc.
Post-Conditions	1. The user's new preference settings are successfully stored in the database. 2. The User interface (UI) is immediately updated according to the selected changes.
Main Scenario	1. The user accesses the Settings panel from the application dashboard. 2. The system displays current configuration options, including Language selection and Theme selection. 3. The user selects a preferred language or changes the visual theme (e.g. switching from Light to Dark mode). 4. The user confirms and saves the changes. 5. The system updates the preferences in the database and applies the new interface settings instantly.

	6. The system displays a success message: “Settings updated successfully.”
Alternative Scenarios	<i>User cancels changes (At step 3-4)</i> 3a, The user exits the settings panel without saving. 3b. The system maintains the previous settings and no changes are applied.
Non-Functional Constraints	NFR-PER-01: The system shall return API responses within 3 seconds for setting updates. NFR-USA-03: The settings interface shall be optimized for desktop web browsers. NFR-REL-02: The system shall not lose personal configurations during unexpected system failures.

Nhấn vào icon “Bánh răng” và vô phần Cài đặt chung, sẽ có nơi điều chỉnh giao diện lẫn ngôn ngữ.





4.2.14 Daily Digest

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

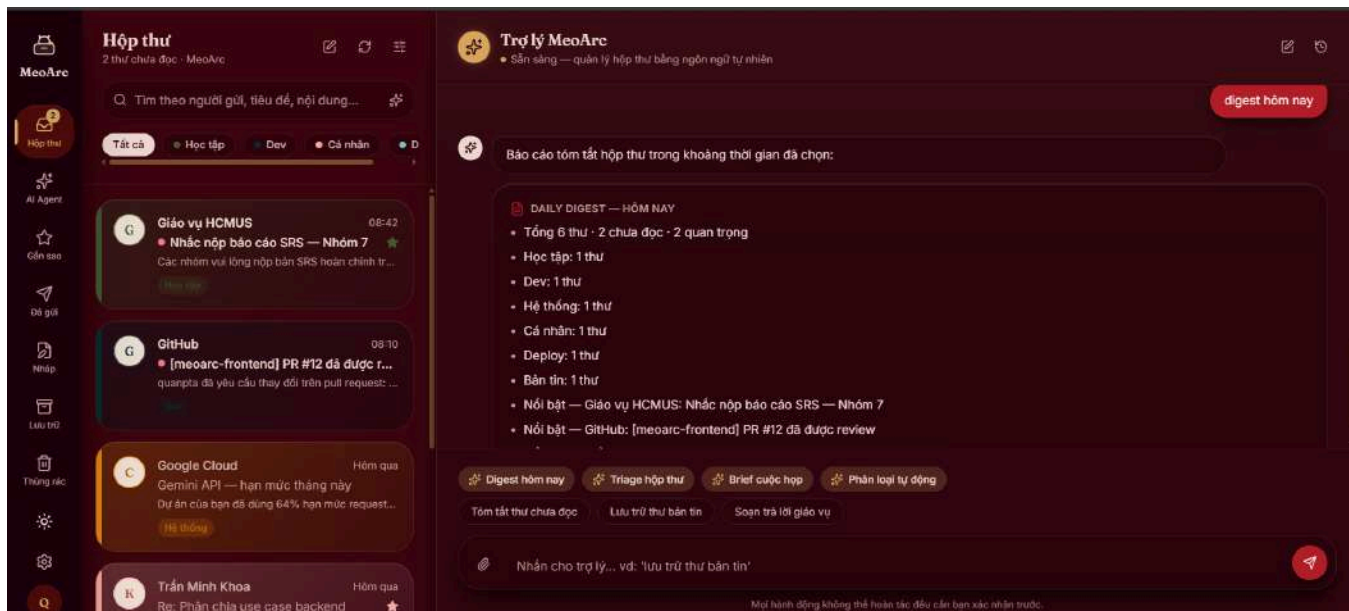
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC014
Use Case Name	Get Daily Email Digest
Brief Description	Describes how a user requests MeoArc to generate an AI-powered daily email digest summarizing important emails received during a selected period. The system automatically retrieves relevant emails, analyzes their importance, groups related messages, extracts key updates and action items, and produces a concise digest that helps users quickly understand the state of their inbox without reading every individual email.
Actor	Primary Actor: User Supporting Services: Gmail API, Gemini API
Pre-Conditions	1. The user has successfully authenticated with Google (UC001).

	- The user has access to the AI assistant interface (UC007). - The Gmail account contains emails available for analysis. - Network connectivity to Gmail API and Gemini API is available.
Post-Conditions	- A structured daily digest summarizing the selected email period is generated and displayed to the user. - Important updates, action items, and deadlines are highlighted. - Conversation history is updated and stored. - No email content or mailbox state is modified during this workflow.
Main Scenario	- The user enters a natural-language request such as: “Give me today's email digest.” “Summarize today's important emails.” “What happened in my inbox today?” - The system receives the request and identifies the intent as a Daily Email Digest workflow. - The system retrieves emails received within the selected period (e.g., today, last 24 hours, or another user-specified timeframe). - The system analyzes email metadata and content to identify: - Important senders - Unread emails - Urgent or time-sensitive messages - Emails that may require follow-up actions - The system groups related emails into logical categories, such as: - Work / Academic - Administrative Notifications - Personal Communications - Promotions and Newsletters - The system extracts key information from each category, including: - Major updates - Important announcements - Deadlines and schedules

	• Action items and requests 7. The system invokes the AI model to synthesize the extracted information into a consolidated daily digest. 8. The AI generates a structured digest containing: • Executive Summary • Important Emails • Key Updates • Pending Action Items • Upcoming Deadlines (if any) 9. The system displays the generated digest to the user through the AI assistant interface. 10. The conversation history is stored for future reference.
Alternative Scenarios	A1. No Emails Available (At Step 3) 3a. No emails are found within the specified period. 3b. The system informs the user: "No new emails were found for the selected period." 3c. The use case ends. A2. Gmail API Unavailable (At Step 3) 3a. The system fails to retrieve emails from Gmail. 3b. The system displays: "Unable to access Gmail at the moment. Please try again later." 3c. The use case ends. A3. AI Service Failure (At Step 6) 6a. Gemini API fails to generate the digest. 6b. The system provides a fallback response containing only a chronological list of retrieved emails. 6c. The use case ends. A4. User Cancels Request

	4a. The user cancels the request before generation is completed. 4b. The system terminates the workflow. 4c. No digest is generated. A5. Large Number of Emails (At Step 3) 3a. The number of retrieved emails exceeds the system processing limit. 3b. The system selects the most recent and highest-priority emails for analysis. 3c. The system informs the user: "Only the most relevant emails were included in the digest due to processing limits." 3d. The workflow continues at Step 4.
Non-Functional Constraints	NFR-PER-01: The system should generate a daily digest within 10 seconds under normal operating conditions. NFR-REL-01: Temporary failures from Gmail API or Gemini API shall be handled gracefully without crashing the system. NFR-SEC-03: All communications with Gmail API and Gemini API shall use HTTPS/TLS encryption. NFR-MAI-04: Digest generation events shall be logged for debugging and monitoring purposes. NFR-SCA-01: The digest generation mechanism shall support future integration with other LLM providers without major architectural changes.



4.2.15 Triage Inbox

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

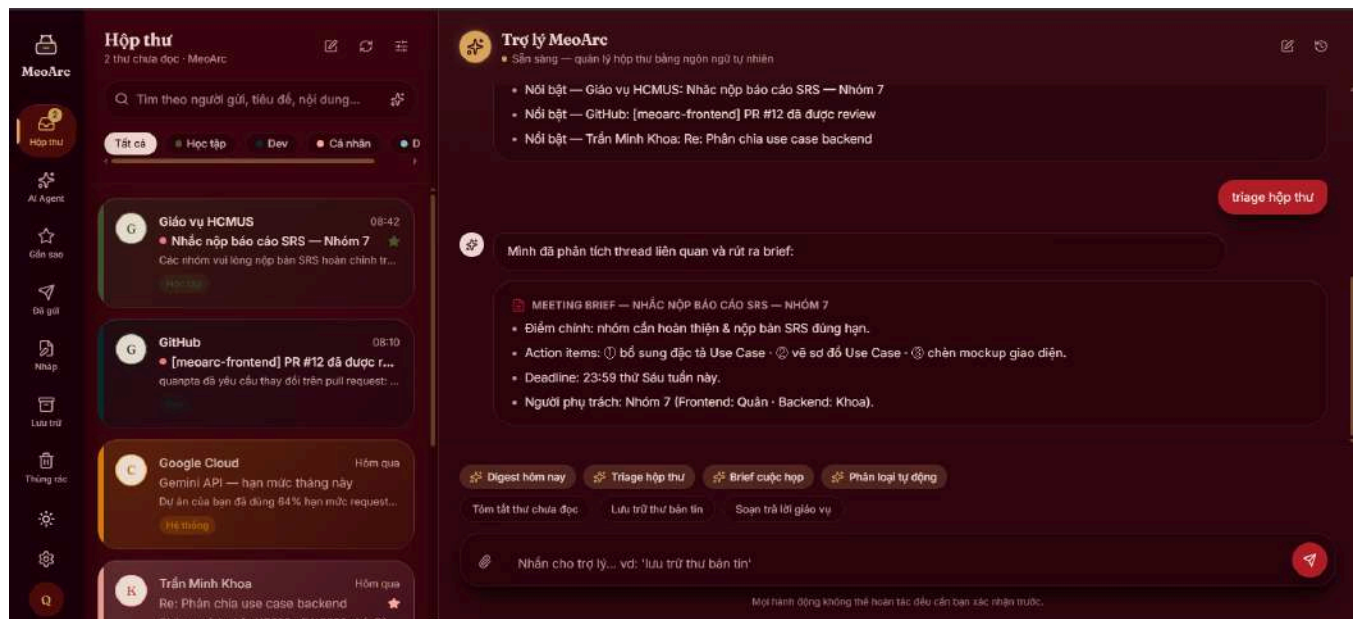
Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC015
Use Case Name	Triage Inbox
Brief Description	Describes how a user requests MeoArc to analyze and organize the inbox by automatically prioritizing emails, identifying important messages, grouping low-priority emails, and presenting actionable recommendations. This workflow helps users quickly understand and manage large volumes of emails.
Actor	Primary Actor: User Supporting Services: Gmail API, Gemini API
Pre-Conditions	1. The user has successfully authenticated with Google (UC001).

	- The user has access to the AI assistant interface (UC007). - The Gmail account contains emails available for analysis. - Gmail API and Gemini API are accessible.
Post-Conditions	- The inbox triage result is generated and displayed. - Conversation history is updated. - No emails are modified automatically unless explicitly requested by the user.
Main Scenario	- The user enters a natural-language request such as: "Help me organize my inbox." "Triage my emails." "What should I deal with first?" - The system recognizes the request as an Inbox Triage workflow. - The system retrieves recent emails from Gmail. - The system analyzes email metadata and content. - The system categorizes emails into priority groups, for example: - High Priority - Medium Priority - Low Priority - Promotional / Newsletters - The system identifies urgent or actionable emails. - The system generates recommendations, including:

	• Emails requiring immediate attention. • Emails that can be postponed. • Emails that may be archived or ignored. 8. The AI generates a structured triage summary. 9. The system displays the triage result to the user. 10. The conversation history is saved.
Alternative Scenarios	A1. No Emails Available (At Step 3) 3a. No emails are found. 3b. The system displays: "No emails are available for inbox triage." 3c. The use case ends. A2. Gmail API Unavailable (At Step 3) 3a. The system cannot retrieve emails from Gmail. 3b. The system displays: "Unable to access Gmail at the moment. Please try again later." 3c. The use case ends. A3. AI Service Failure (At Step 8) 8a. Gemini API fails to generate the triage summary. 8b. The system falls back to displaying emails sorted chronologically. 8c. The use case ends. A4. User Cancels Request 4a. The user cancels the request.

	4b. The system terminates the workflow. 4c. No triage result is generated.
Non-Functional Constraints	NFR-PER-01: The inbox triage result should be generated within 10 seconds under normal operating conditions. NFR-REL-01: Temporary failures from Gmail API or Gemini API shall be handled gracefully without crashing the system. NFR-SEC-03: All communications with external services shall be protected using HTTPS/TLS encryption. NFR-MAI-04: Inbox triage operations shall be logged for monitoring and debugging purposes. NFR-SCA-01: The workflow shall support future integration with additional LLM providers without requiring major architectural changes.



4.2.16 Meeting Brief

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

Reviewed by: 24127226 – Phạm Trần Anh Quân

Use case ID	UC016
Use Case Name	Prepare Meeting Brief
Brief Description	Describes how a user requests MeoArc to automatically prepare a concise meeting brief by collecting and analyzing meeting-related emails. The system summarizes important discussions, identifies key participants, extracts actionable items, and presents a structured overview to help the user prepare efficiently for upcoming meetings.
Actor	Primary Actor: User Supporting Services: Gmail API, Gemini API
Pre-Conditions	1. The user has successfully authenticated with Google (UC001). 2. The user has access to the AI assistant interface (UC007). 3. Meeting-related emails are available in the user's mailbox. 4. Gmail API and Gemini API are accessible.
Post-Conditions	1. A meeting brief is generated and displayed to the user. 2. Conversation history is updated. 3. No email content is modified.
Main Scenario	1. The user enters a request such as "Prepare me for today's meeting" or "Summarize emails related to tomorrow's meeting". 2. The system recognizes the request as a Meeting Brief workflow.

	3. The system retrieves meeting-related emails from Gmail. 4. The system analyzes email threads and identifies relevant information. 5. The system extracts key meeting details, including participants, important discussions, dates, and action items. 6. The system invokes the AI model to generate a structured meeting brief. 7. The generated brief includes: • Meeting topic • Participants • Important updates • Pending issues • Action items 8. The system displays the meeting brief to the user. 9. The conversation history is saved.
Alternative Scenarios	A1. No Relevant Emails Found (At Step 3) 3a. No meeting-related emails are found. 3b. The system displays: "No meeting-related emails were found." 3c. The use case ends. A2. Gmail API Unavailable (At Step 3) 3a. The system cannot retrieve emails from Gmail.

	3b. The system displays: "Unable to access Gmail at the moment. Please try again later." 3c. The use case ends. A3. AI Service Failure (At Step 6) 6a. Gemini API fails to generate the meeting brief. 6b. The system falls back to displaying the list of retrieved emails. 6c. The use case ends. A4. User Cancels Request 4a. The user cancels the request before completion. 4b. The system terminates the workflow. 4c. No meeting brief is generated.
Non-Functional Constraints	NFR-PER-01: The meeting brief should be generated within 10 seconds under normal operating conditions. NFR-REL-01: Temporary failures from Gmail API or Gemini API shall be handled gracefully without interrupting the system. NFR-SEC-03: All communications with external services shall be protected using HTTPS/TLS encryption. NFR-MAI-04: Meeting brief generation events shall be logged for monitoring and debugging purposes. NFR-SCA-01: The workflow shall support future integration with additional LLM providers without major architectural changes.

The screenshot displays the MeoArc application interface, which is designed for managing emails and meeting summaries. The interface is divided into two main sections: a left sidebar for navigation and a main content area.

Left Sidebar (Navigation):

- Hộp thư (Mailbox):** 2 thư chưa đọc - MeoArc. Includes a search bar and filters for 'Tất cả' (All), 'Học tập' (Study), 'Dev', 'Cá nhân' (Personal), and 'D'.
- AI Agent:** A section for AI-related tasks.
- Gắn sao (Star):** A section for starred items.
- Tải gửi (Download):** A section for downloaded items.
- Nhập (Import):** A section for imported items.
- Lưu trữ (Archive):** A section for archived items.
- Thông tin (Info):** A section for information.
- Trần Minh Khoa:** A section for a specific user or contact.

Main Content Area:

Trợ lý MeoArc (MeoArc Assistant): A section for managing meeting summaries. It includes a search bar and a list of action items:

- Action Items: ① bổ sung đặc tả Use Case - ② vẽ sơ đồ Use Case - ③ chèn mockup giao diện.
- Deadline: 23:59 thứ Sáu tuần này.
- Người phụ trách: Nhóm 7 (Frontend: Quân - Backend: Khoa).

Meeting Summary (Brief cuộc họp): A section for summarizing meeting discussions. It includes a search bar and a list of action items:

- Meeting Brief: NHẮC NỘP BÁO CÁO SRS - NHÓM 7
- Điểm chính: nhóm cần hoàn thiện & nộp bản SRS đúng hạn.
- Action Items: ① bổ sung đặc tả Use Case - ② vẽ sơ đồ Use Case - ③ chèn mockup giao diện.
- Deadline: 23:59 thứ Sáu tuần này.
- Người phụ trách: Nhóm 7 (Frontend: Quân - Backend: Khoa).

Bottom Section:

- Digest hôm nay (Today's Digest):** A section for daily summaries.
- Triage hộp thư (Mailbox Triage):** A section for managing the mailbox.
- Brief cuộc họp (Meeting Brief):** A section for meeting summaries.
- Phản loại tự động (Automatic Classification):** A section for automatic classification.
- Tóm tắt thư chưa đọc (Unread Email Summary):** A section for summarizing unread emails.
- Lưu trữ thư bản tin (Archive Newsletters):** A section for archiving newsletters.
- Soạn trả lời giáo vụ (Draft Reply to Administration):** A section for drafting replies to the administration.

Search Bar: A search bar at the bottom of the main content area with the placeholder text "Nhấn cho trợ lý... ví: 'lưu trữ thư bản tin'".

5

AI Usage Declaration

Based on the provided AI Usage Guideline document, teams must submit their AI usage declaration here. Students are encouraged to leverage AI tools creatively and effectively in this project.

ChatGPT. GPT-4o, chatgpt.com, accessed 19:59 on June 17, 2026, prompt: “tôi đảm nhận viết phần 3.2.1 viết báo cáo bằng chữ, nêu rõ chi tiết về các yêu cầu chức năng của hệ thống, với báo cáo phân tích yêu cầu như này thì có lưu ý gì trong việc viết và triển khai k (attached Template1)”, this is used to know and clearly understand the standard technical writing; AI told me to be careful not to include NFC, the FCs should have it, important notes for numbering <gpt-fc-req>

ChatGPT. GPT-4o, chatgpt.com, accessed 10:07 on June 18, 2026, prompt: “chuyển đoạn sau thành tiếng anh chuẩn cho các báo cáo requirement analysis ...(nội dung mục 3.2.1 bằng tiếng Việt)”, this is used to translate to English; AI provides accurate translations of requirement reports. <gpt-translate>

Gemini, 3.1 Pro, gemini.google.com, accessed 22:06 on June 18,2026, prompt:”tôi đang định làm dự án và file trên là ý tưởng của dự án, tôi muốn thiết kế một web thật ấn tượng, bạn hãy gợi ý cho tôi”, this is used to ask AI for recommendation of web layout, color; AI provide me some of optimal layouts and also the architecture should have in the website <gemini-layout>

Gemini, 3.1 Pro, gemini.google.com, accessed 22:15 on June 18,2026, prompt:”tôi cần bạn chỉ đánh giá, soi lỗi chưa hoàn thiện để cải thiện hình ảnh giao diện có thêm chiều sâu và màu sắc nổi bật hơn. Đồng thời nêu ra hướng giải quyết”, this is used to ask AI for feedbacks based on the pictures of my website UI; AI provide me the unqualified details such as the boring colors, depth & layering, subtle effects <gemini-layout>

Gemini, 3.1 Pro, gemini.google.com, accessed 22:32 on June 18,2026, prompt:”tôi cần cặp màu trẻ trung nhưng vẫn giữ đúng tinh thần như file md tôi gửi thì nên chọn gì”, this is used to ask AI for recommendation of the outstanding color combination; AI provide me some of combinations such as Powder-Blue & Hermes Orange or Luxury Noir & Gold <gemini-layout>

Gemini, 3.1 Pro, gemini.google.com, accessed 22:45 on June 18,2026, prompt:”bạn đưa ra nhận xét về giao diện light và dark”, this is used to ask AI for the feedbacks based on the latest edition of the website; AI provide me compliments and also suggest me some action to raise the layout’s quality <gemini-layout>

Gemini, 3.1 Pro, gemini.google.com, accessed 22:45 on June 18, 2026, prompt: "tôi muốn hơi hướng màu như hình cho trẻ trung nhưng mình đang để bảng vị màu light mode như `/* === LIGHT – "Summer Holiday" (nắng, biển, cát) === */ --background: #f8f0e5; /* cát nắng */ --surface: #fdf9f2; ..... --shadow-lg: 0 2px 4px rgba(58, 46, 40, 0.06), 0 10px 20px -6px rgba(58, 46, 40, 0.14), 0 30px 56px -16px rgba(58, 46, 40, 0.24); }` tôi chỉ sợ kết hợp màu của hình sao cho không bị chói mắt nhưng vẫn đủ tinh tế. Bạn hãy hướng dẫn và gợi ý thêm", this is used to ask AI for the recommendation based on the picture I sent to apply its color into the website; AI provide me the strategy of combining colors and also supply me the CSS code instead of not asking [<gemini-layout>](#)

6

Presentation

Teams are required to record presentation videos of their work using the provided template.

Every team member must participate in the presentation, specifically covering the sections their contributed to the project.

Videos must not exceed 30 minutes.

Please upload your videos to Youtube (set to either Unlisted or Public).

Provide the Youtube links to the field below.

[NHÓM 7 - REQUIREMENT ANALYSIS](#)

7 Reflective Report

Which sections of this template are most helpful for implementation, and which are unnecessary? Provide specific examples to support your reasoning.

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Edited by: 24127529 – Nguyễn Chí Tài

Reviewed by: 24127545 – Nguyễn Ngọc Thiên

- Sections are helpful for implementation:
 - **Section 2 – Problem Statement** is also highly practical, as it forces the team to clarify the operating environment and technical constraints from the outset, preventing mid-project changes to the technology stack.
 - **Section 3.2.1 – Functional Requirements** is useful because it serves as a basis for verifying the completeness of the Use Case Model – every functional requirement should map to at least one use case.
 - **Section 4 – Requirements Analysis (Use Case Model & Use Case Specification)** is the most practically valuable part of this template. The Use Case Diagram gives the entire team a clear picture of the system's scope and the relationships between actors, while the Use Case Specification table – covering Pre-Conditions, Main Scenarios, Alternative Scenarios, and Non-Functional Constraints – provides a direct foundation for writing test cases and designing interfaces. For example, UC003 (View Email List) and UC005 (Search & Filter Emails), when fully specified, allow developers to immediately identify which APIs need to be integrated (e.g., Gmail API, LLM API) without needing further clarification.

- Sections are unnecessary:
 - **Section 1 – Member Contribution Assessment**, while necessary for team evaluation, requires detailed Jira/Git screenshots that generate significant paperwork without directly contributing to software quality. A concise summary table with a repository link would be sufficient.
 - **Section 6 – Presentation** does not belong in a traditional SRS document and could be separated into a standalone deliverable. Embedding a YouTube link inside a technical specification reduces its professionalism when presented to real-world stakeholders.
- Conclusion: Overall, this template effectively introduces students to real-world requirements analysis processes. However, in an actual project context, the Presentation section and individual contribution evidence should be separated from the SRS so that the document remains focused purely on technical specification./.